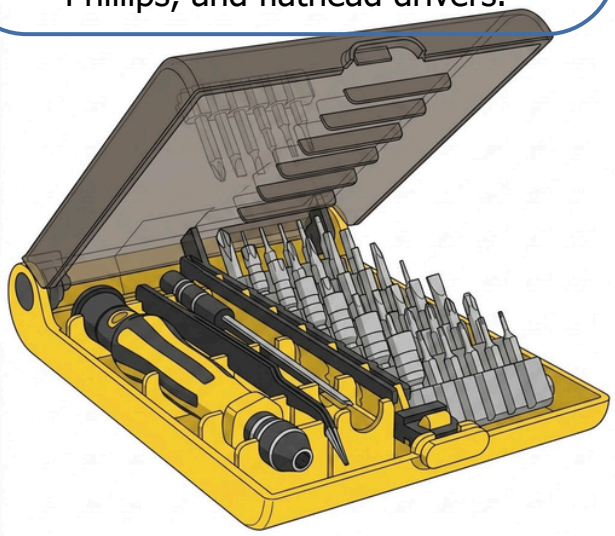
Robot Assembly Guide

Version 1.0

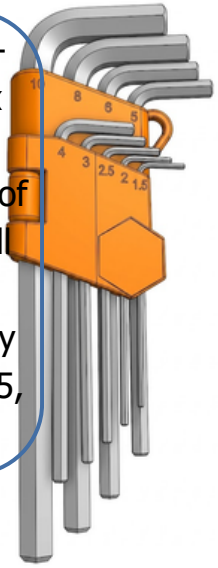
It is recommended to use a cordless drill/driver, as the assembly requires driving many screws; this will significantly reduce effort. For driver functionality, choose a model with a slip clutch to effectively control torque. In this case, all screws should be tightened to 1 Nm using the drill, then, if necessary, additional torque can be applied manually with an L-key to ensure that the 3D-printed parts are not damaged.

Also, make sure to use a screwdriver set that is compatible with the drill/driver.

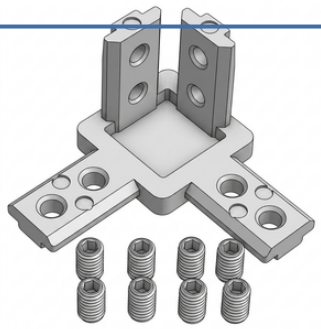
Screwdriver tool set: Any screwdriver set is acceptable, as long as it includes hex, Phillips, and flathead drivers.



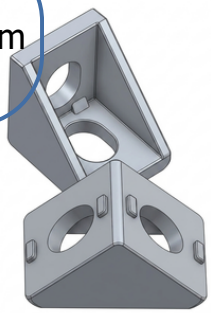
It is suggested to use a 9-piece or larger L-type hex key set (or something similar), and various types of screws, nuts, and bolts will be necessary during this robot build. Make sure they include 1.2, 2.0, 2.5, 3, 4, 5, and 6 mm sizes.

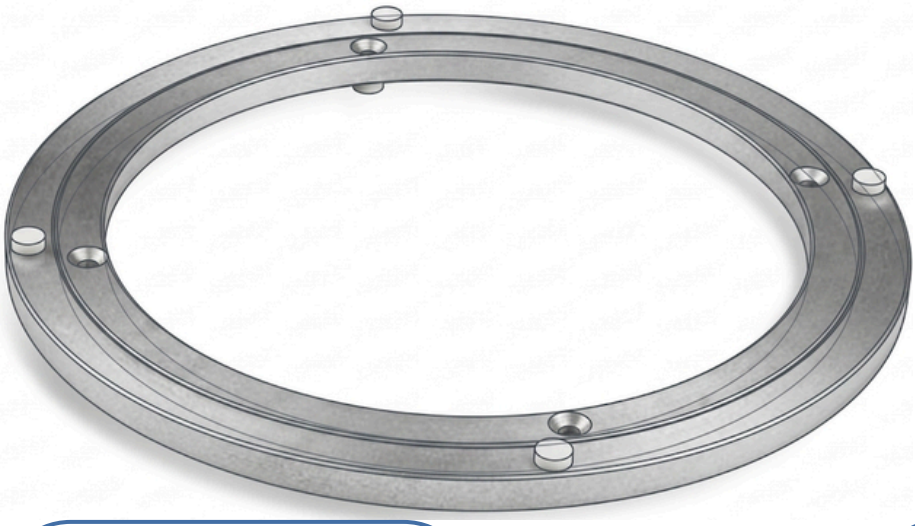


Corner Bracket 3-Way Connector for 2020 V-Slot Aluminium Extrusion Profile



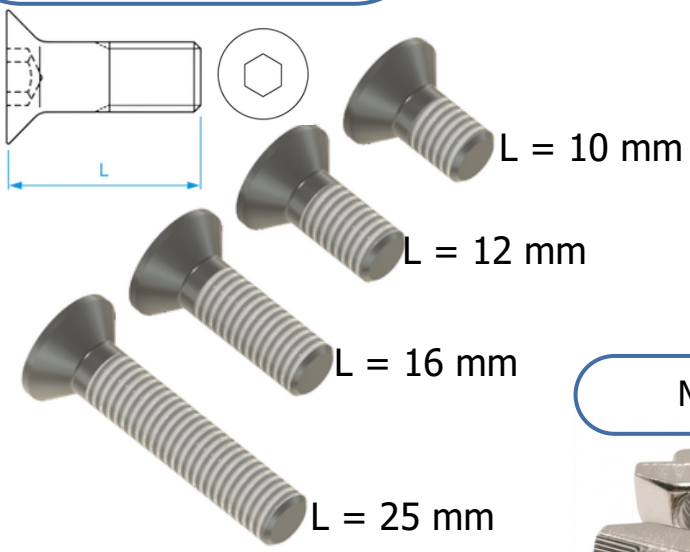
2020 Aluminium Corner Bracket for V-Slot Aluminium Extrusion Profile



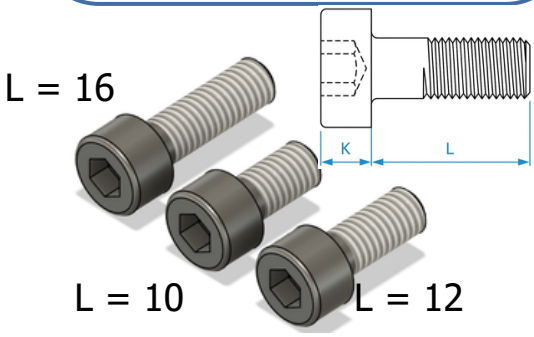


Inch Aluminium Rotating Turntable Bearing Swivel

M5 Hexagon Socket Flat Countersunk Head Cap Screw

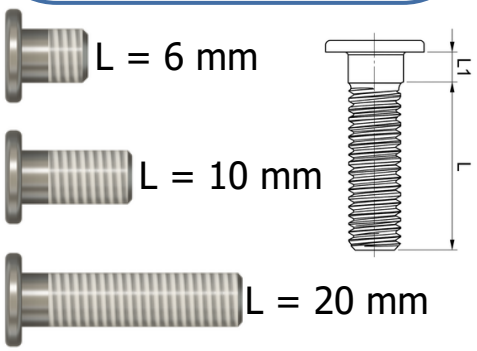


M5 Socket Head Cap Screw
K ~ 5mm

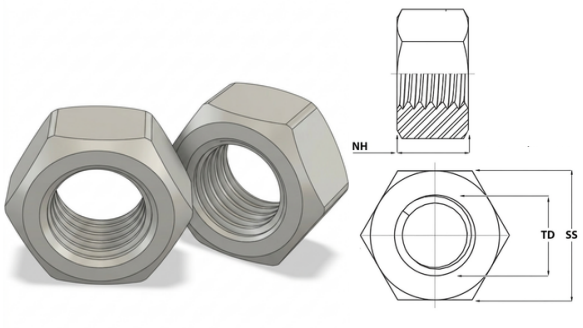


M5 Drop-in T-Nut

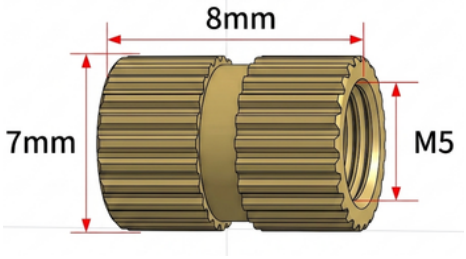
M5 Low Profile Flat Head Carbon Steel Hex Cap Screw
L1 = 1.5 mm



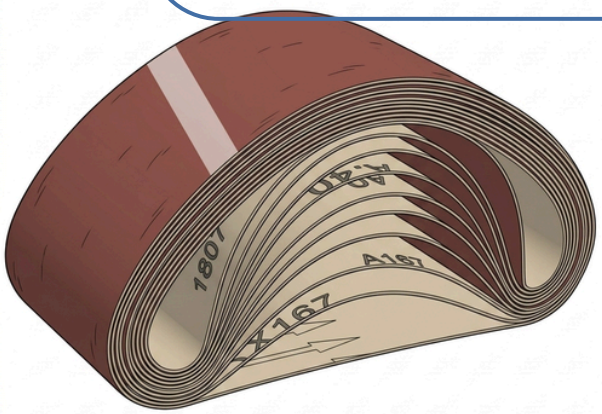
M5 Hex Nut,
TD: 5mm, SS: 8mm, NH: 4mm



M5 Inserts
Height: 8mm, Outer Diameter: 7mm

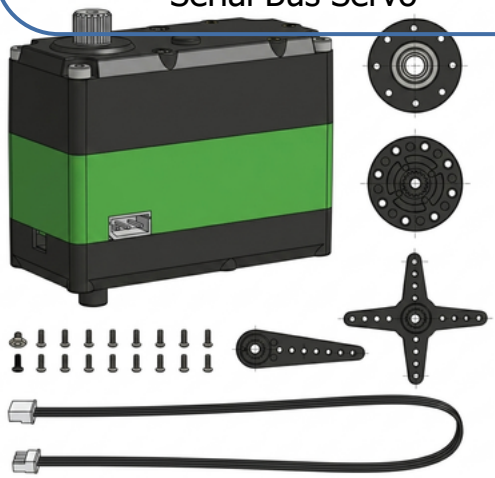


Sandpaper is also required if you want to smooth out rough surfaces on the 3D-printed parts.

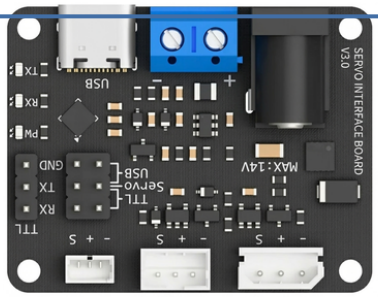


Three Hiwonder HX-35HM Serial Bus Servo

One Hiwonder HTD-85H Intelligent Serial Bus Servo

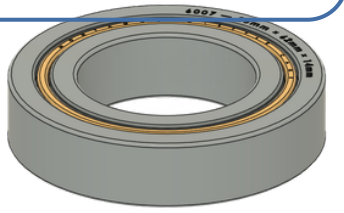


One Hiwonder BusLinker V3.0 Servo Debugging Board, Embedded Robotic Arm Servo Driver Module Controller

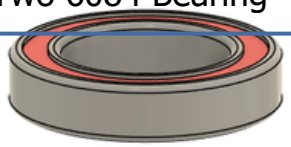


One 12V 10A SMPS Metal Case Power Supply

One 6007 Bearing



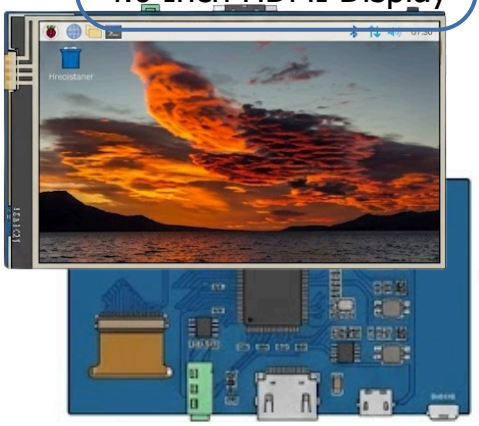
Two 6084 Bearing



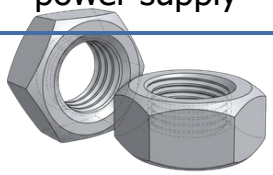
Note: Before starting the build test the motors and make sure they work properly, give them their unique ID (1-Hip Motor, 4-Tilt Motor, 5-Pan Motor) and after the testing is done, put them in servo mode with value 500 to avoid any mechanical incidents in future.

Viewsonic TD1655 Display

4.0 Inch HDMI Display

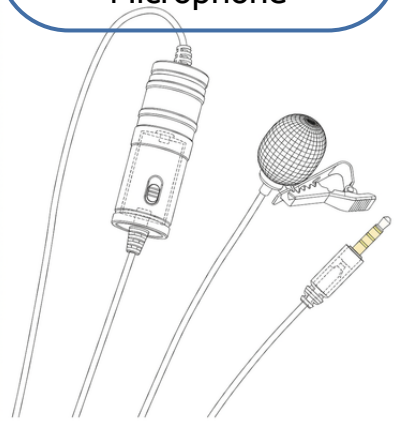


M4 Hex Nuts for the power supply



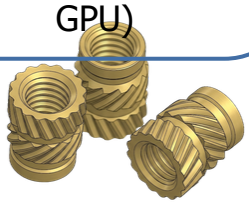
8mm/10mm M4 Screws for the Power Supply Unit and Speaker

Boya BY-M1 Omnidirectional Lavalier Microphone

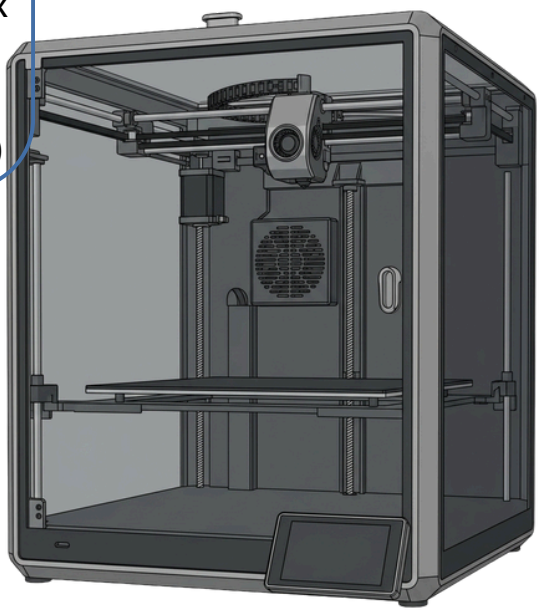


52mm speaker driver (Here the one from Havit SK717 was used along with the controller)

M3*5.7-OD4.6 Inserts (For Motherboard and GPU)

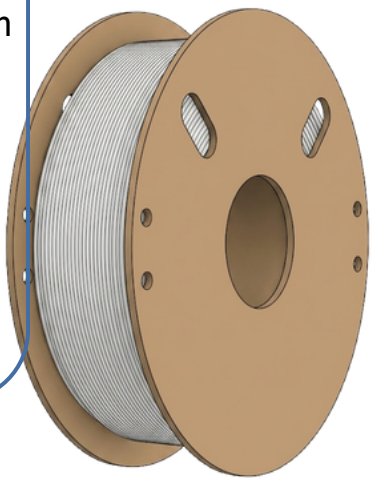


A good 3D printer with build volume of atleast 300 x 300 x 300 mm
(Here we will be using the Creality K1 Max without CFS)

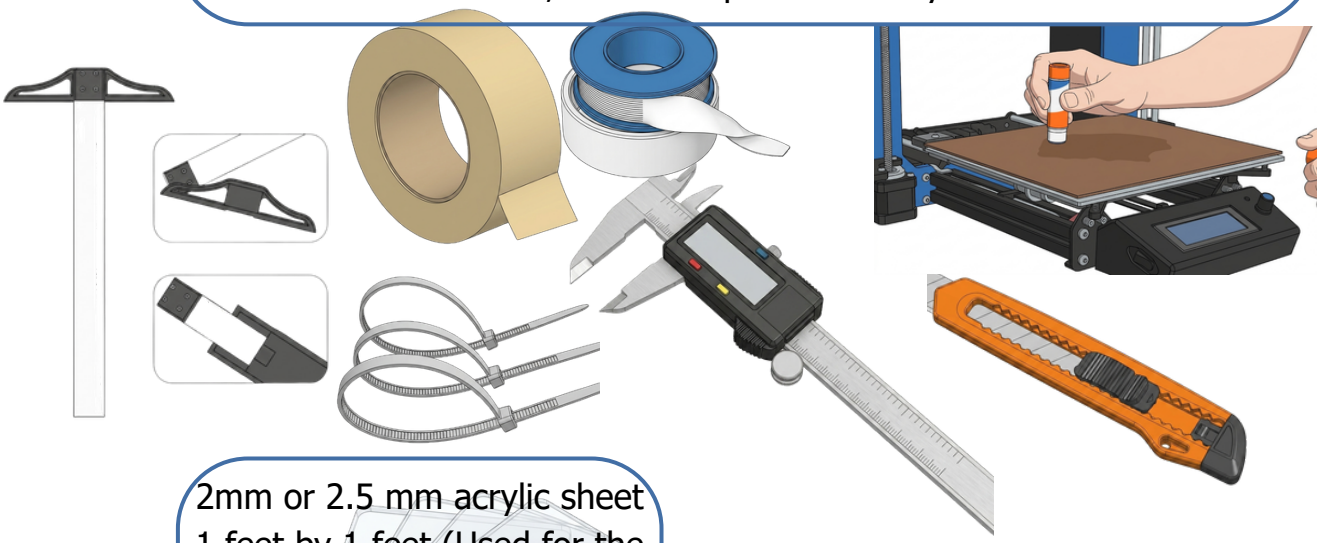


3D printer filament. We suggest using PLA (white) for this but make sure to use from same company to get a consistent color across whole build.

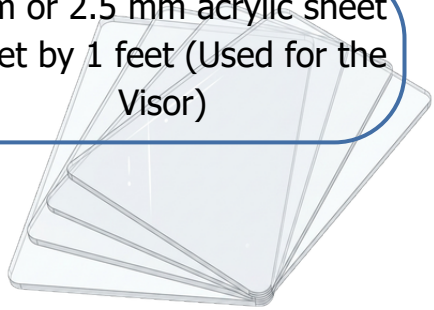
In our build we used Creality Hyper PLA (White) For all the parts.



You may also need additional tools, such as pliers, wire cutters, a utility knife, a 3D printer scraper, adhesive for 3D-printed parts, calipers for accurate measurement, and a T-square. Use any other tools as needed.

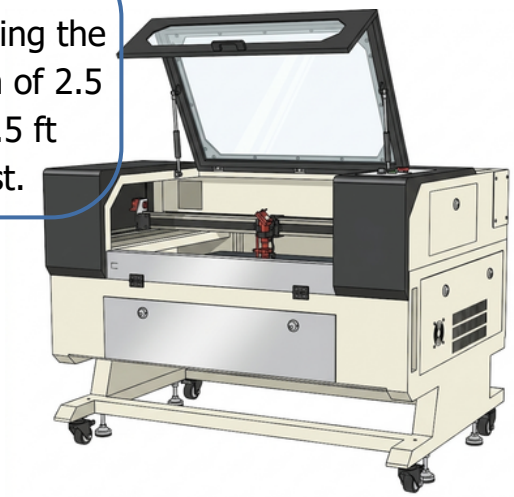


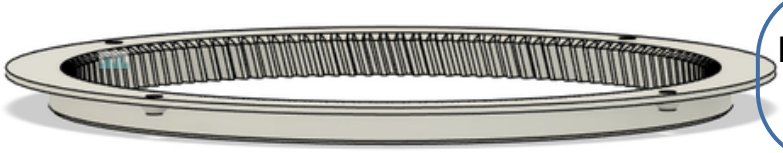
2mm or 2.5 mm acrylic sheet 1 feet by 1 feet (Used for the Visor)



3mm Frosted Acrylic Sheet (2.5 feet by 2.5 feet) for the outer part of the base. You can use any color for you base.

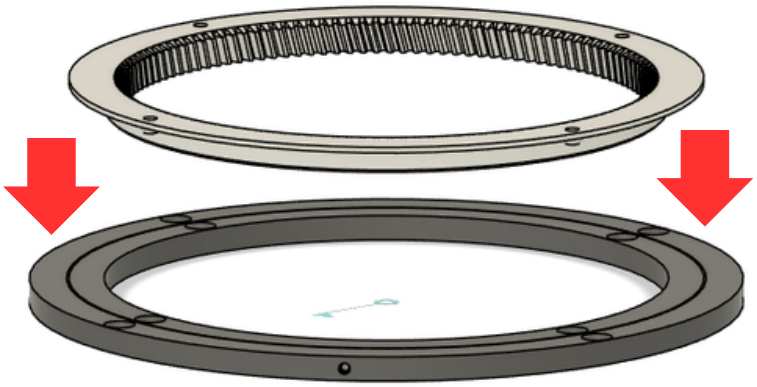
Access to a laser cutter having the dimension of 2.5 ft by 2.5 ft atleast.

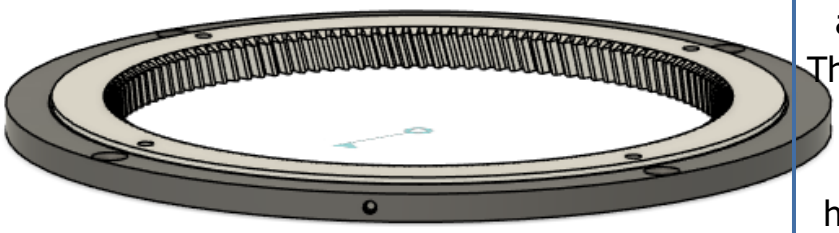




Print the "Inner Hip Gear," which will control the hip rotation.

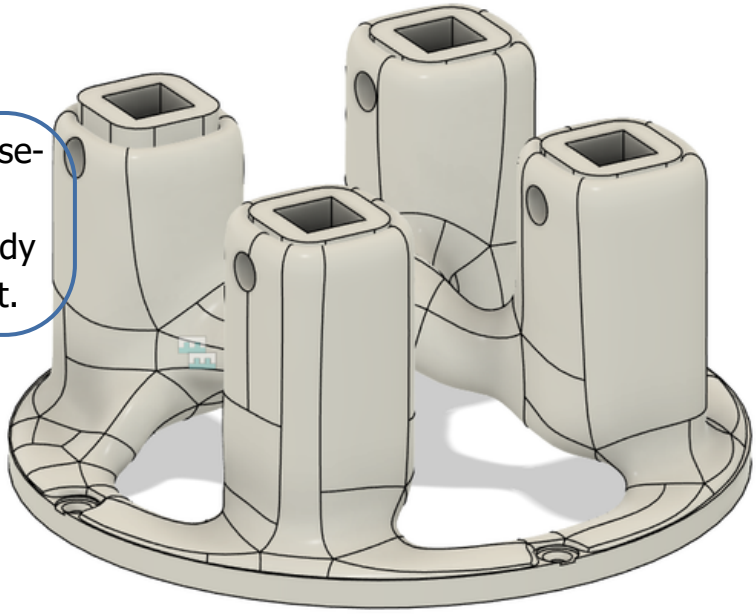
Place the gear inside the Lazy Susan Bearing (10-Inch Size)



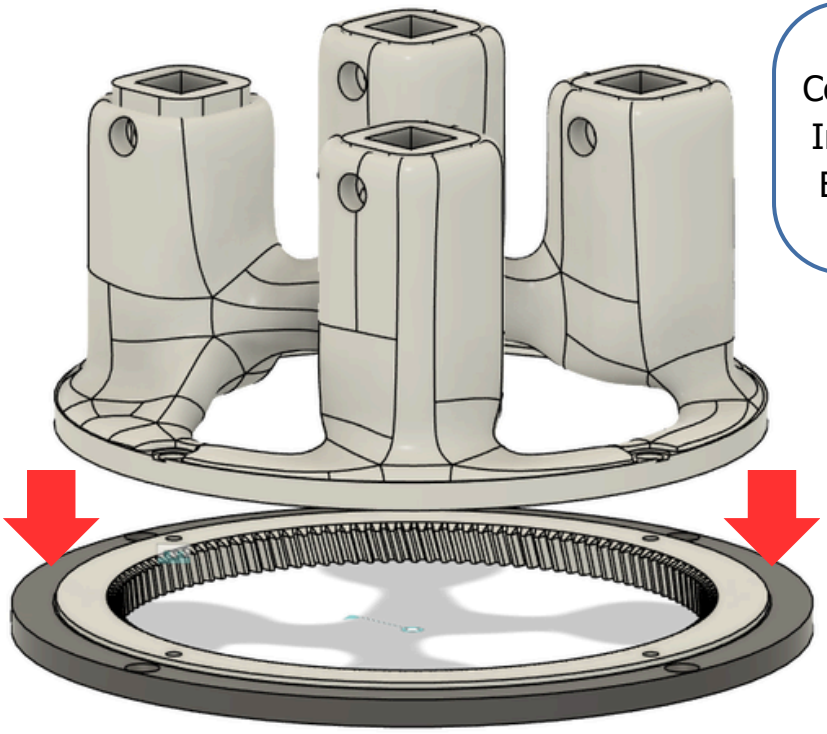


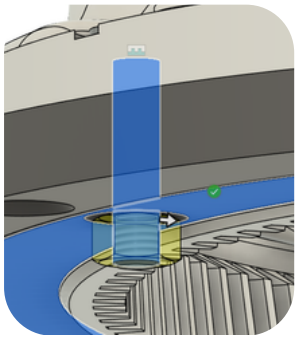
This is how it should look once it is complete. Make sure the holes align with those on the gear. The gear is called the "Inner Hip Gear." The tolerance is extremely tight, so you may have some difficulty placing it, but this is intentional.

Now print the part called the "Base-to-Core Connector," which will connect the lower part of the body with the upper part of the robot.

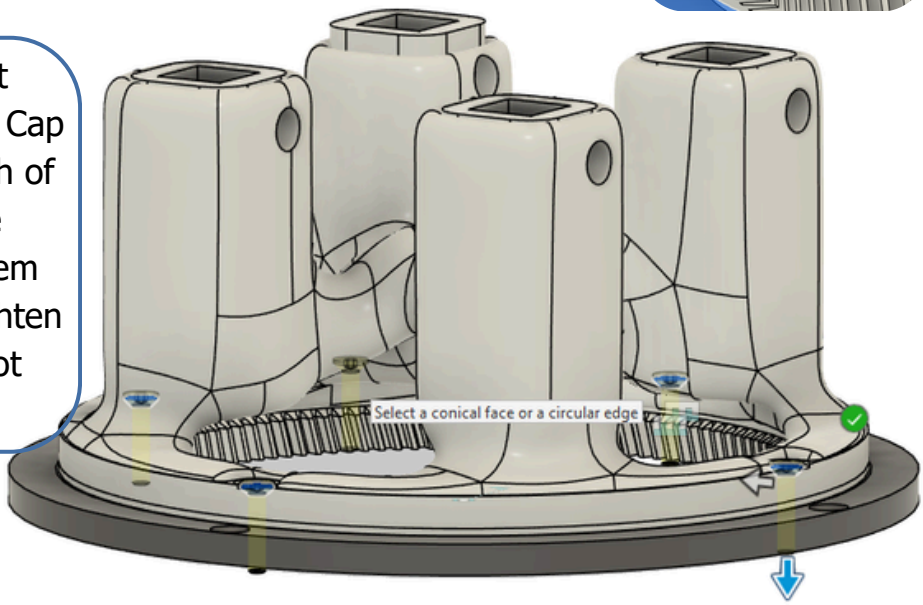


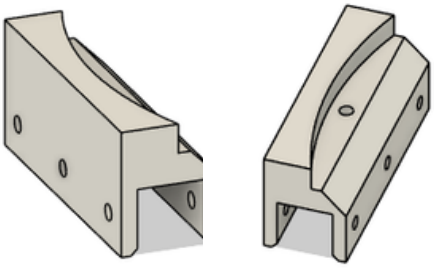
Now place the Base-to-Core Connector on top of the existing Inner Hip Gear and Lazy Susan Bearing setup, making sure to align the holes.



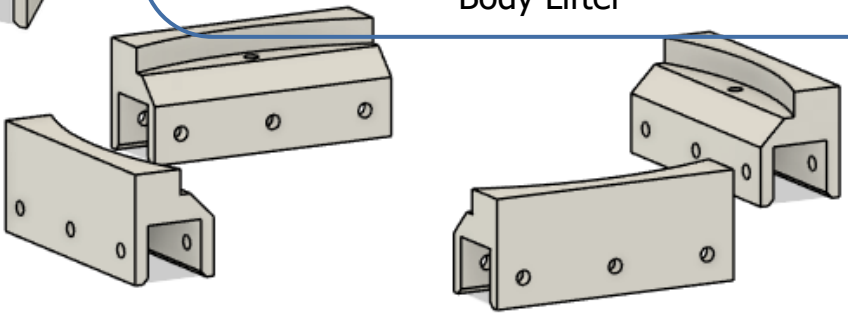


Then place the M5 Flat Countersunk Head Socket Cap Screws (25 mm) into each of the four holes. On the opposite side, secure them with M5 hex nuts, and tighten firmly so that hey do not loosen easily.



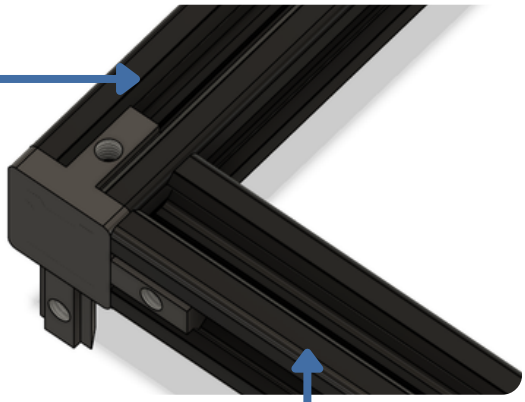


Now print the "Main Body Lifter" You will need to print four of these. This part will support the entire body of the robot. Together they are named "Main Body Lifter"



Now we will start building the lower aluminum structure.

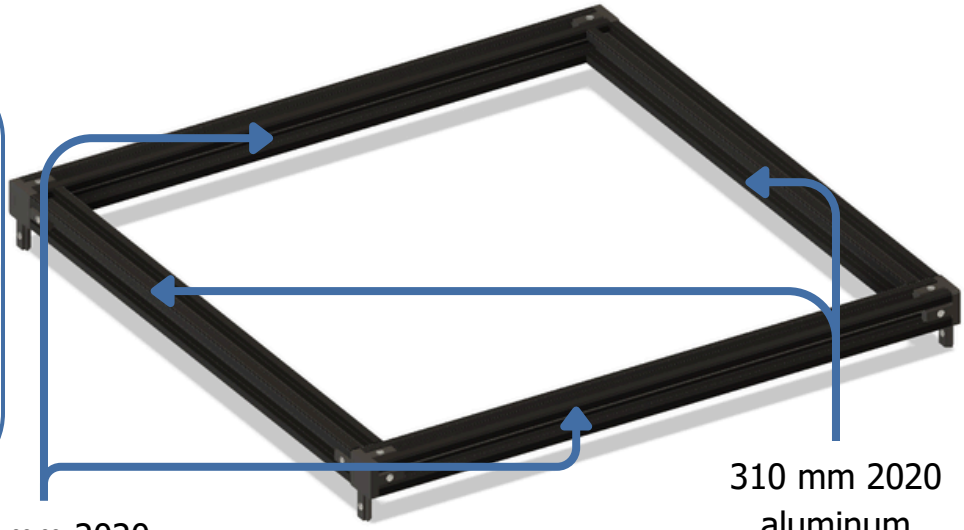
340 mm 2020 aluminum profile



310 mm 2020 aluminum profile

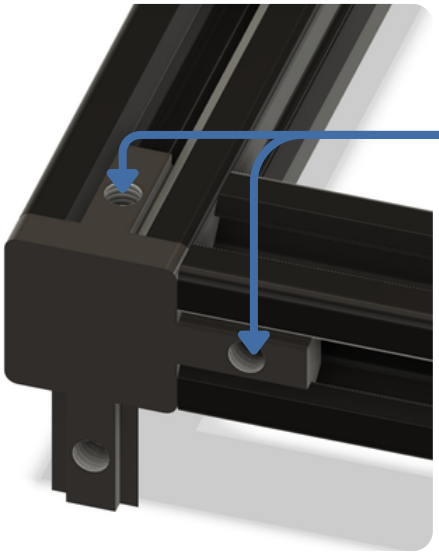
Now grab two 340 mm 2020 aluminum profiles and two 310 mm 2020 aluminum profiles, and join them as shown. The 340 mm profiles should be placed on opposite sides, and the 310 mm profiles should also be placed on opposite sides, forming a square.

Then install three other Corner Bracket 3-Way Connectors for 2020 profiles, one on each corner, ensuring that the protruding part of each corner bracket faces downward. Finally it should look like this.

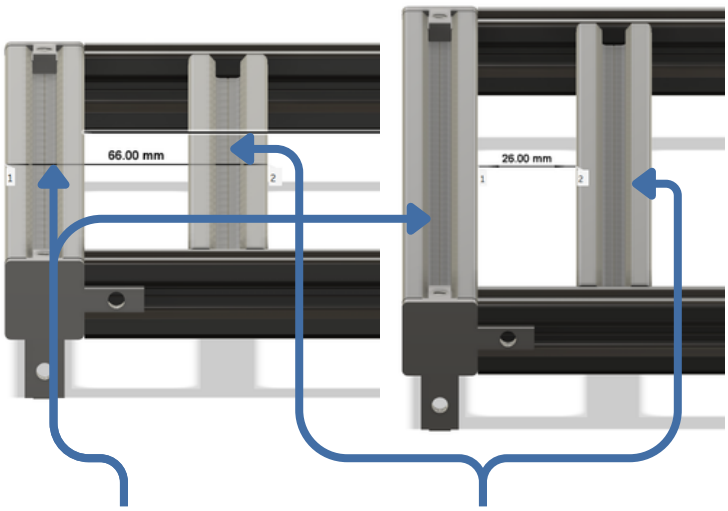


340 mm 2020 aluminum profile

310 mm 2020 aluminum profile



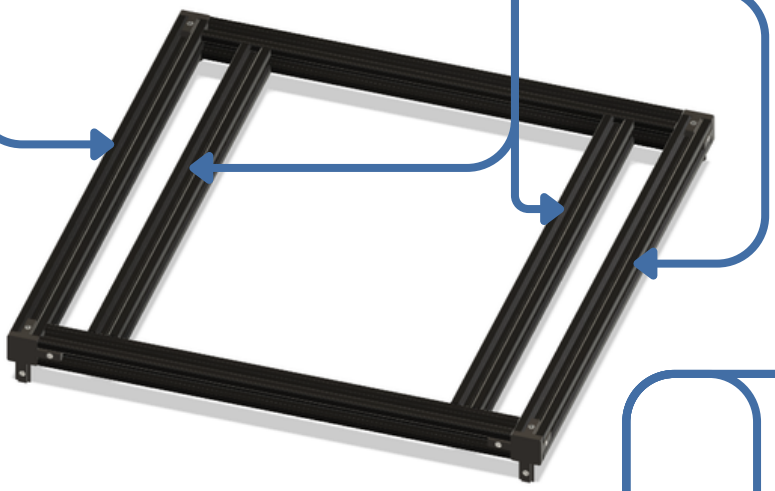
Now use the proper L-type hex key set to tighten the screws of the corner bracket 3-way connectors (do this for all four of them) so that it becomes a rigid square.



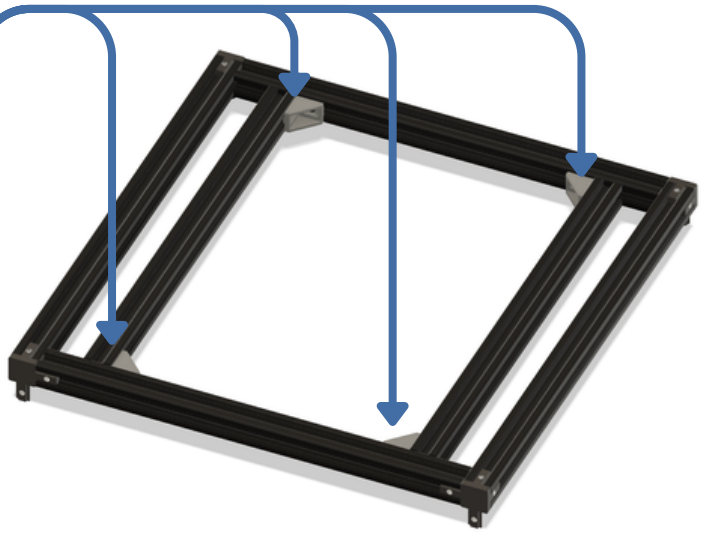
Now insert the 310.00 mm 2020 aluminum profiles into the newly created frame. Ensure that the distance between the outer faces of the profiles (between the 340 mm and 310 mm profiles) is 66 mm, and that the distance between the inner faces is 26.00 mm. Repeat this on the other side (between the other 340 mm and 310 mm profiles) so that the final setup matches the one shown in the image.

340 mm 2020 aluminum profile

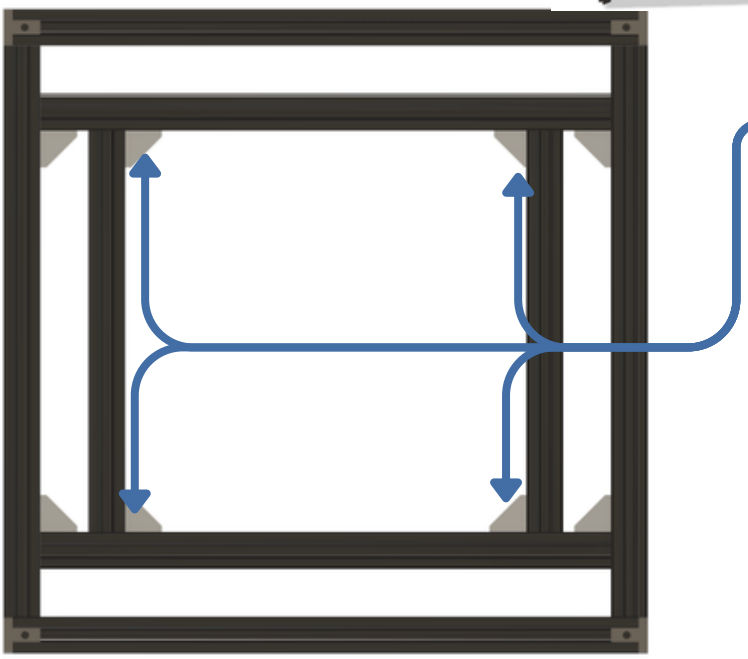
310 mm 2020 aluminum profile



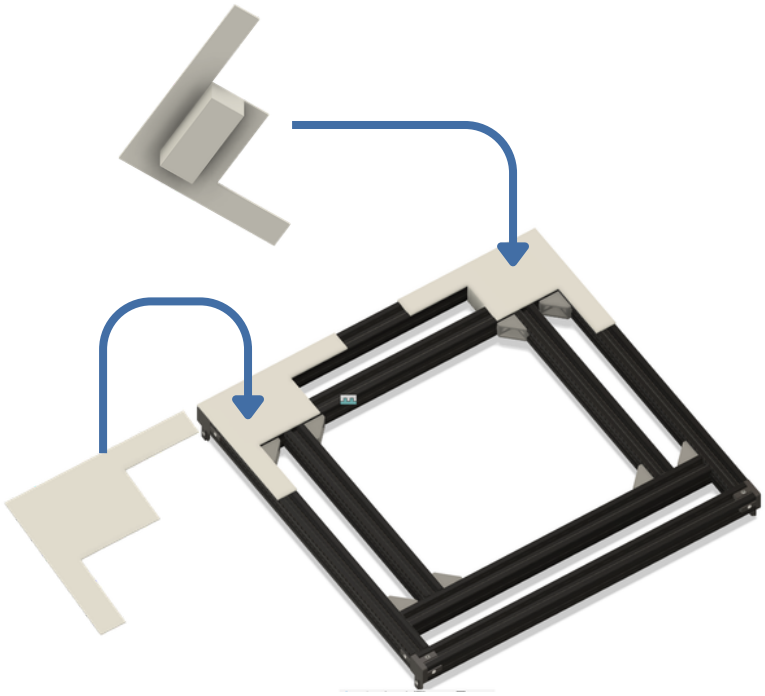
Now use the Aluminium Corner Bracket to secure the two 310mm aluminum profiles with the existing structure. Use proper screw and T nut for this. Here 6mm Socket Flathead was used.



Now attach the two 218.00 mm 2020 aluminum profiles to the previously installed 310.00 mm profiles within the frame, maintaining the same spacing as before: 26 mm between the inner faces and 66 mm between the outer faces.

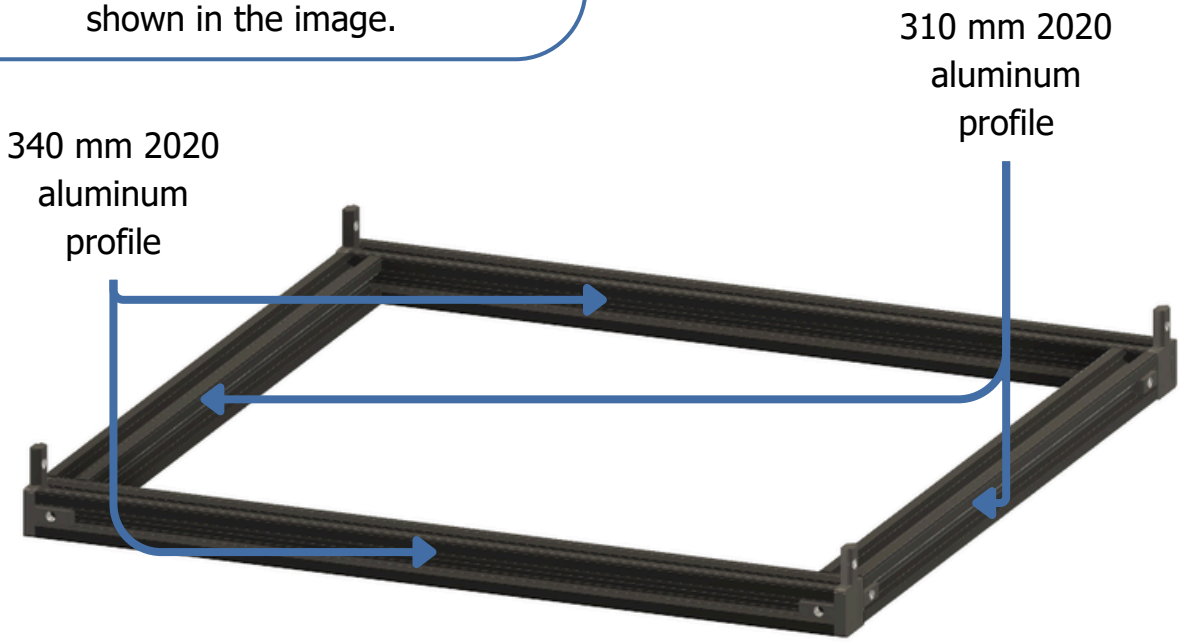


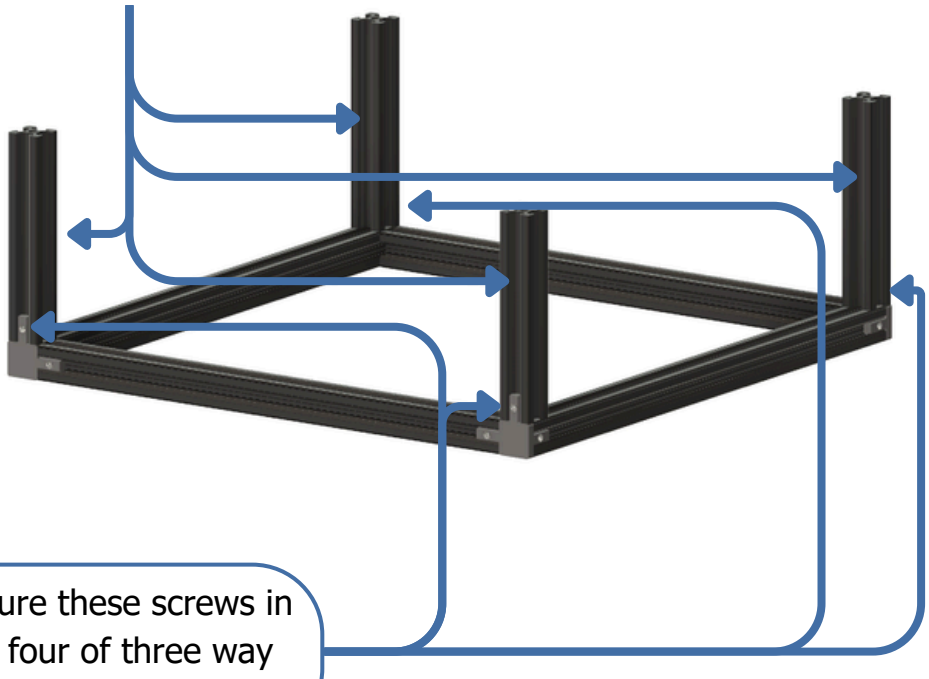
Now, as before, use two corner brackets to connect the profiles. Secure them tightly using appropriate screws and T-nuts.



If you are having difficulty accurately setting the 26 mm spacing, you can print the files "Helping with Measurements 1 (Spacing between profiles on the top side of the box)" and "Helping with Measurements 2 (Spacing between profiles on the top side of the box)." These files will save time and help ensure correct spacing during assembly. Once all components are secured using screws, these can be removed easily. After tightening the screws, recheck the spacing and adjust for any tolerances if necessary. These parts also help ensure that all profiles are properly aligned.

Now use the remaining 340 mm 2020 aluminum profiles and two 310 mm 2020 aluminum profiles to assemble the lower part of the robot base. Ensure that the protruding side of each corner bracket faces upward. Secure the three way corner brackets tightly using appropriate screws and T-nuts, taking care not to overtighten and damage them. The assembled structure should match the configuration shown in the image.



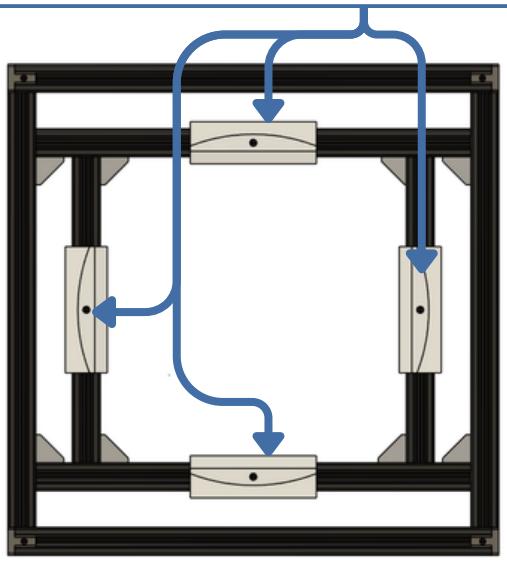


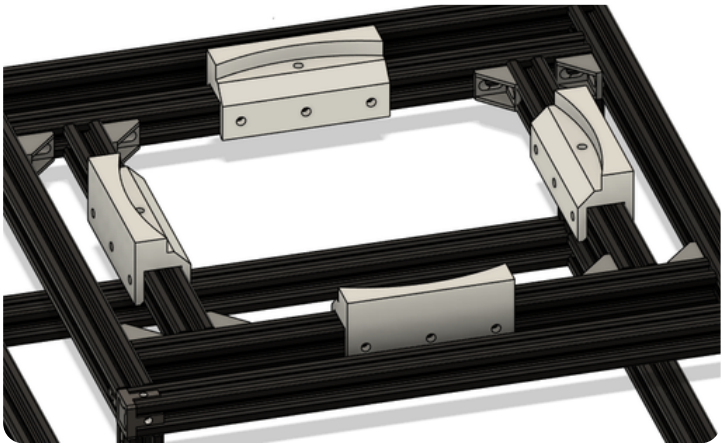
Now attach the four 118.00 mm 2020 aluminum profiles to the newly assembled lower part of the robot base, specifically to the protruding sections of the three-way corner connectors. Secure them using the screws provided with the three-way connectors.

Secure these screws in all four of three way connectors.

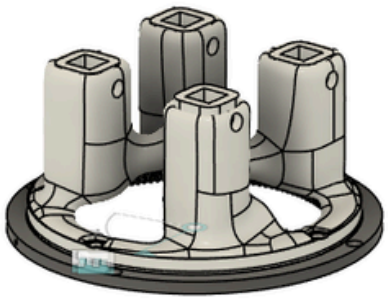
Once this is complete, place the previously assembled top part of the robot base onto the four 118.00 mm profiles, aligning the protruding sections of the top frame with the ends of the profiles. Join them carefully, then secure the connections by tightening the three-way corner connectors. The base frame of the robot is now complete, and you can proceed to assembling the main body.

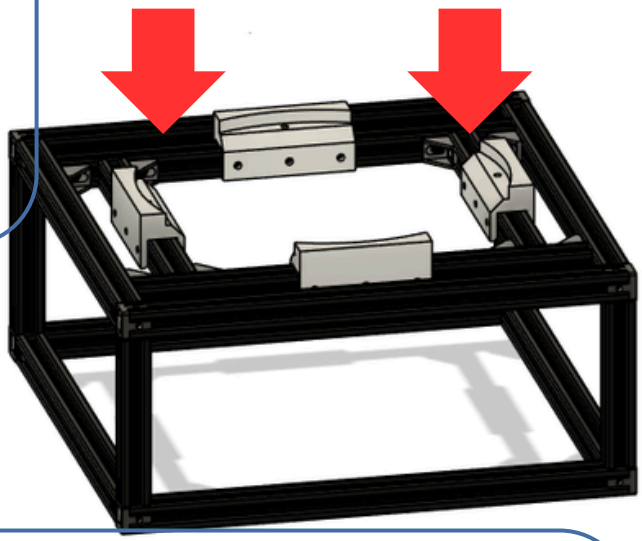
Place the four ”Main Body Lifter” parts that you 3D-printed earlier inside the inner square of the top frame of the base. Do not secure them with screws at this stage. Arrange them so that they form a circular pattern.

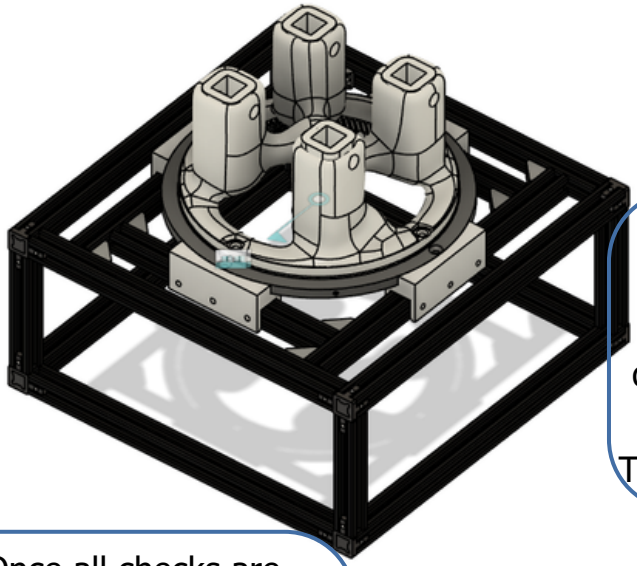




Now place the previously assembled bottom unit (including the gear and rotary bearing) on top of the "Main Body Lifter" parts. Ensure that the holes in the Main Body Lifters align properly with the outer mounting holes of the rotary bearing. Then use M5 hexagon socket flat countersunk head screws (25 mm) to secure the assembly to the Main Body Lifters. Do not use T-nuts at this stage, as the screws are only intended to provide temporary support and to help verify that the frame dimensions are correctly set.

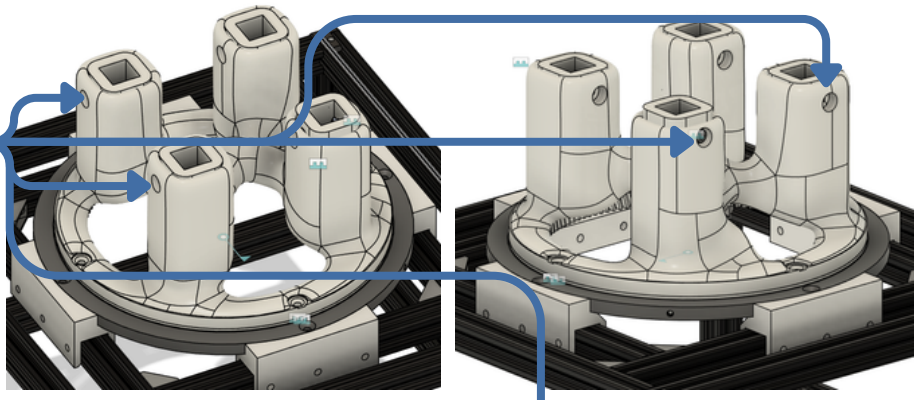




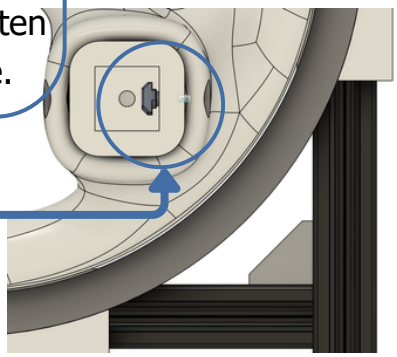


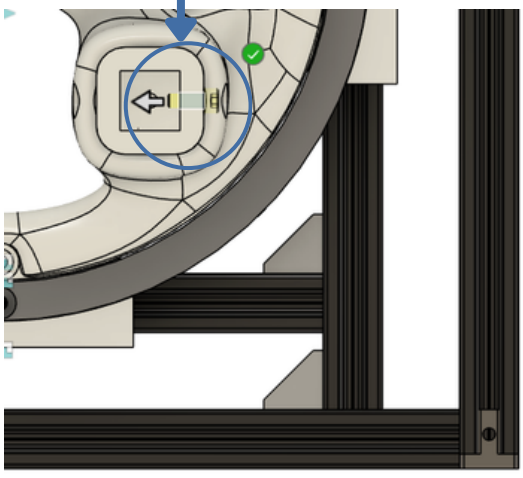
This step may be difficult on the first attempt. If you are unable to position the parts correctly, recheck the alignment and dimensions of the assembled profiles before proceeding to the next step.
This is how it should look like after it is done.

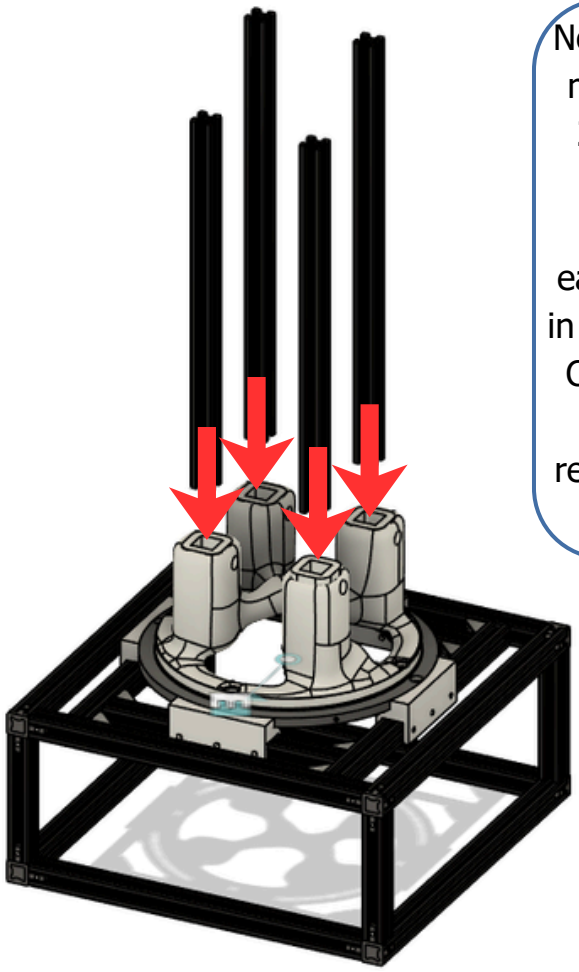
Once all checks are complete, add the four core 2020 aluminum profiles that will support the main body. First, insert M5 × 12 mm bolts through the outer holes of the main part.



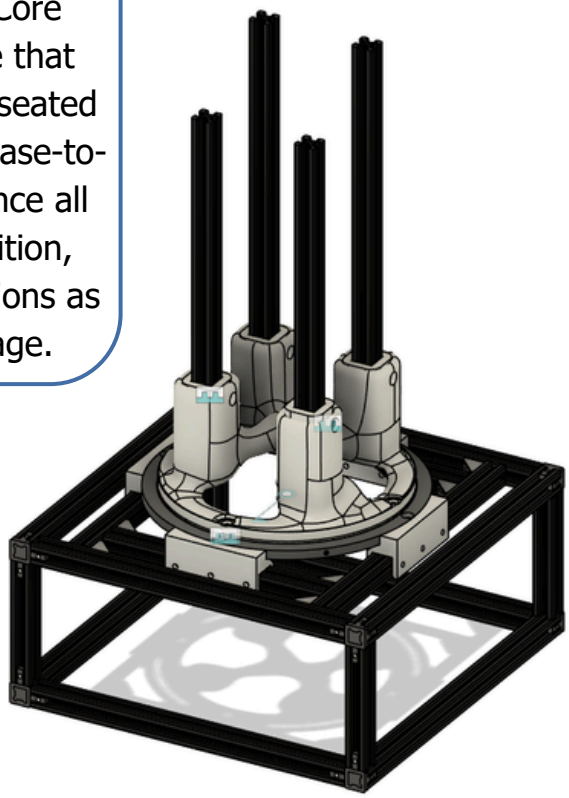
Then, from the inside, thread T-slot nuts onto these bolts. Do not tighten anything at this stage.

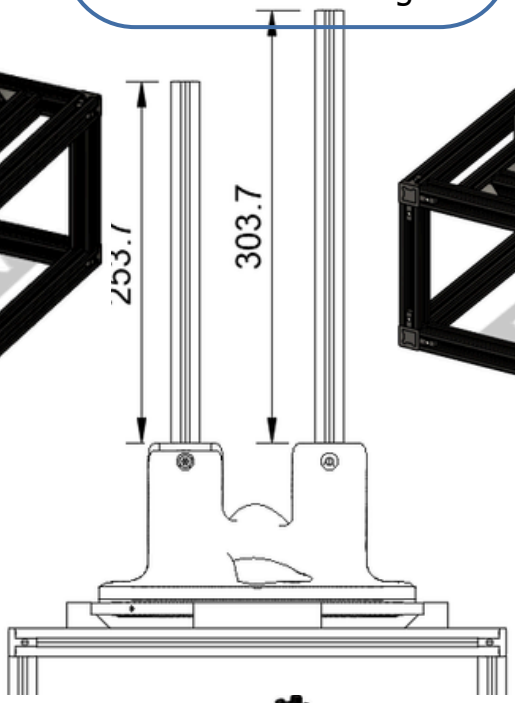




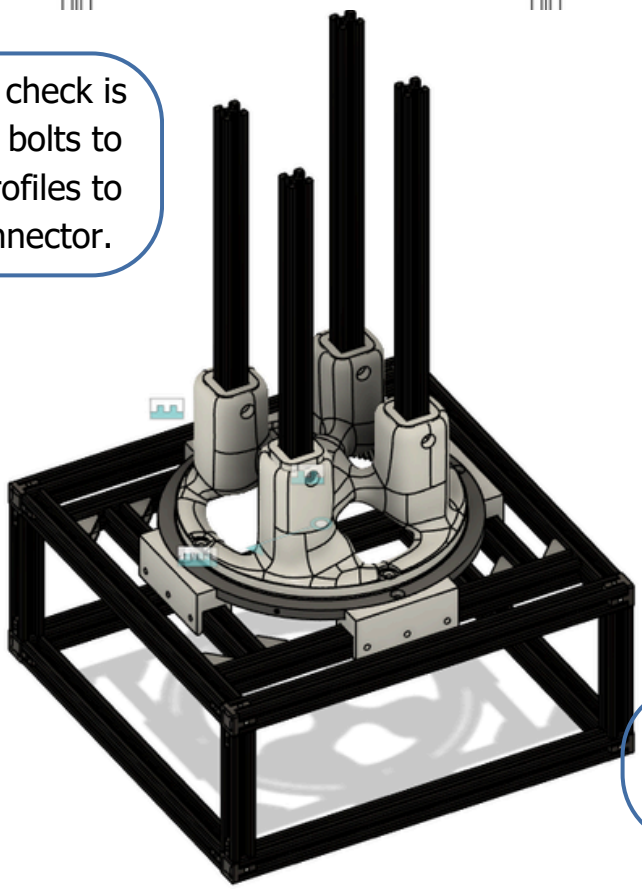


Now insert the two 343.00 mm and two 393.00 mm 2020 aluminum profiles into the Base-to-Core Connector. Ensure that each profile is fully seated in the holes at the Base-to-Core Connector. Once all profiles are in position, recheck the dimensions as shown in the image.



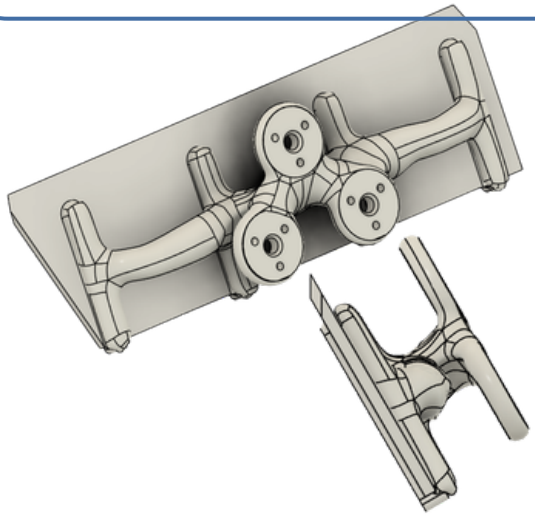


Once the dimensional check is complete, tighten the bolts to securely fasten the profiles to the Base-to-Core Connector.

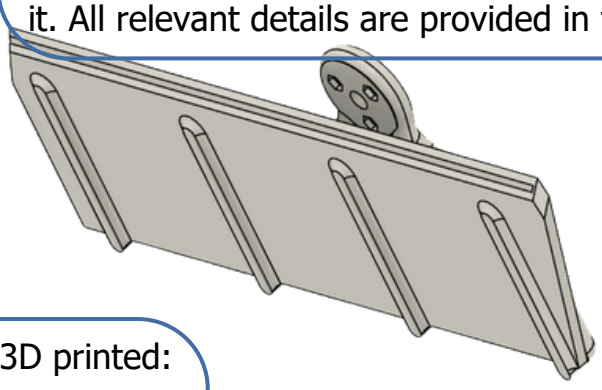


Now our core structure is ready.

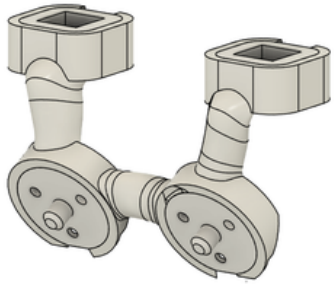
Now we will build the holders for the display. In this build, we will use the TD1655 16” USB-C Multi-Touch Portable Monitor. However, if you want to use a different monitor, you can easily do so, as the display mount is highly modular.



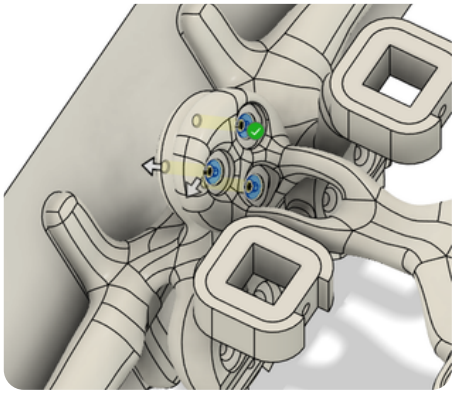
If you are using the TD1655, you need to 3D print the “TD1655 Mount”. If you are using a different display, you will need to design a custom mount for it. All relevant details are provided in the CAD file.

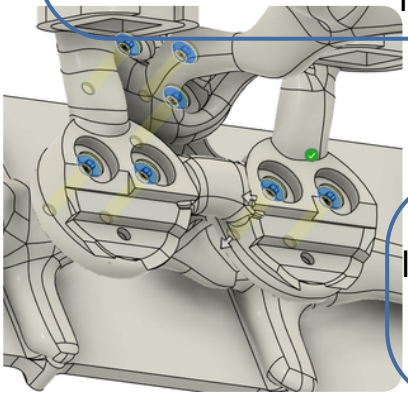


Next, two additional parts need to be 3D printed: “Display Connector 1 With Core Body” and “Display Connector 2 With Core Body”. These parts connect the display mount (TD1655 Mount in this case) to the core structure.



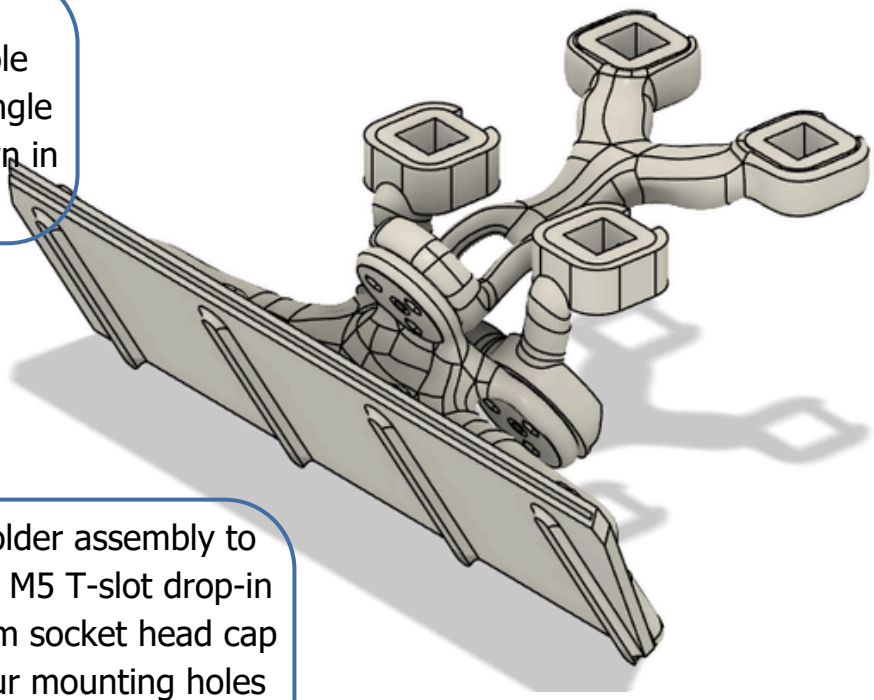
When all three parts (“Display Connector 1 With Core Body”, “Display Connector 2 With Core Body”, and “TD1655 Mount”) are printed, connect them using M5 hexagon socket flat countersunk head cap screws 25mm and M5 hex nuts, as shown in the image.



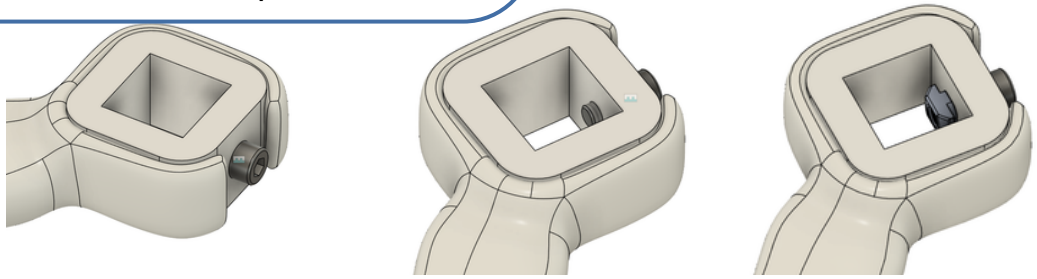


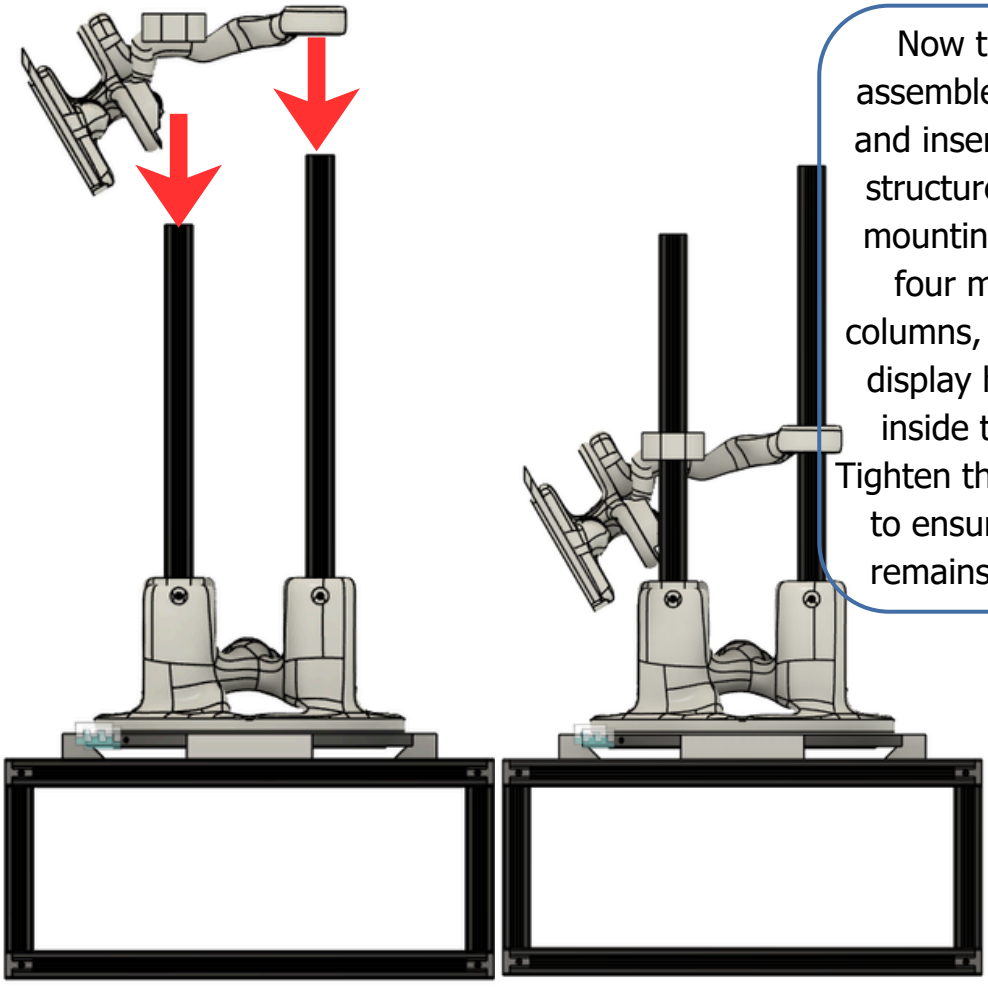
Do not use screws in the lower two holes (as shown in the image). These holes are reserved for future updates.

If everything is done correctly, now the whole setup will be one one single body like done one shown in the image.



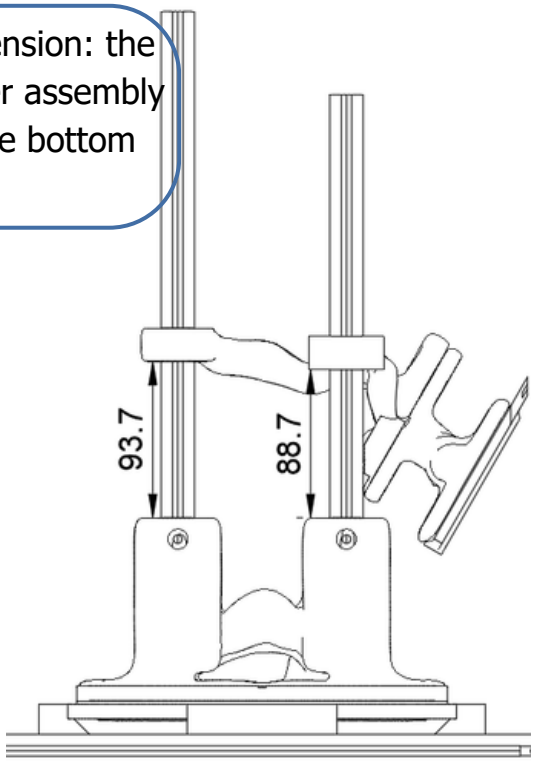
To mount the display holder assembly to the core structure, install M5 T-slot drop-in nuts and use M5 × 10 mm socket head cap screws in each of the four mounting holes on the aluminum profiles.





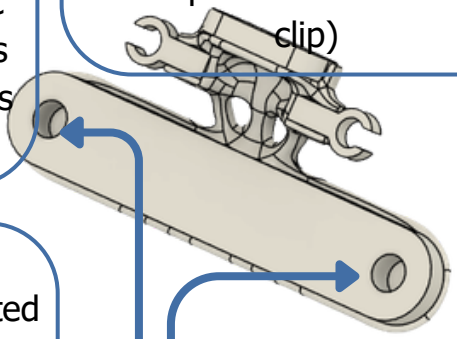
Now take the entire assembled display holder and insert it into the core structure. Align the four mounting holes with the four main aluminum columns, then position the display holder assembly inside the core frame. Tighten the screws securely to ensure the assembly remains firmly in place.

Make sure to maintain the following dimension: the distance between the bottom of the holder assembly mounting hole in the core profile and the bottom frame.

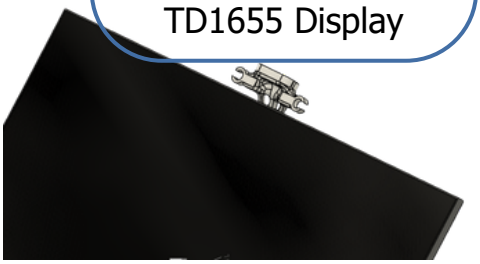


We have to build the "Magnetic Microphone Mount" that will go on top of the ViewSonic TD1655 Display. If you are not using ViewSonic TD1655 Display then this step is not necessary and you have a microphone holder for your own display. Also in this build we will be using a Boya BY-M1 and the mount for the microphone is based on that specefic model. So, if you want to use this mount make sure that your microphone is the same size as this one. Also on top of the magnetic mount there is a 5mm holes for any external attachments you want to add.

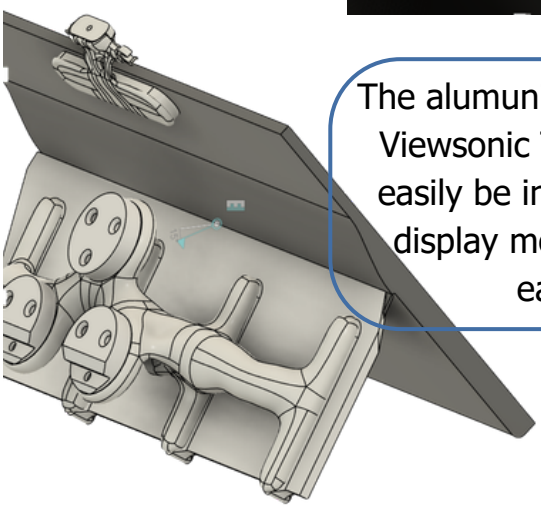
In the "Magnetic Microphone Holder" the Boya BY-M1 will easily fit with a small amount of force (the same way the microphone fits to it's own clip)



You can set the "Magnetic Microphone Holder" to the left, right or center of the top of Viewsonic TD1655 Display



When the "Magnetic Microphone Holder" is printed attach a Nyubudium 8mm diameter and 7.00mm height in the holes. You have to press a bit insert those magnets. We suggest a bench press or vice to do this. Also add a little bit of glue to secure the magnets.

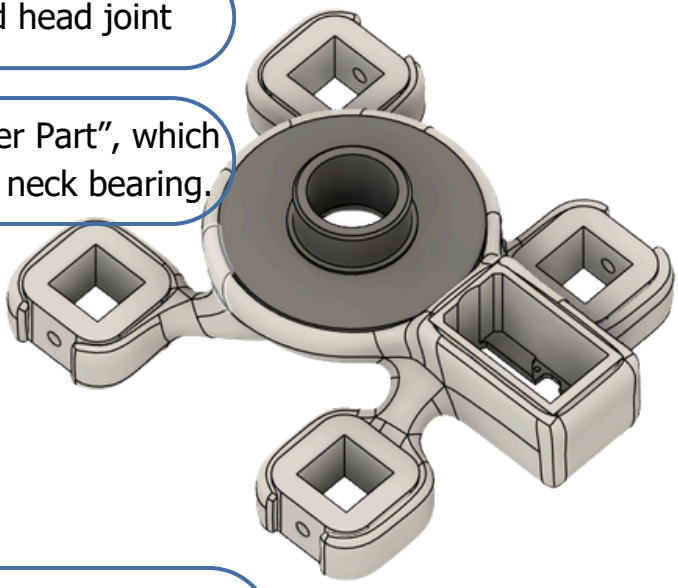


The alumunium backplate of Viewsonic TD1655 should easily be inserted into the display mount we setup earlier.

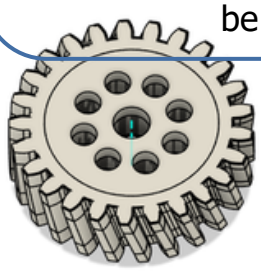
Now our display and microphone is ready too.

Now we will assemble the neck and head joint

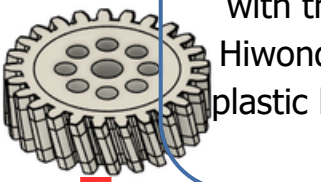
Now print the "Neck Joint Body – Lower Part", which will hold the pan servo motor and the neck bearing.

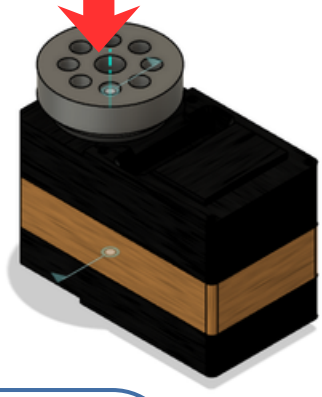


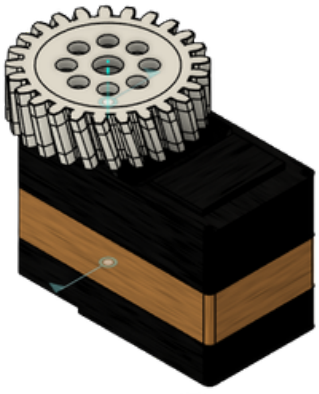
Also 3D print the "Pan Gear Driver" part which will be connected to servo.



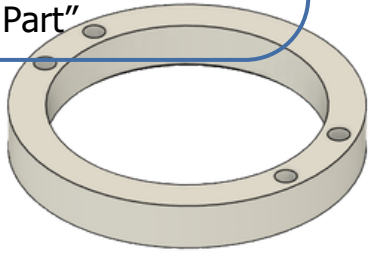
When "Pan Gear Driver" part is printed, connect it with the servo using the screws provided with Hiwonder HX-35HM. (Make sure to use circular plastic horn for this application). The final setup should look like this:



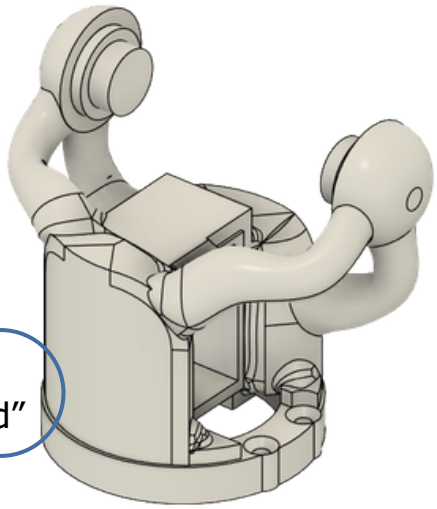




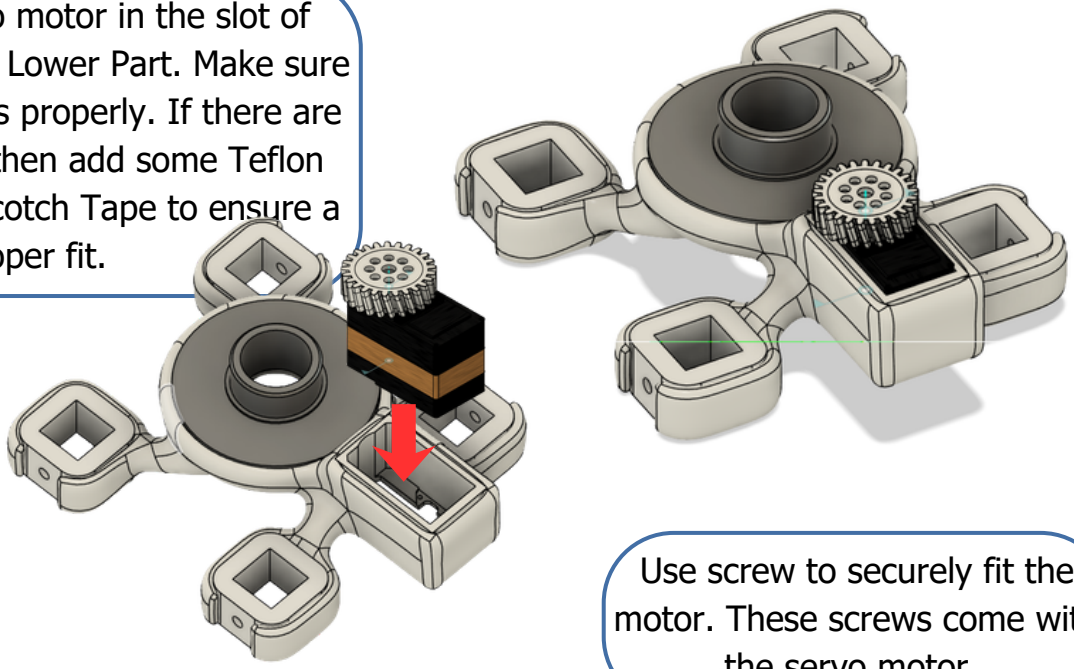
Now 3D print the "Bearing Outer Part"



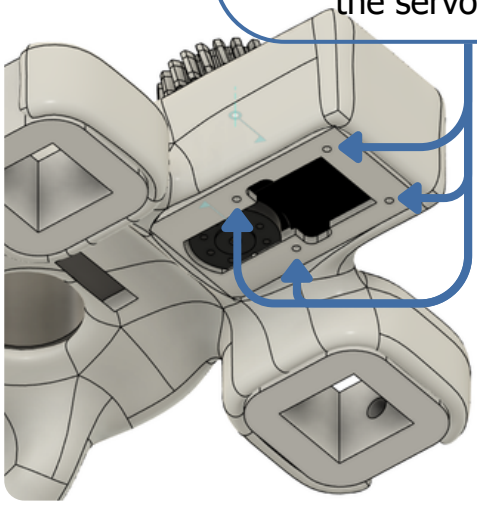
Then 3D print the "Neck Joint Conenction Between Body and Head"



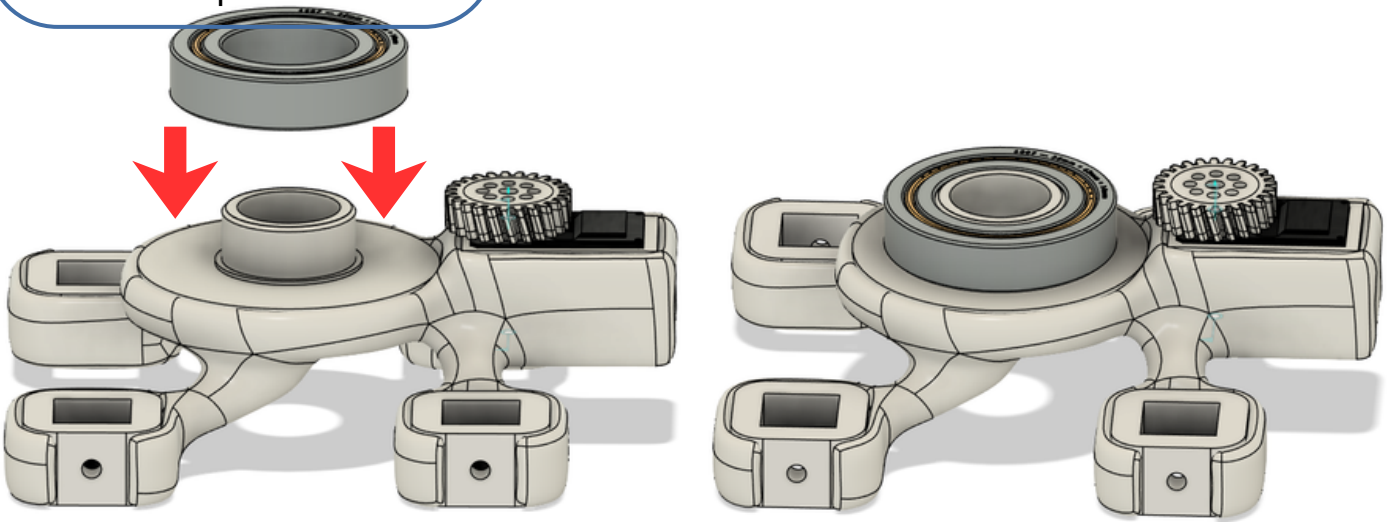
Insert the servo motor in the slot of Neck Joint Body – Lower Part. Make sure that the motor fits properly. If there are more tolerance then add some Teflon Tape/Standard Scotch Tape to ensure a proper fit.



Use screw to securely fit the motor. These screws come with the servo motor.

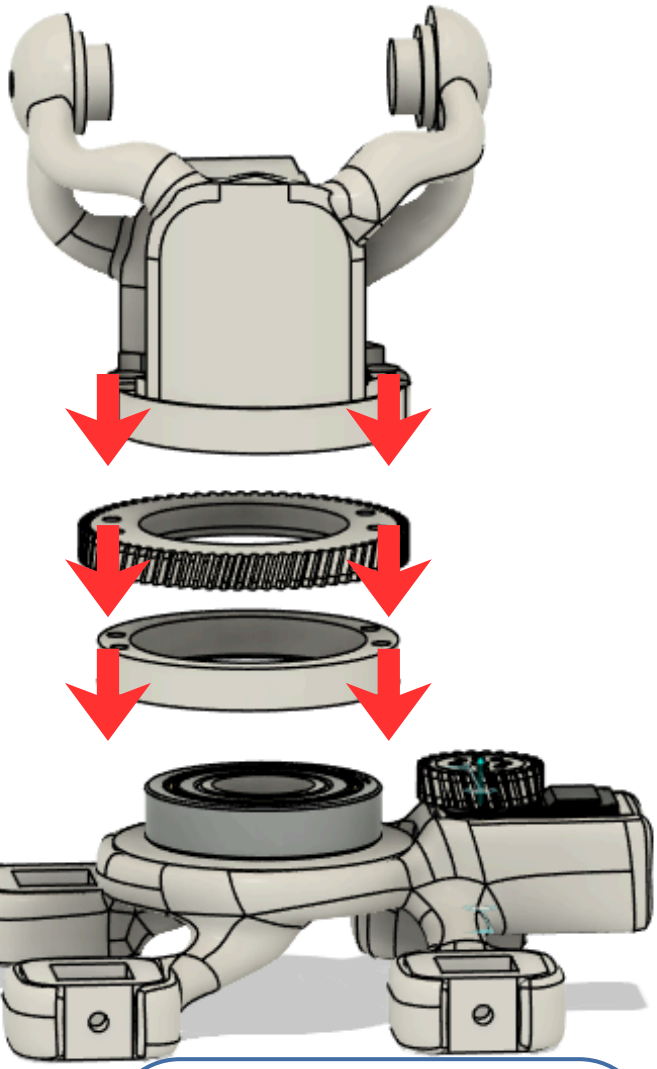


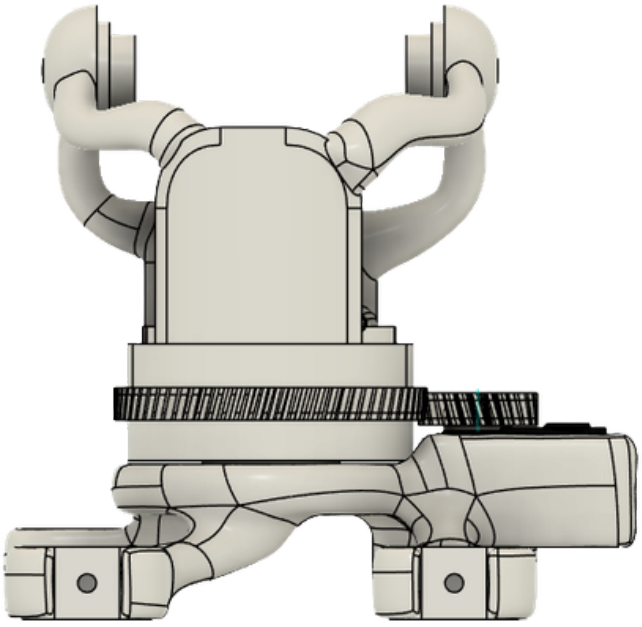
Now add the 6007 Bearing in the slot and like before, you can add a little bit of Teflon Tape/Standard Scotch Tape to ensure profer Fit.



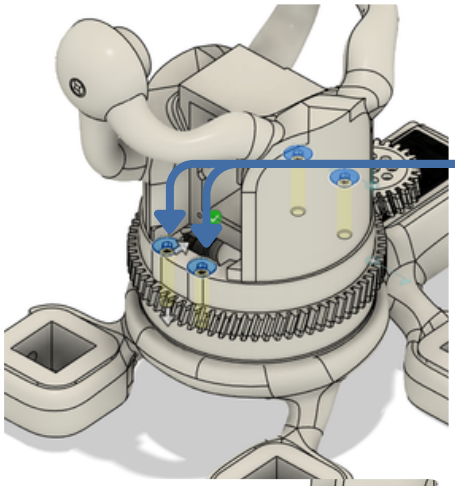
After installing both the servo and the bearing, gently check for any play to ensure a proper fit. If any looseness is observed, take appropriate steps to compensate for the tolerance.

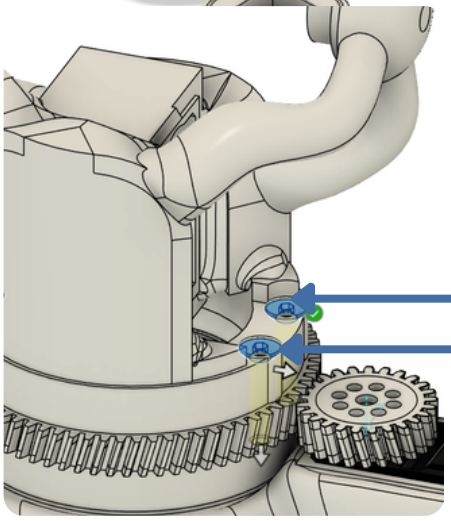
Once the "Neck Joint Connection Between Body and Head", "Neck Pan Driven Gear", and "Bearing Outer Part" are printed, stack them on top of each other and position them as shown in the image.



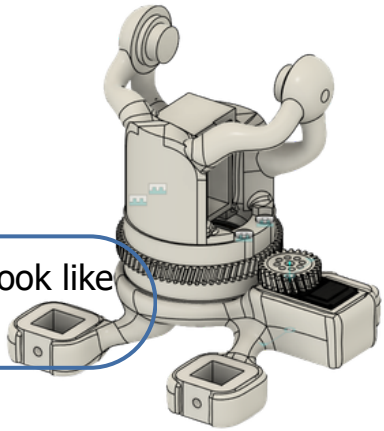


After stacking the components, secure them using M5 × 25 mm hexagon socket flat countersunk head cap screws. Ensure that the "Bearing Outer Part" fits tightly with the 6007 bearing. If necessary, apply PTFE (Teflon)/Standard Scotch Tape tape to achieve a snug fit. This is important, as the joint relies on friction to support the head.

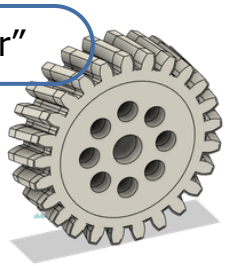




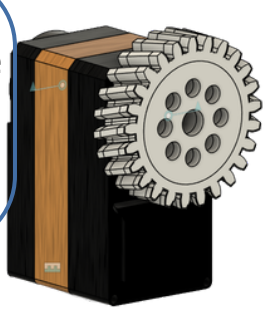
The final setup shold look like this:



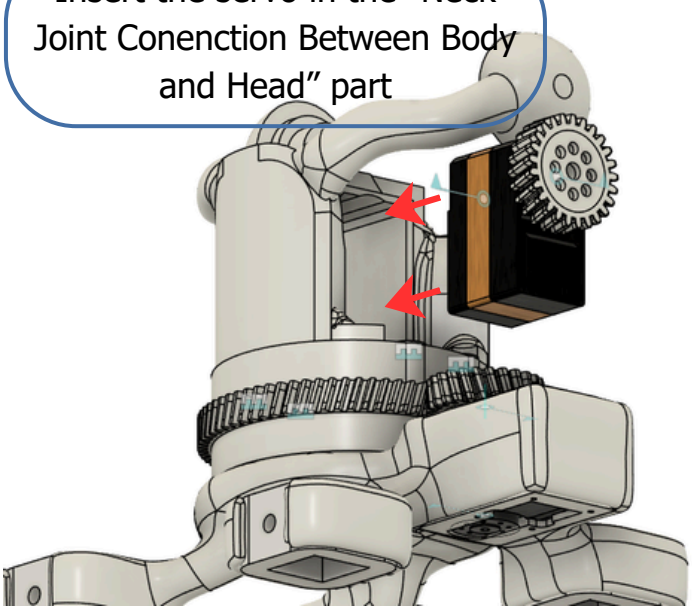
Print the “Tilt Gear Driver”

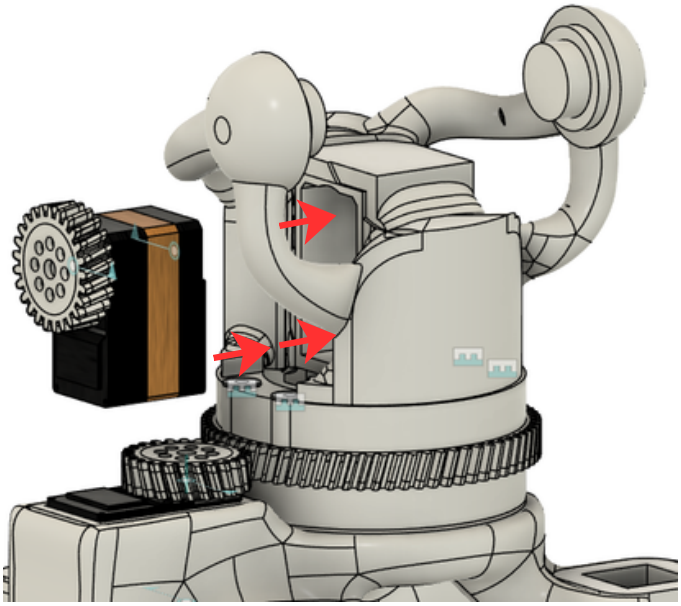


Attach the “Tilt Gear Driver” with the Hiwonder HX-35HM using the circular plastic horn povided with the servo itself. (Just like we did before).

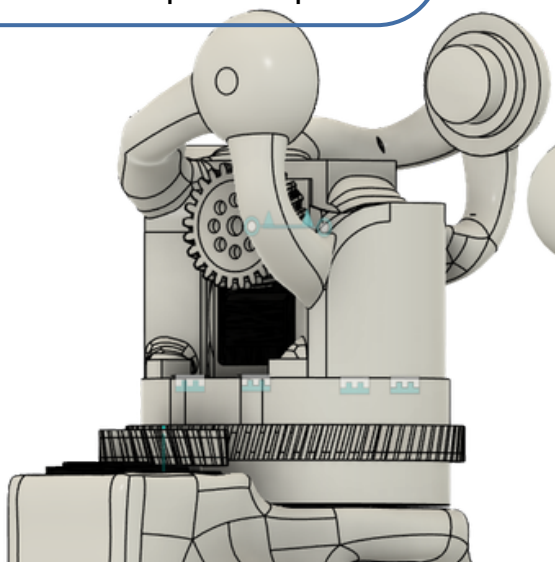


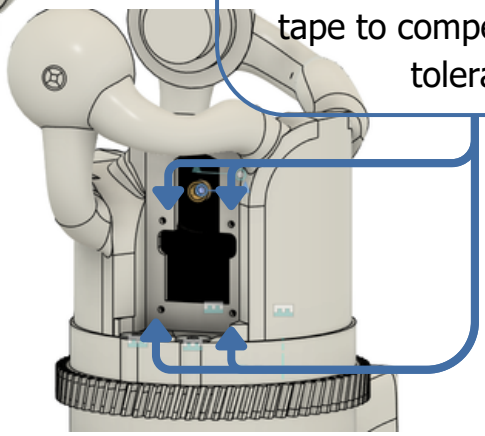
Insert the servo in the “Neck Joint Conenction Between Body and Head” part





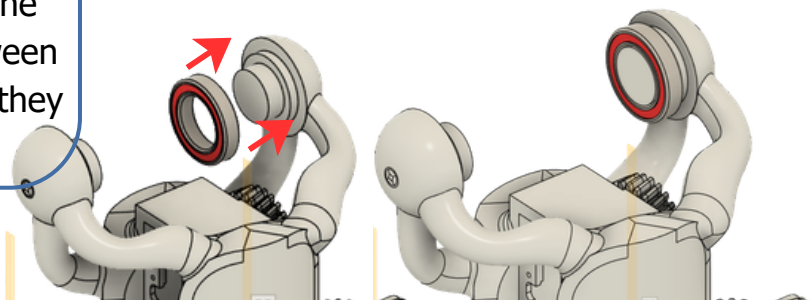
Finally it should look like this after the setup is complete:

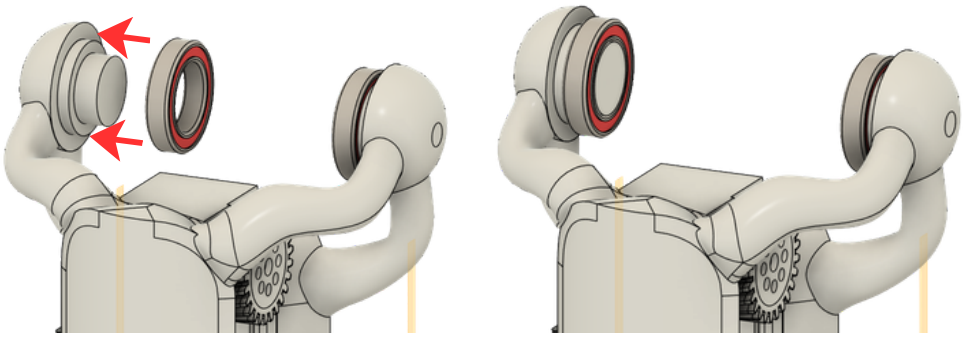




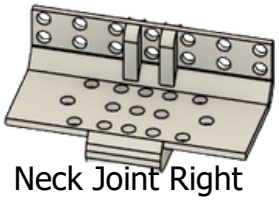
Use screws to securely attach the motor to the part. If the motor wobbles or does not fit properly, use PTFE (Teflon) tape or Scotch tape to compensate for the tolerance.

Add two 6804 bearings to the upper extruded section of the "Neck Joint Connection Between Body and Head" and ensure they fit properly.

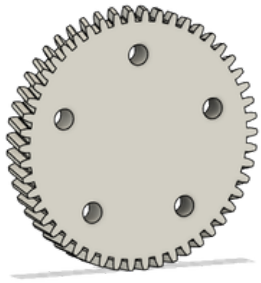




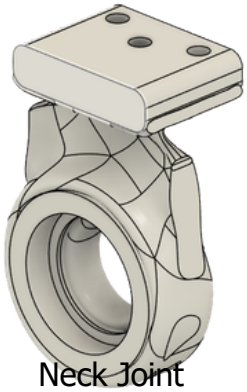
Now 3D print the following parts:
"Tilt Gear Driven",
"Neck Joint Head Right",
"Neck Joint Head Left",
"Neck Joint Right Left Connecting Part"



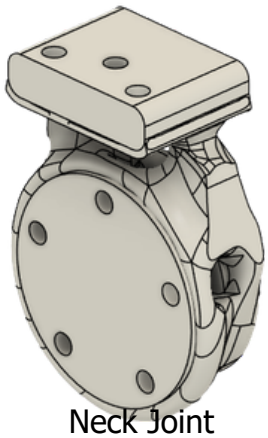
Neck Joint Right Left Connecting Part



Tilt Gear Driven

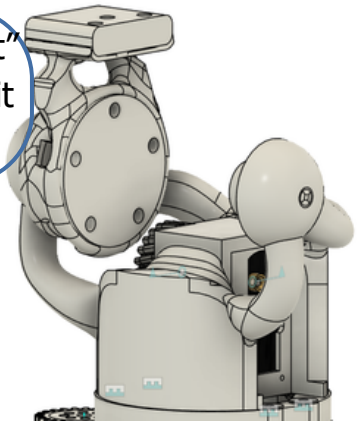


Neck Joint Head Right

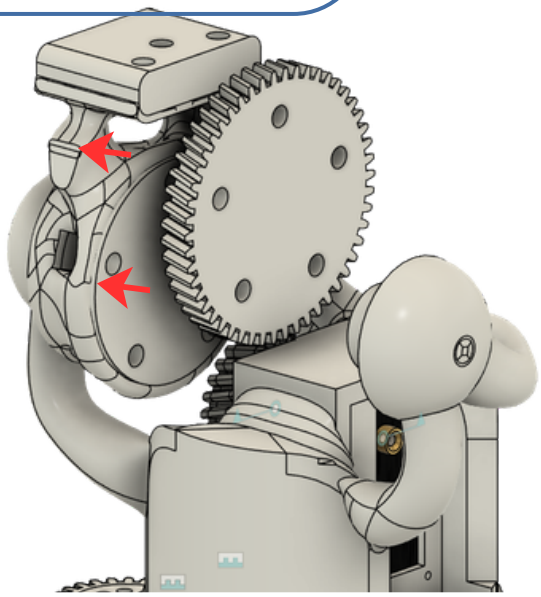


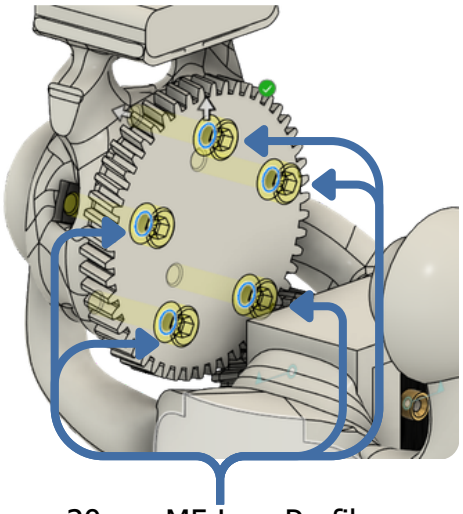
Neck Joint Head Left

Attach the "Neck Joint Head Left" to the left bearing. Ensure that it fits properly.

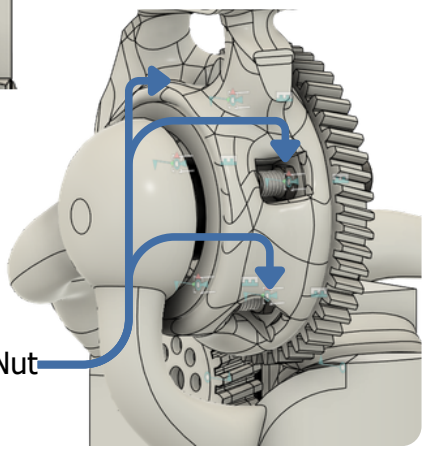


Add the "Tilt Grear Driven" with the "Neck Joint Head Left". Use 20mm M5 Low Profile Flat Head Hex Cap Screw and M5 Hex Nut for a proper fit.



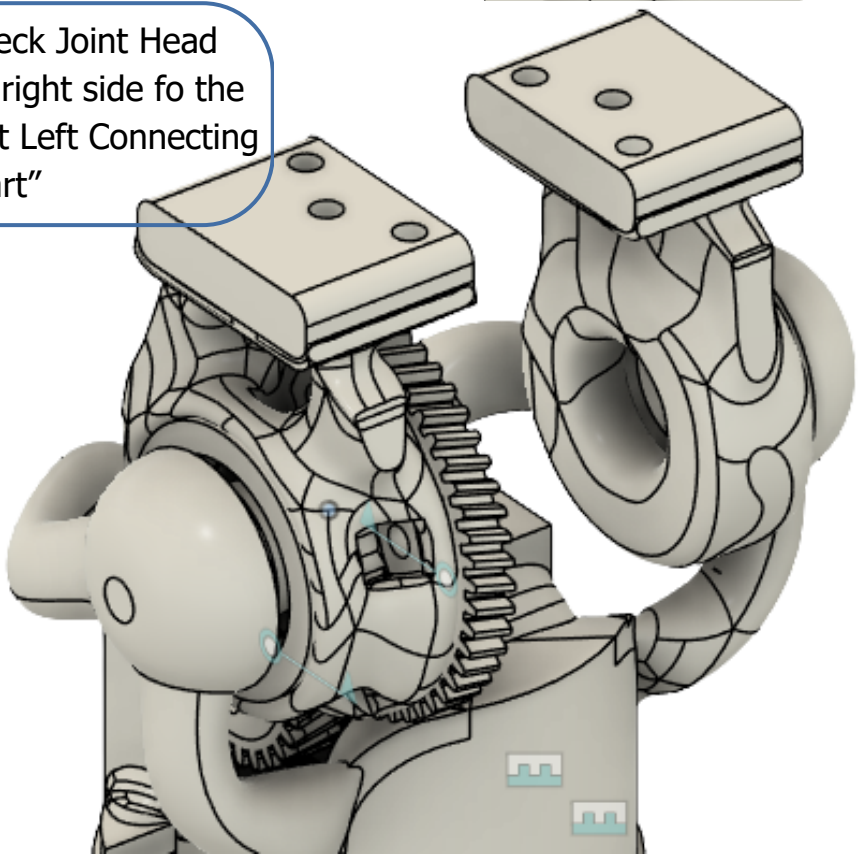


20mm M5 Low Profile Flat Head Hex Cap Screw

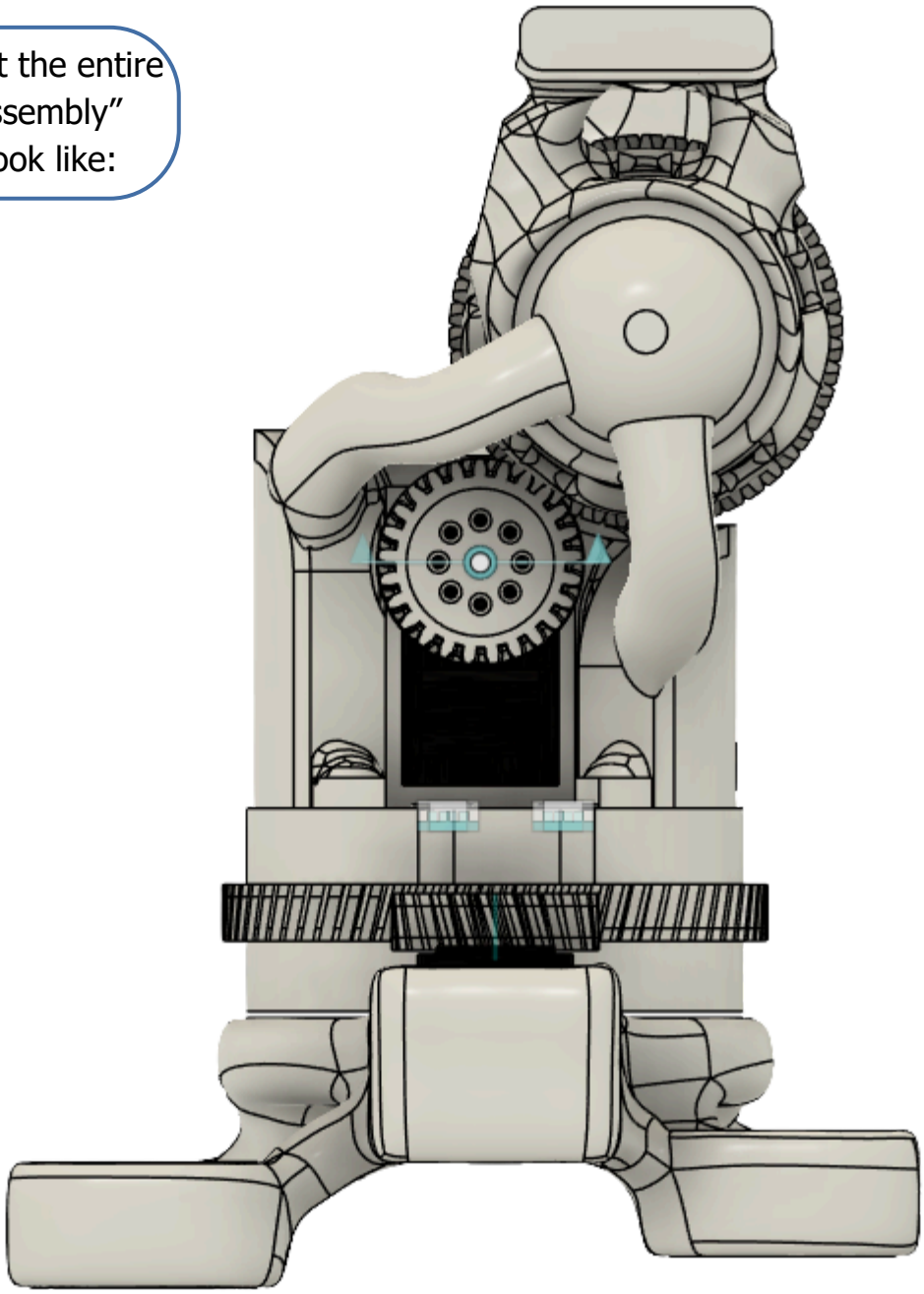


M5 Hex Nut

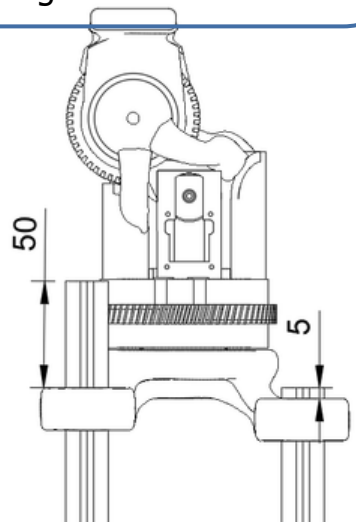
Attach the "Neck Joint Head Right" with the right side fo the "Neck Joint Right Left Connecting Part"

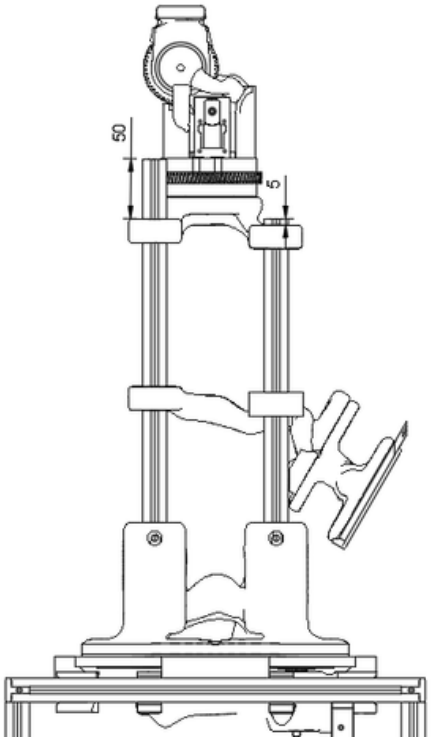


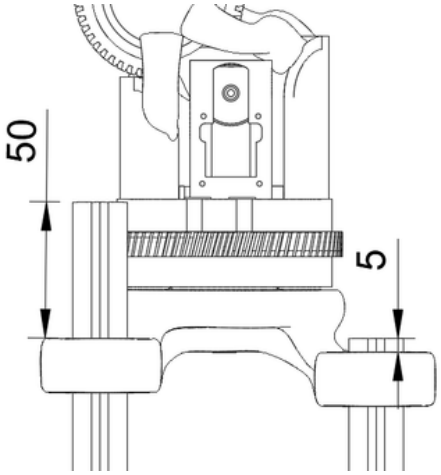
This is what the entire "Neck Assembly" should look like:



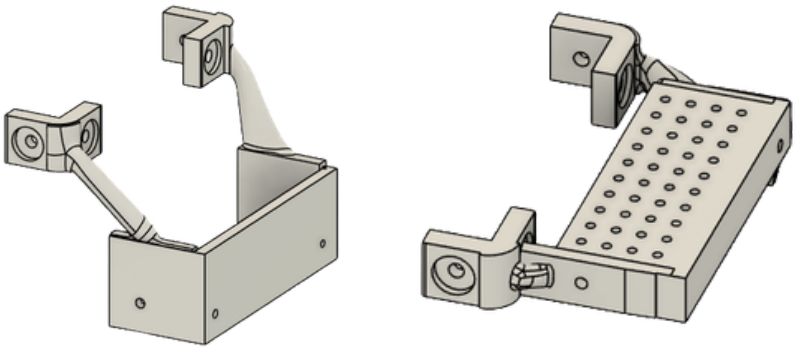
Now using proper screw and with the help of Drop-in T-Nut install this setup with the Core Profile following this measurements:

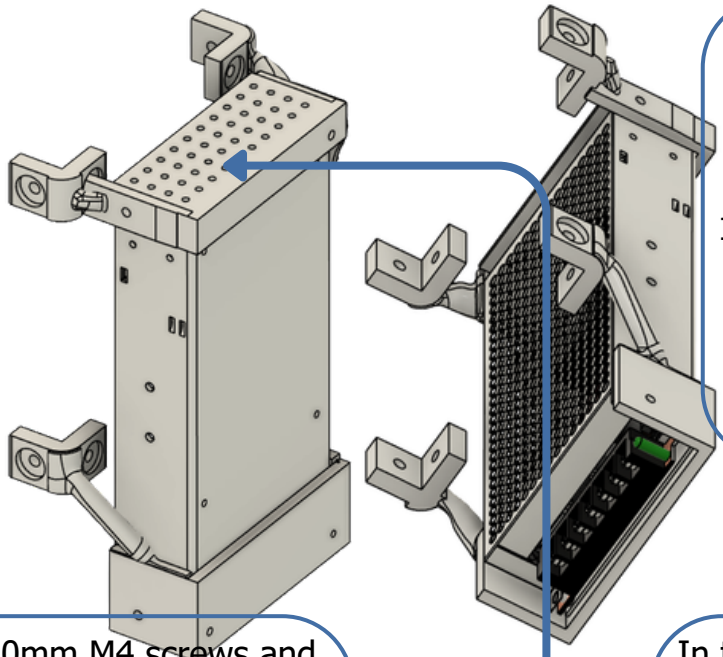






3D Print the "12V SMPS Holder Lower" and "12V SMPS Holder Upper"





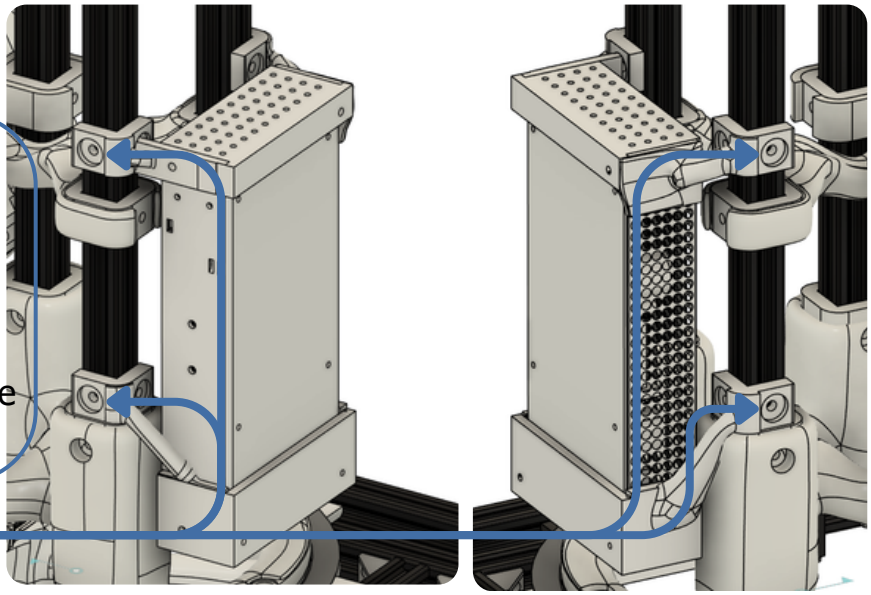
Attach the "12V SMPS Holder Lower" and "12V SMPS Holder Upper" part with the 12V SMPS, but make sure to connect about 1 feet of AC input and DC Output cable to the SMPS as after attacking the SMPS to Core Profile, you can't add or remove cables.

Use any 10mm M4 screws and M4 nuts to attack the SMPS Hoder parts with the smps. A small (6mm to 8mm) M3 screw is needed too to screw them properly.

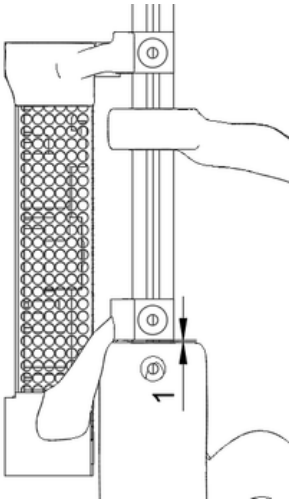
In the "12V SMPS Holder Upper" a lot of 3mm holes are there so that any electrical connections can be screwed here.

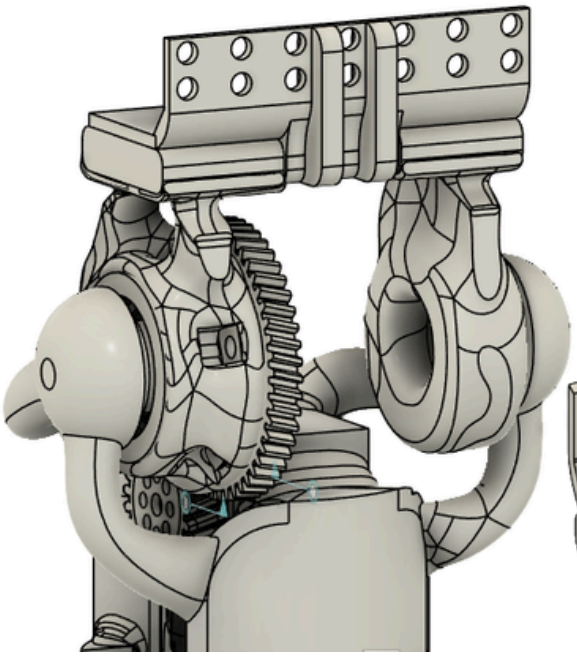
Now insert 4 M5 screws (10mm) and Drop-in T-Nut on the four outer holes.

Then attach it to the core frame. You might apply a little bit of force to do so.

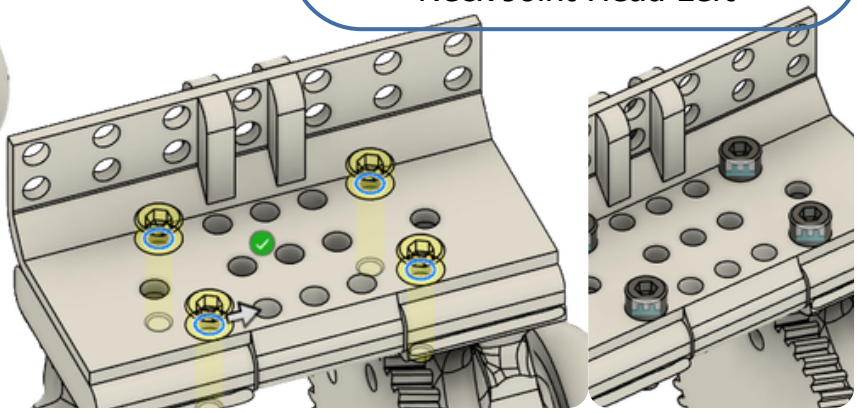


Once done setting up SMPS in the Core Frames, tighten the screws but make sure to adjust a bit to achieve dimensions shown on the image (This 1 mm clearance is intended to compensate for tolerance.):

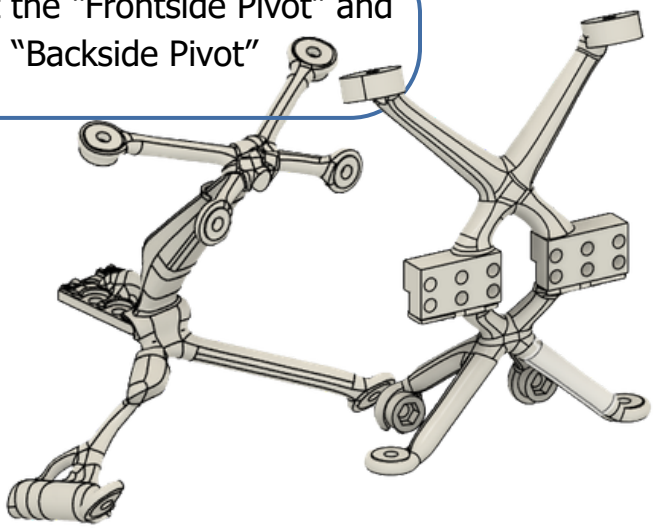




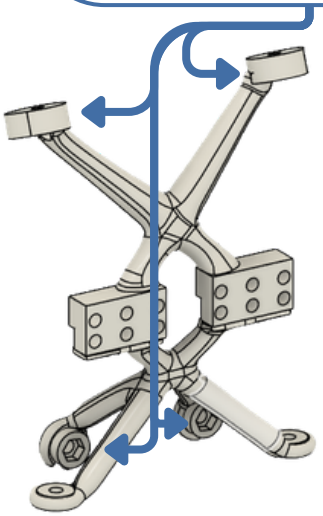
Now connect the "Neck Joint Right Left Connecting Part" joning the left and right part. Use 20mm M5 screw and Hex Nut to secure it with the "Neck Joint Head Right", "Neck Joint Head Left"



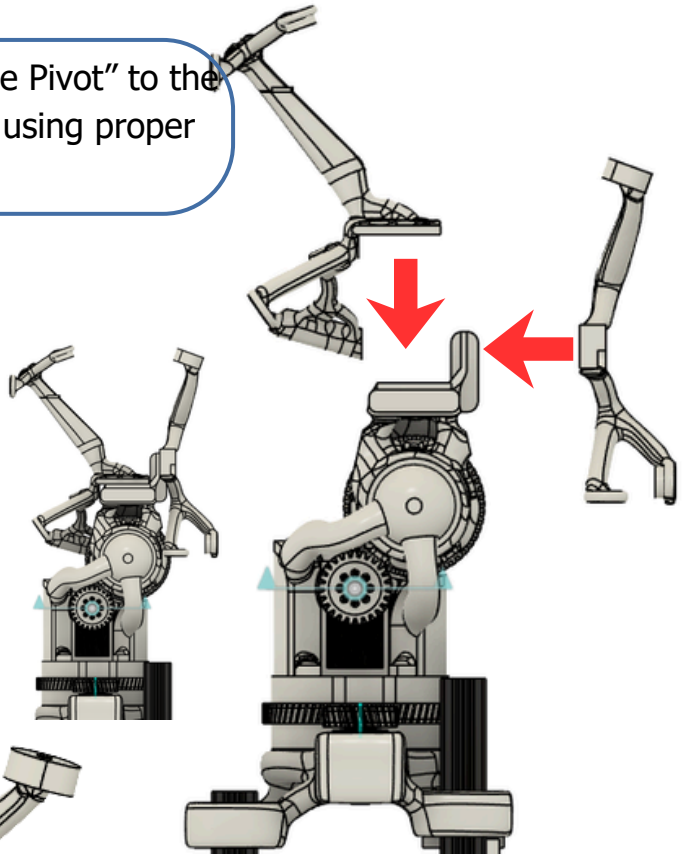
3D Print the "Frontside Pivot" and "Backside Pivot"



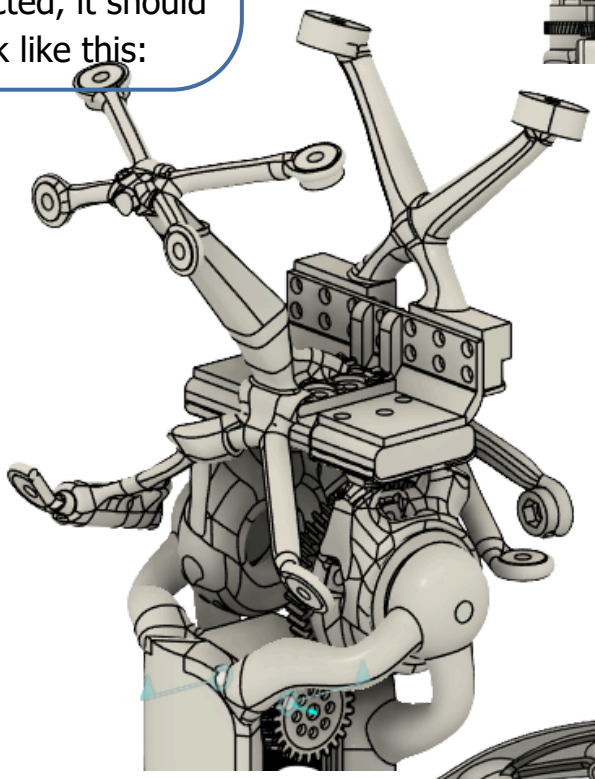
In the "Backside Pivot" you will see some place for Hex Huts. Insert Hex Nuts there with a little bit of glue so that it holds the postion.



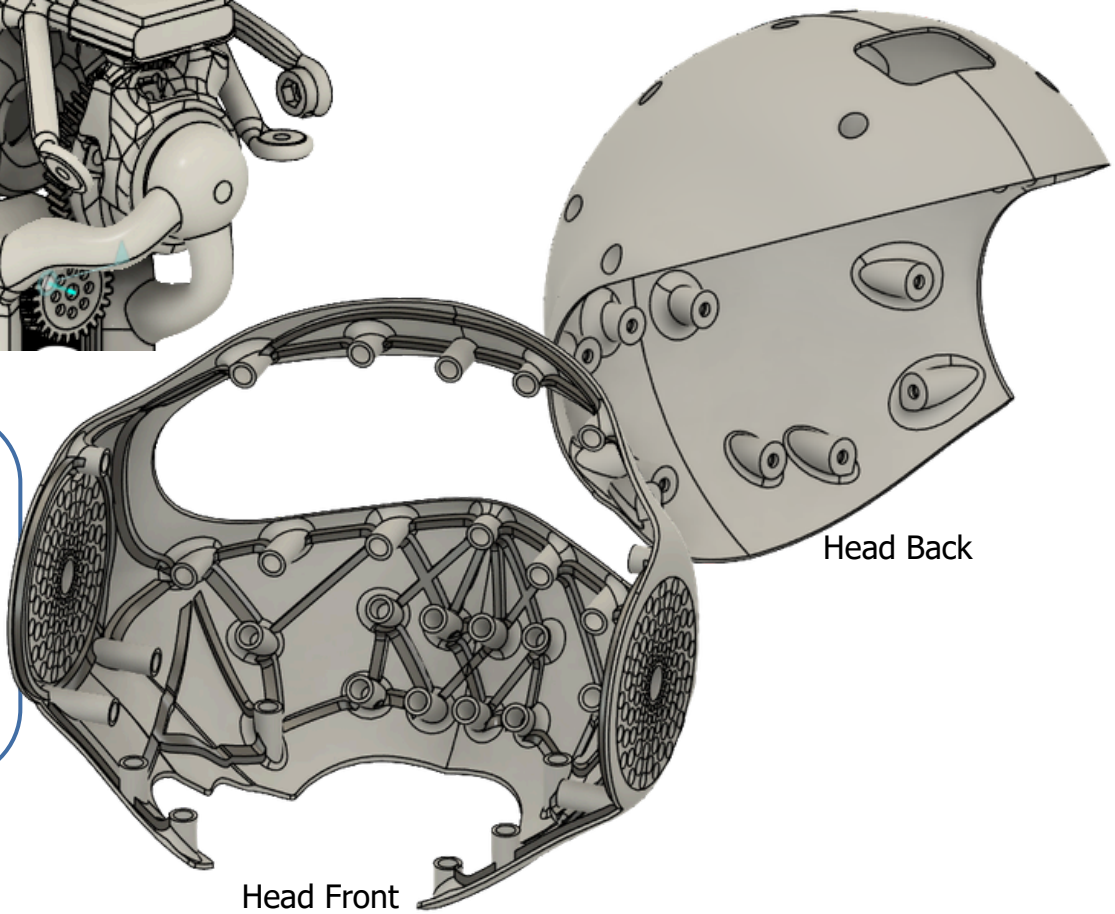
Attach the "Frontside Pivot" and "Backside Pivot" to the "Neck Joint Right Left Connecting Part" using proper screw and Hex Nut.



After they are connected, it should look like this:



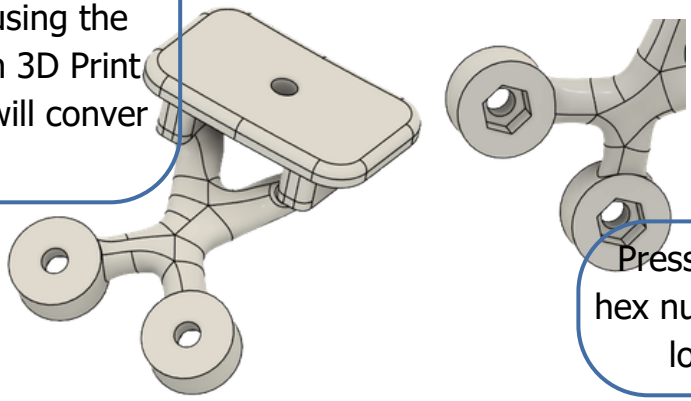
3D Print "Head Font" and "Head Back" Part which are two main structure of the robot head and these will hold all the other parts.



Head Back

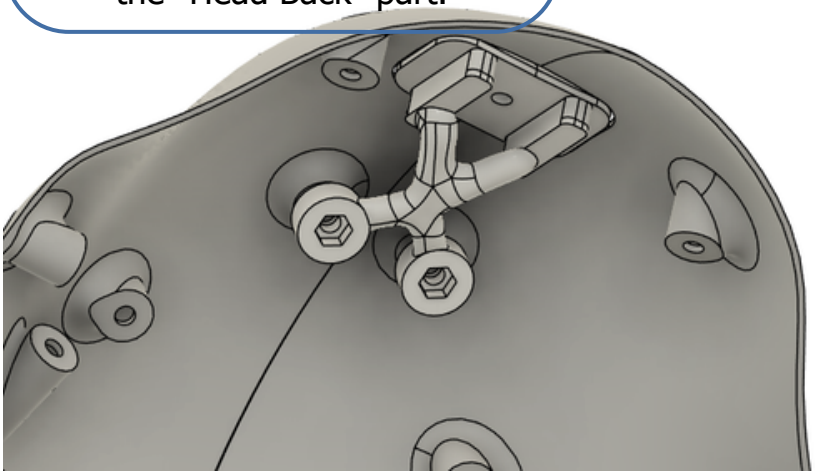
Head Front

The holes in head back (which is in the forehead in the whole head) can be used to add additional tracker or camera. But if you are not using any or using the cheast mounted camera then 3D Print "Top Camera Holder" which will conver the hole.

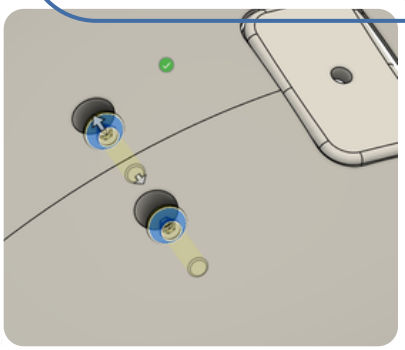


Press-fit two M5 hex nuts into these locations.

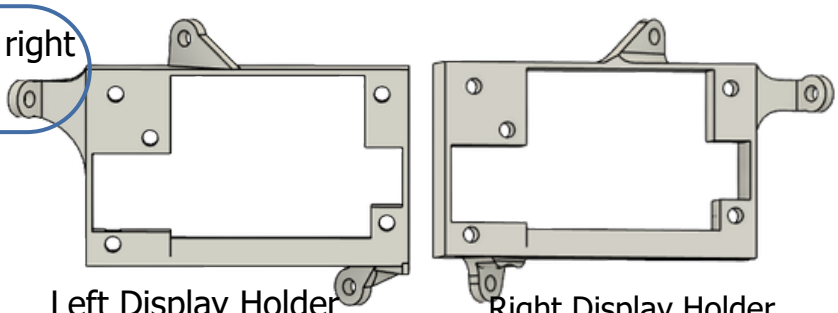
Attach "Top Camera Holder" on the "Head Back" part.



Insert the countersunk screws from the top (16mm or 25mm both will work)



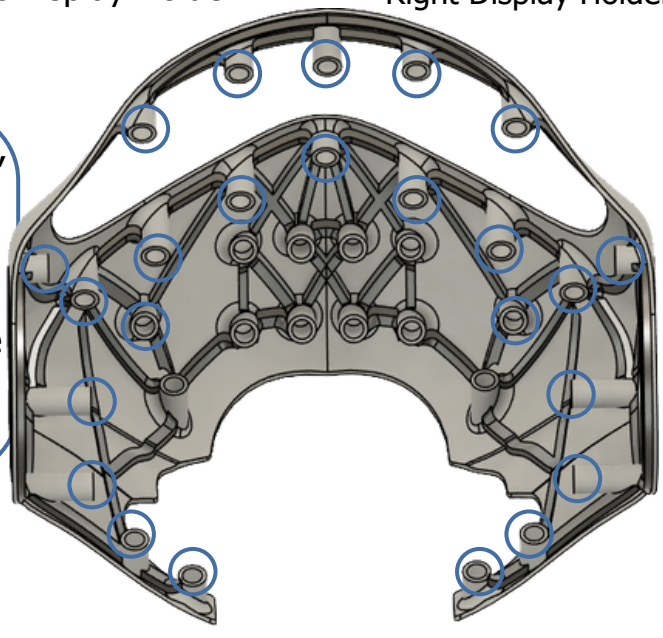
Now 3D print the left and right dispaly holder.

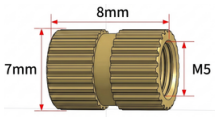


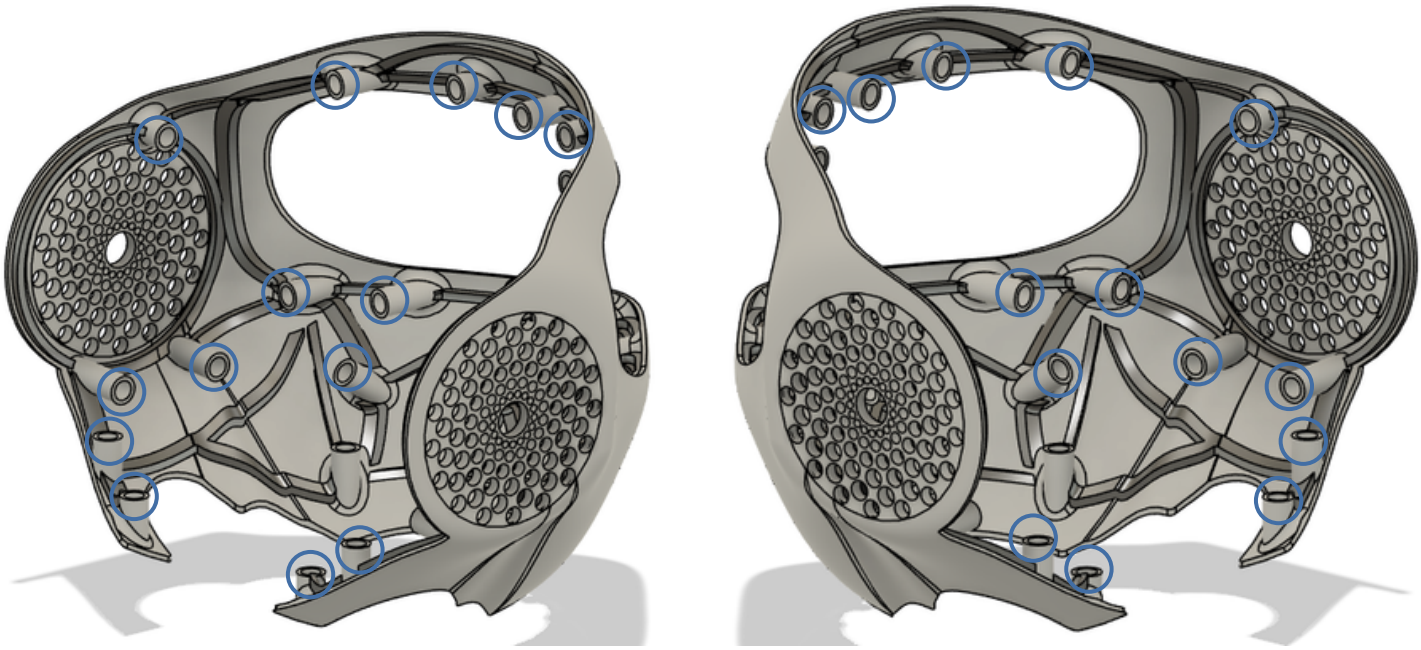
Left Display Holder

Right Display Holder

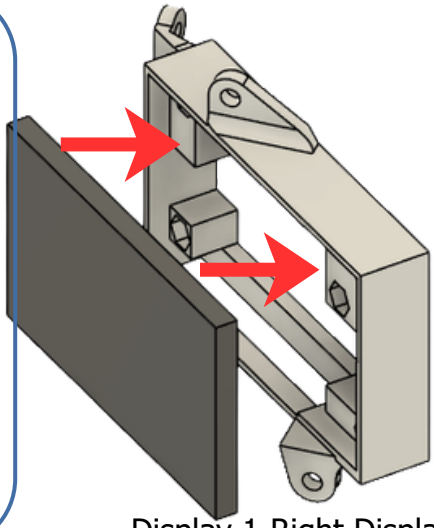
Once the "Head Front" is printed, insert the M5 Inserts H:8mm, OD:7mm in the red marked spots of this image:
(The other insert holes are there to use in future if any additional device is added to the head)



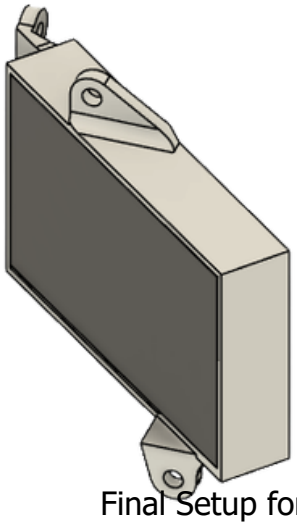




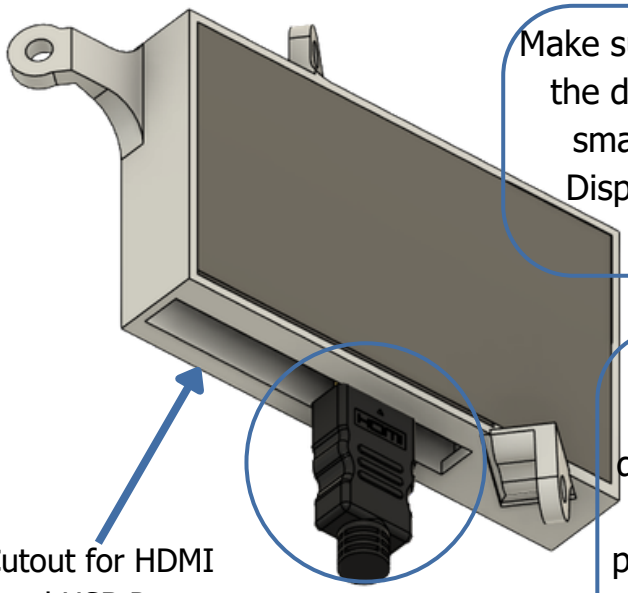
Now we will add the displays to the "Left Display Holder" and "Right Display Holder". For display we will be using two "4.0 inch HDMI display". But you can use any display as long as they follow the same holes in the "Head Front" and dimentionally fits into position but of course you have to make your own Display Holder Setup



Display 1 Right Display Holder



Final Setup for the Right Display



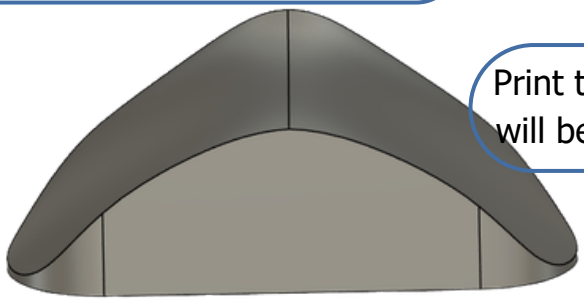
Cutout for HDMI and USB Port

Make sure that the HDMI Port of the display is oriented at the small opening in the "Left Display Holder" and "Right Display Holder"

Use double sided tapes or scotch tapes to secure the display with the mount. Make sure they fit and sticks properly otherwise the motion of the head will throw them off.

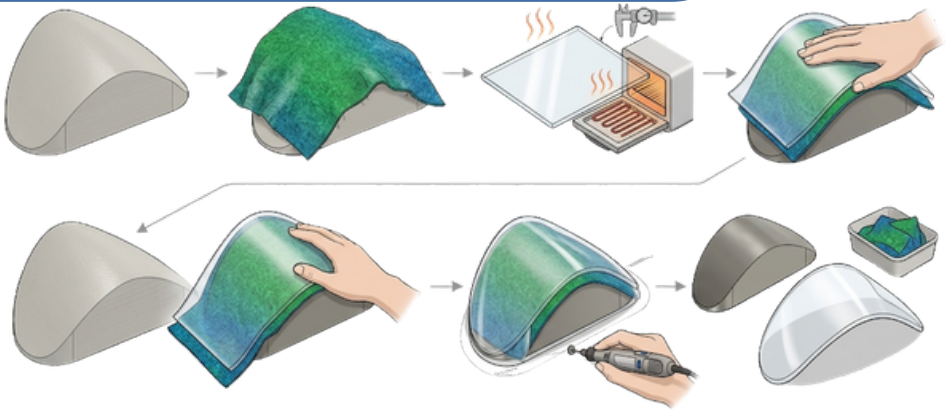
Now do the same for the Display 2 and "Right Display Holder"

Now we will add the Visor in the robot head.

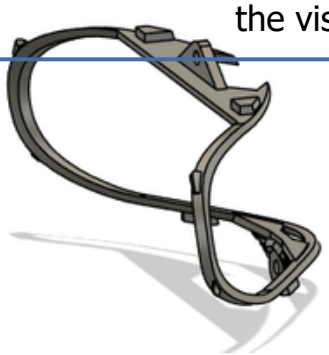


Print the part "Eye Mold" which will be used to create the Visor

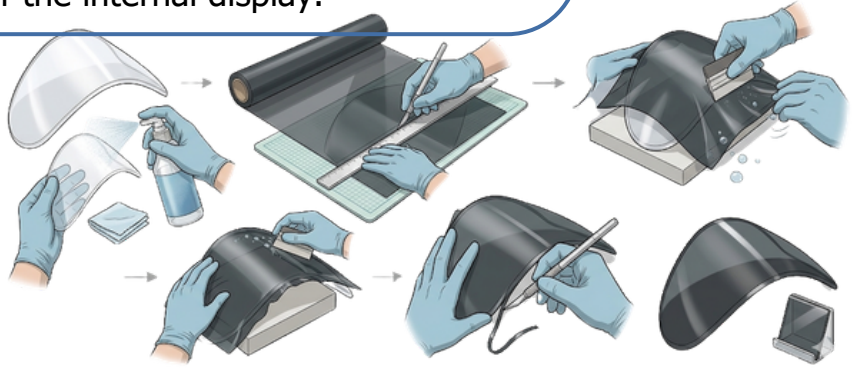
Print the "Eye Mold" part. Then take a 2 mm or 2.5 mm clear acrylic sheet, heat it (using a heat gun), and press it over the Eye Mold. Place a thin fabric layer between the mold and the acrylic to prevent white residue or surface marks from transferring to the acrylic. Once formed, trim the acrylic to the correct shape to obtain the visor.



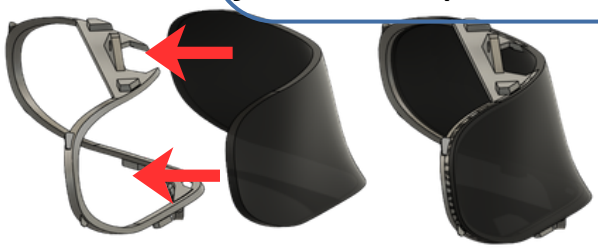
In the meantime print the part "Visor Holder" which will hold the visor.



In our case, we tinted the acrylic, but it became too dark after tinting. As a result, we removed the tint to ensure clear visibility of the internal display.

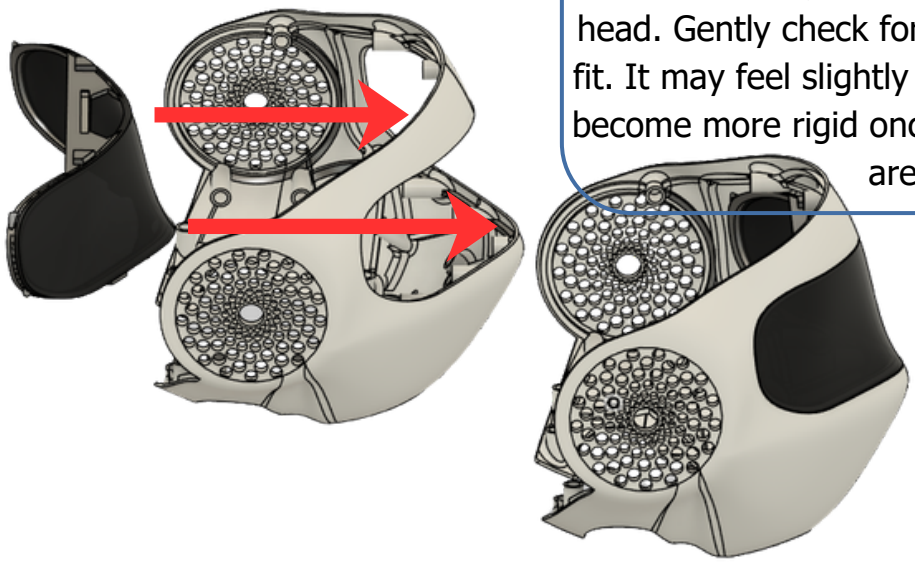


Using glue attach the “Visor Holder” to the acrylic visor we just created. (Don’t use super glue as it may causes frosting)

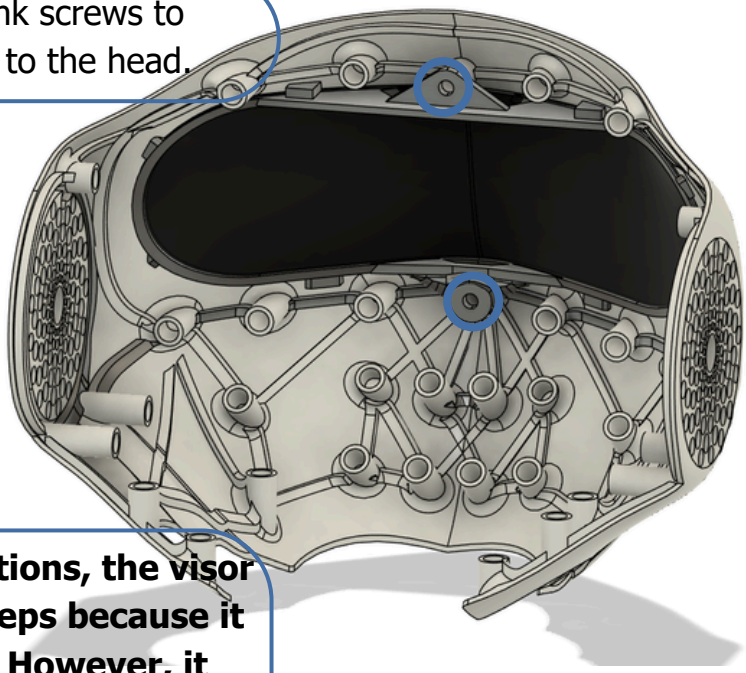


Grab the “Head Front” where we just put the inserts.

Press and attach the visor assembly (visor and visor holder glued together) to the front of the head. Gently check for any play to ensure a proper fit. It may feel slightly loose at this stage, but it will become more rigid once the remaining components are installed.

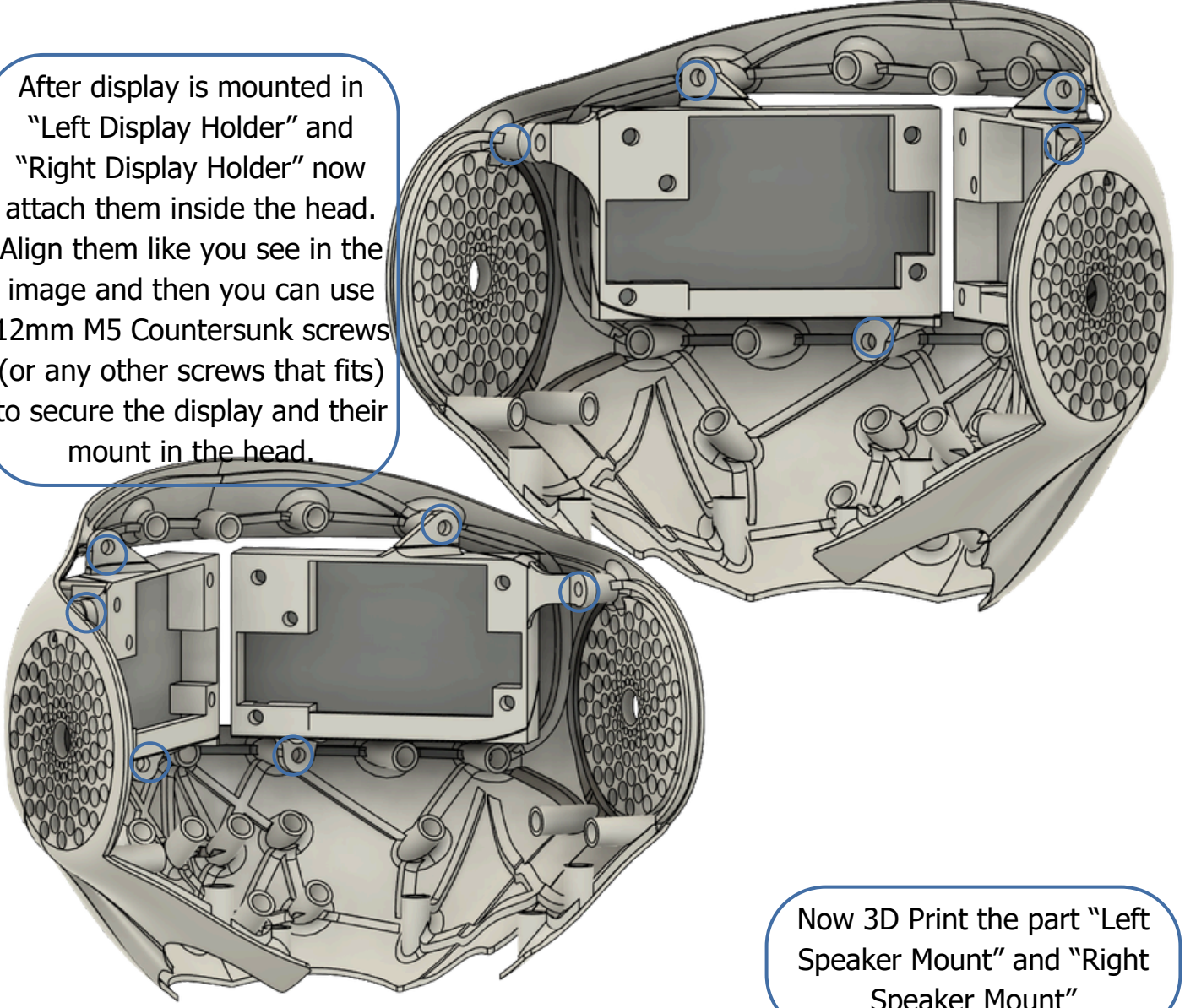


Use M5 × 12 mm countersunk screws to permanently attach the visor to the head.

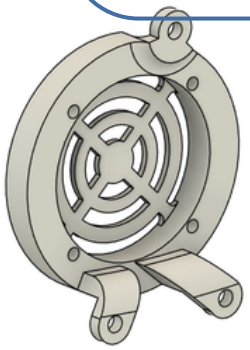


Note: In the following instructions, the visor may not be shown in some steps because it obscures the internal view. However, it should be assumed to be installed during assembly.

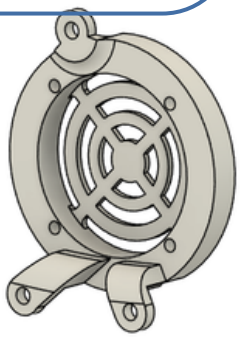
After display is mounted in "Left Display Holder" and "Right Display Holder" now attach them inside the head. Align them like you see in the image and then you can use 12mm M5 Countersunk screws (or any other screws that fits) to secure the display and their mount in the head.



Now 3D Print the part "Left Speaker Mount" and "Right Speaker Mount"

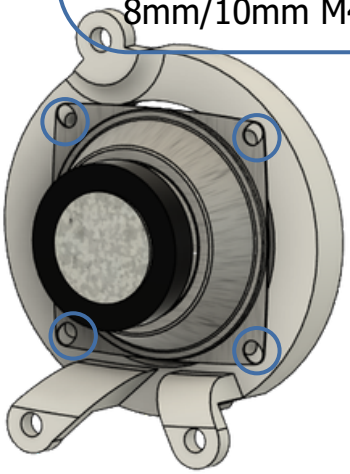


Right Speaker Mount

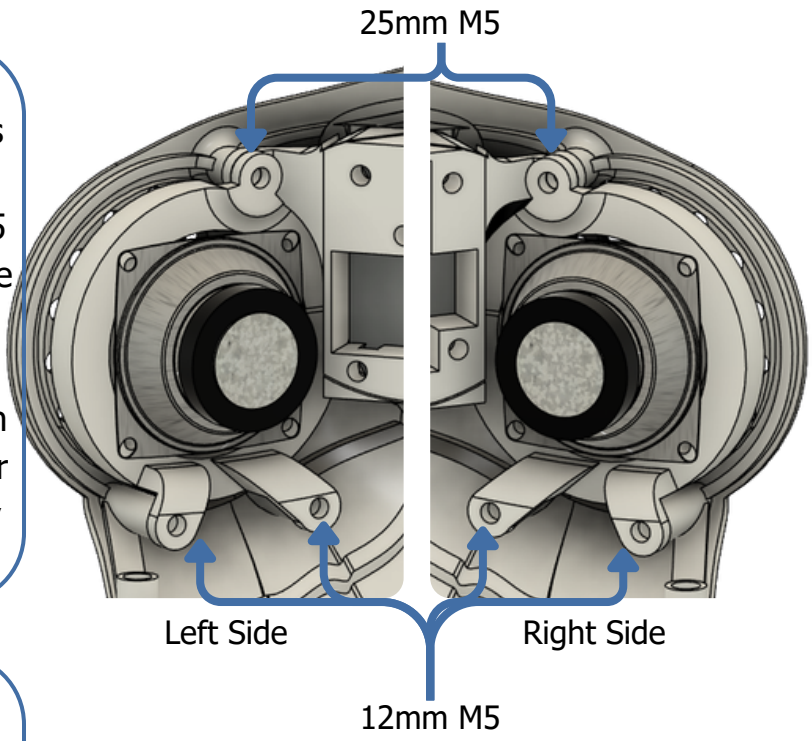


Left Speaker Mount

when "Left Speaker Mount" and "Right Speaker Mount" are printed, attach the 52mm driver into them using 8mm/10mm M4 Screws



Now attach the speaker mount with the main head as shown in the image. Like the display holders use 12mm M5 Countersunk screws to get the job done. But in two cases you need 25mm M5 Countersunk Screws to attach them. There both a connector from the speaker and display will join.

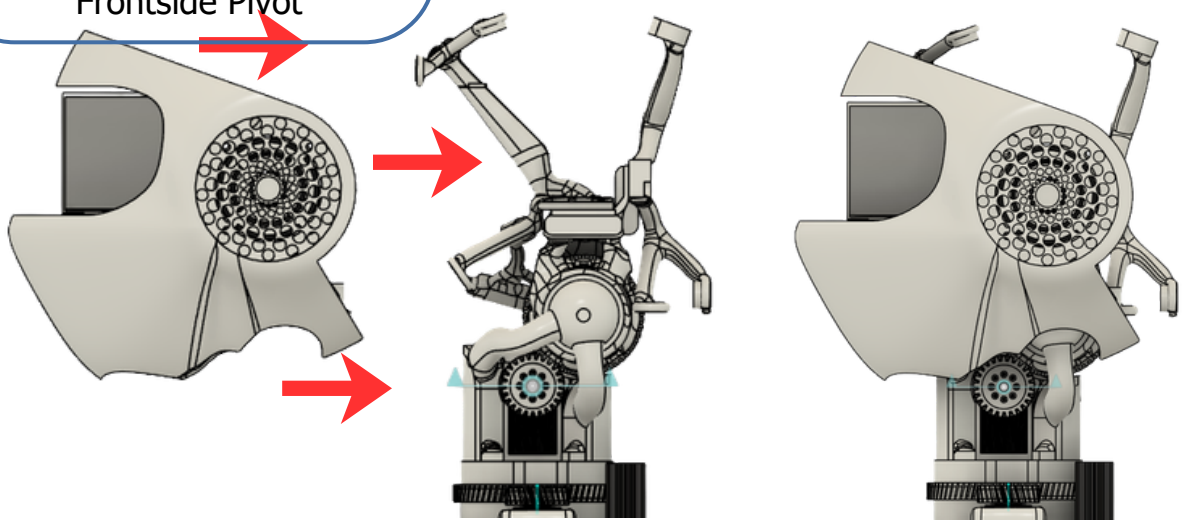


We suggest attacing the necessary cables (2 HDMI in two display, 2 USB in two display, the soldered cables in the speaker (driver).

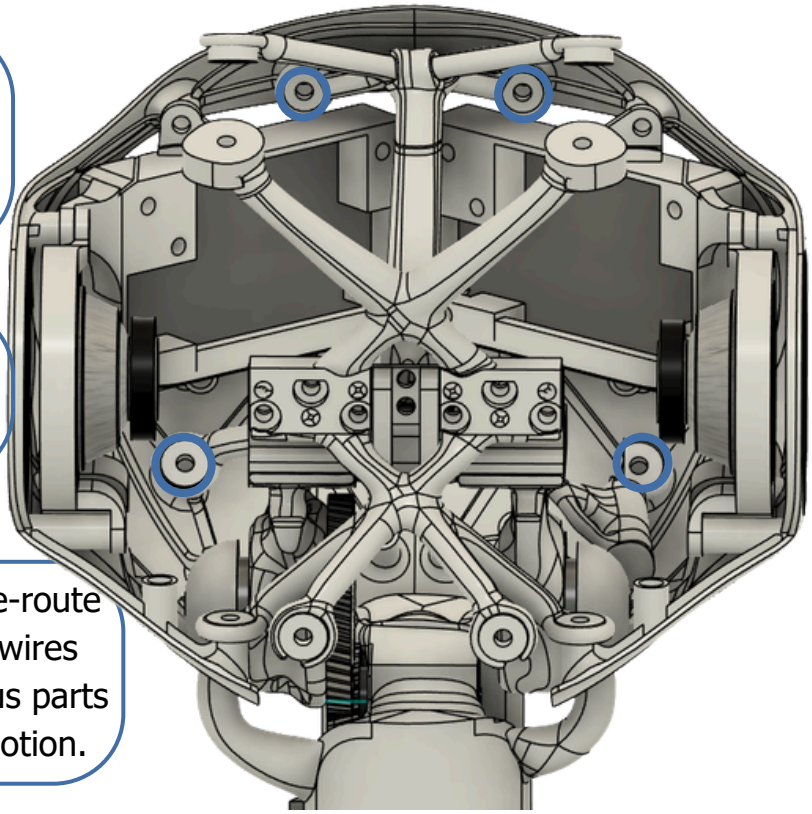
Route the cable in a way so that they don't get on the way of head internal movements.

Use small zip ties/tapes to secure them

When the cable routing is done slide the entire Head Setup (where we just attached the displays and speakers along with their mounts) so that you can properly join it with the "Frontside Pivot"



Connect the four screws of "Frontside Pivot" and "Head Front" (using 12mm M5 Countersunk Screws)

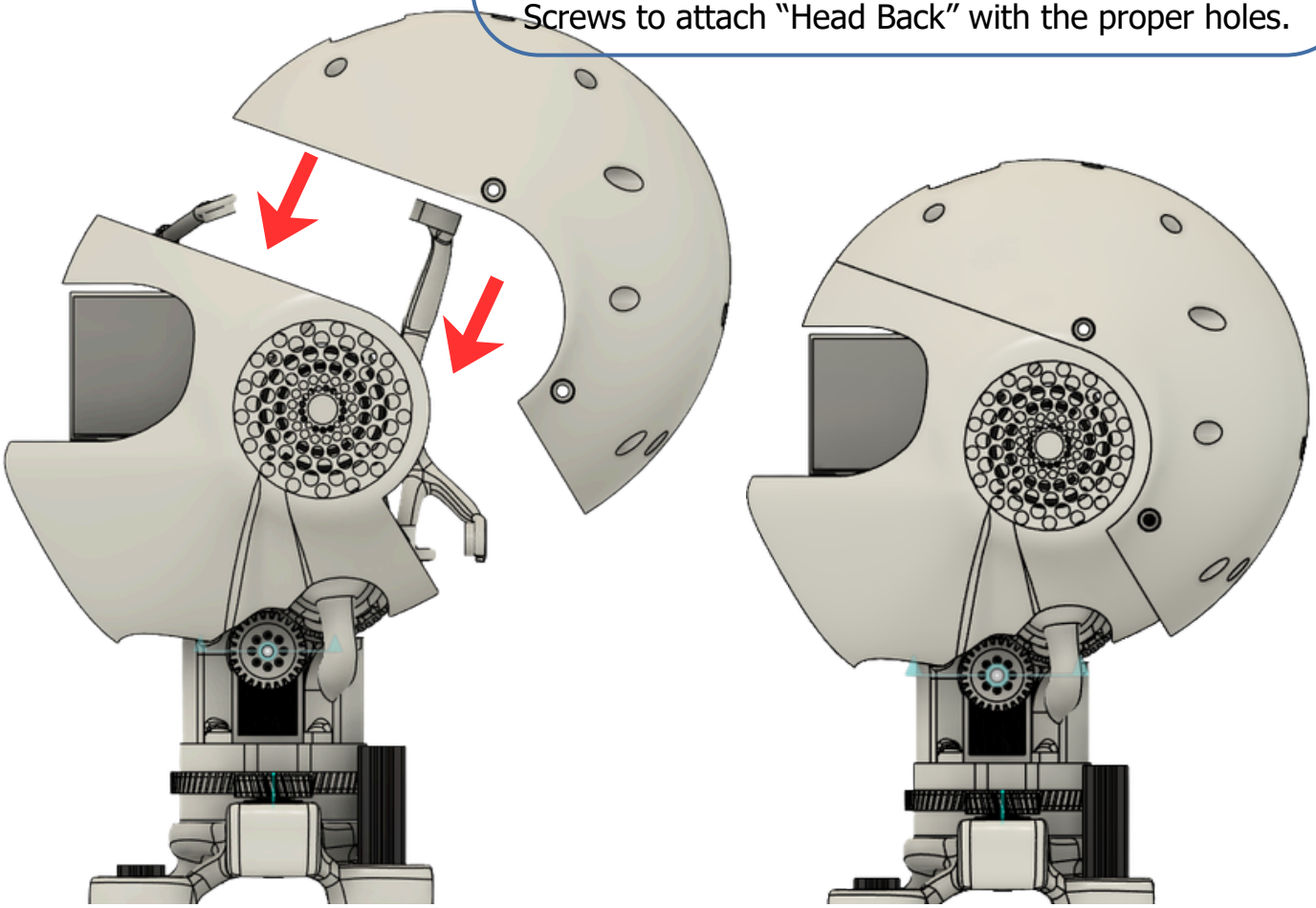


After the screws are connected, route the cables on the spacing at the back

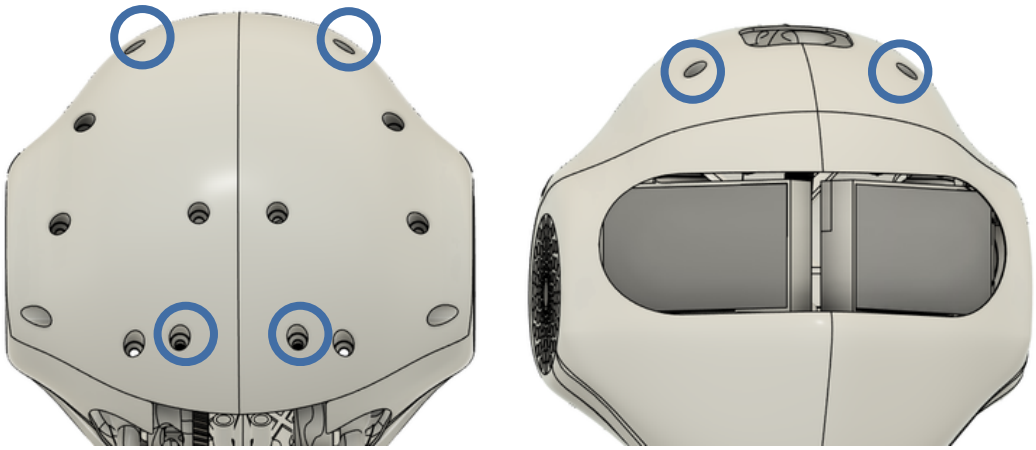
Check the wiring again and re-route if needed to make sure the wires don't get on the way of various parts when the head is doing tilt motion.

Secure the wires using zip ties

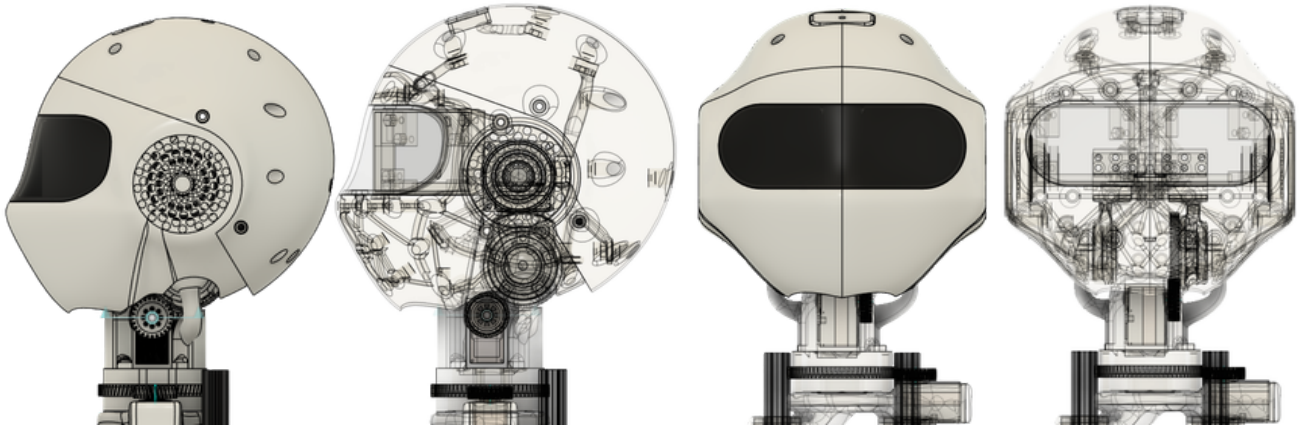
After the "Head Front" is secured now at the same way attach the "Head Back". Use 16mm M5 Countersunk Screws to attach "Head Back" with the proper holes.



These holes will support 16mm M5 Countersunk Screws to attach “Head Back” with the proper holes.

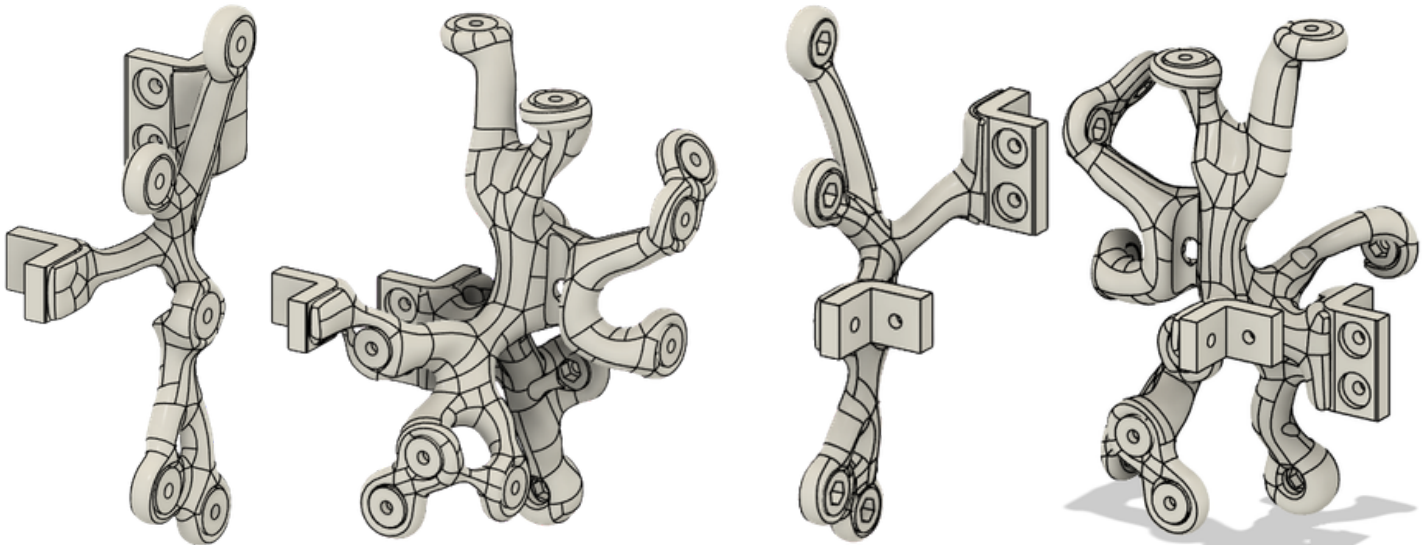


So our head is complete.



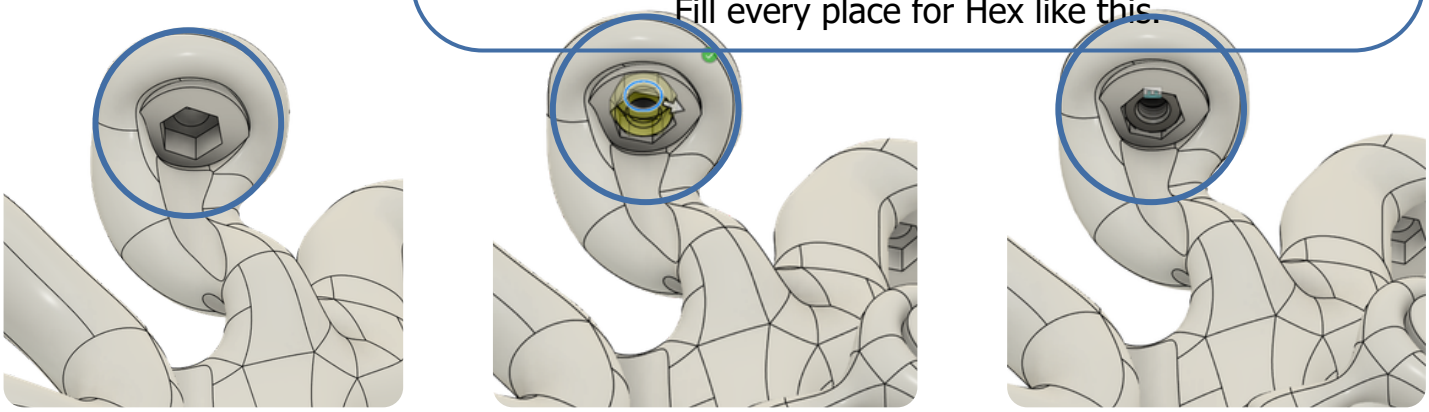
Now we will move on to the main mody (the outer shell of the robot)

First 3D print the "Left Upper Generated Connector", "Right Upper Generated Connector", "Left Lower Generated Connector", "Right Lower Generated Connector". These parts will hold all the body pannels with the robot.



After successful printing of "Left Upper Generated Connector", "Right Upper Generated Connector", "Left Lower Generated Connector", "Right Lower Generated Connector", you will see a lot of hex slots where you can fit M5 Hex Nuts (TD: 5mm, SS: 8mm, NH: 4mm). Presse fit them and add a little bit of glue around the M5 Hex Nuts to make sure that they don't come off easily from the printed parts.

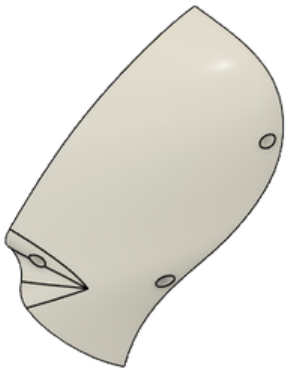
Fill every place for Hex like this.



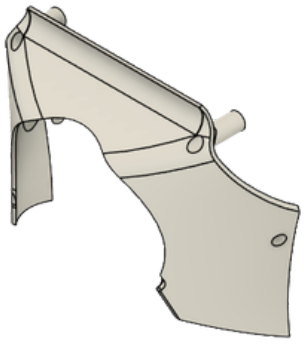
These M5 Hext Nuts will grab the pannesl using 16mm M5 Countersunk screws.

Now 3D print the main body pannels, don't attach them right now because we have to build the base of robot. But 3D printing body parts early is important as it takes a lot of time to be printed.

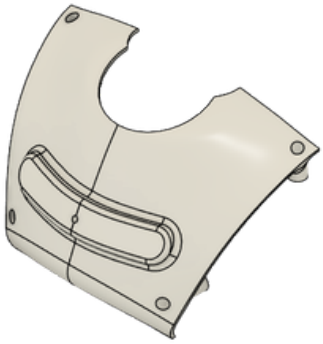
Left Sholder Back



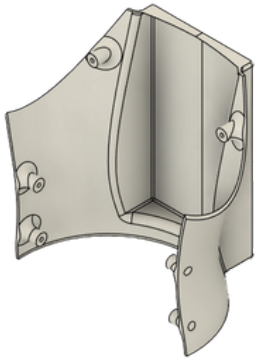
Left Sholder Front



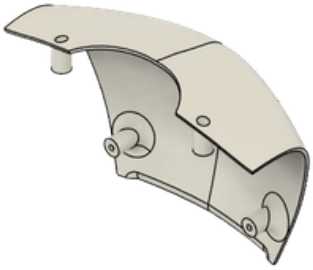
Abdomen



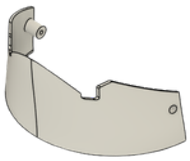
Chest



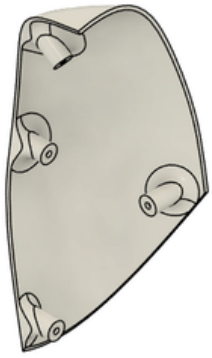
Back



Back Top



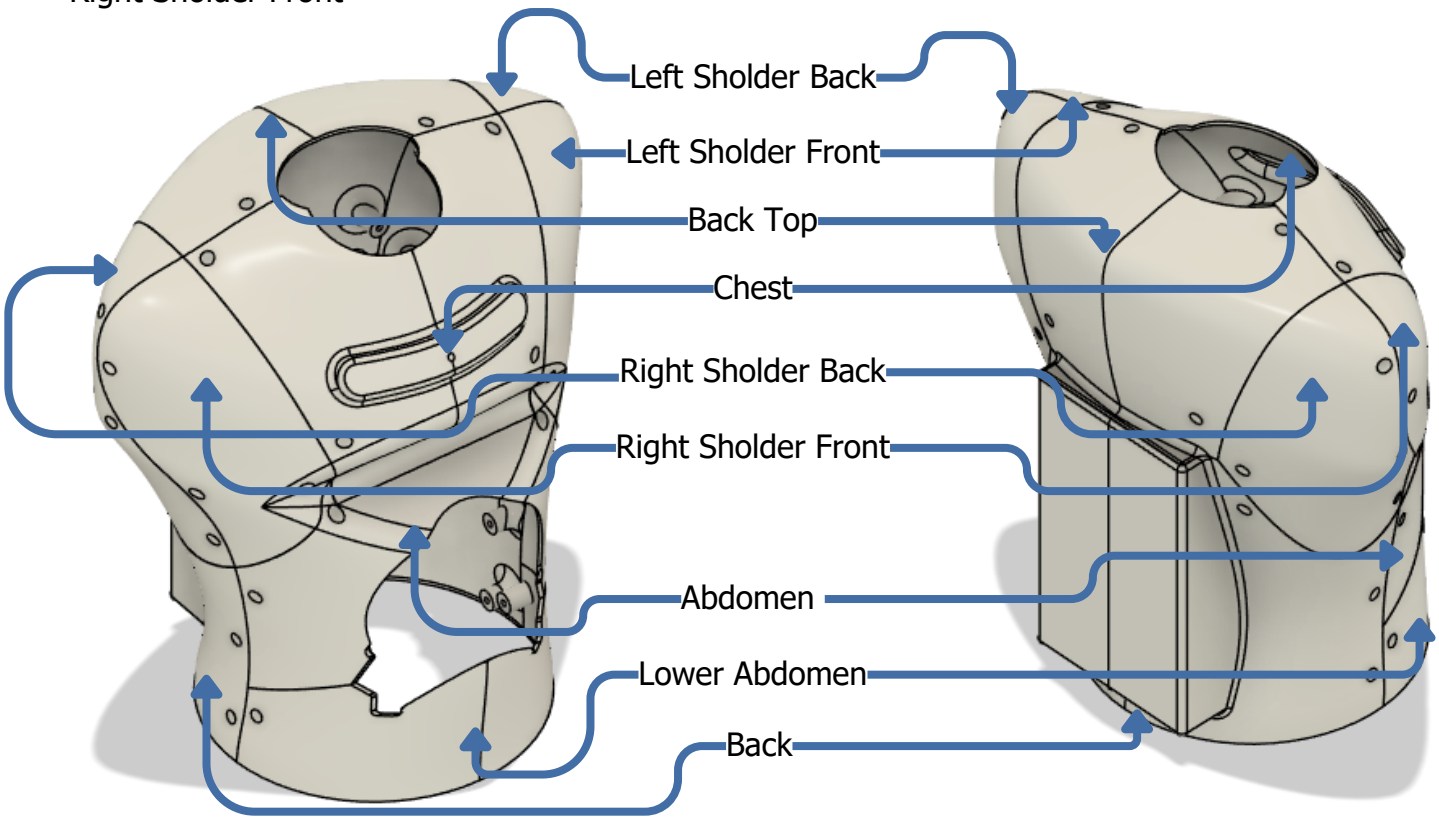
Lower Abdomen



Right Sholder Back

Right Sholder Front

"Left Shoulder Back", "Left Shoulder Front", "Abdomen", "Chest", "Back", "Back Top", "Lower Abdomen", "Right Shoulder Back", "Right Shoulder Front", these nine parts will create the outer perimeter of the main body.



Here we will discuss the computer that goes inside the base of the robot. You can choose any computing module you like, for which you have to 3D print the necessary setup for that.

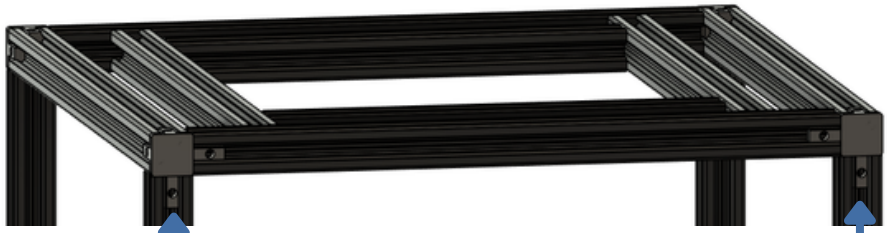
So to fit the PC inside the robot, first we have to remove/separate the entire stucture of the robot again and separate the bace from the main body of the robot.

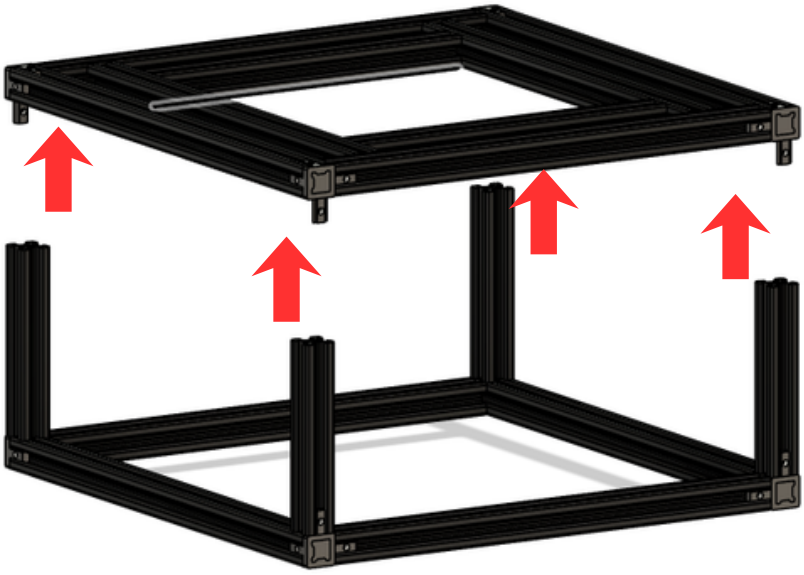
As we we instructed earlier not to attack the bearing, "Main Body Lifter" parts permanently with the base of the robot. So now if you lift the main body it should easily separate from the base.

Loosen the screw of three way connectors that are attached with 118.0mm 2020 Alumunium Profiles and separate the top from assembly from the base. Now we will attach all the computer accessories here and the complete the base, and set the whole body again.

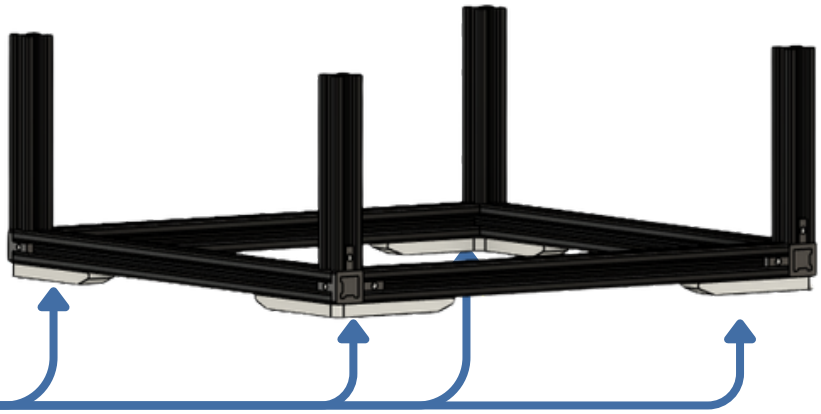
In our build, we used an ITX system with the following specifications:

- AMD Ryzen 5 7500F
- MAXSUN B850ITX ICE (Wi-Fi 6)
- Kingston FURY Beast 8GB DDR5 5600 MHz CL36
- Team MP44L 500GB M.2 PCIe Gen4 NVMe
- DeepCool CH170 Digital Mini Tower (only the power button and USB ports from the case were used; the rest of the chassis was not used)
- DeepCool AN600 Low-Profile 120 mm CPU cooler
- Gigabyte GeForce RTX 3050 OC Low Profile 6G 6GB GDDR6
- Cooler Master PCIe 4.0 x16 300 mm riser cable
- DeepCool RF 120 W White LED fan
- CORSAIR CX Series CX650 650W 80 Plus Bronze



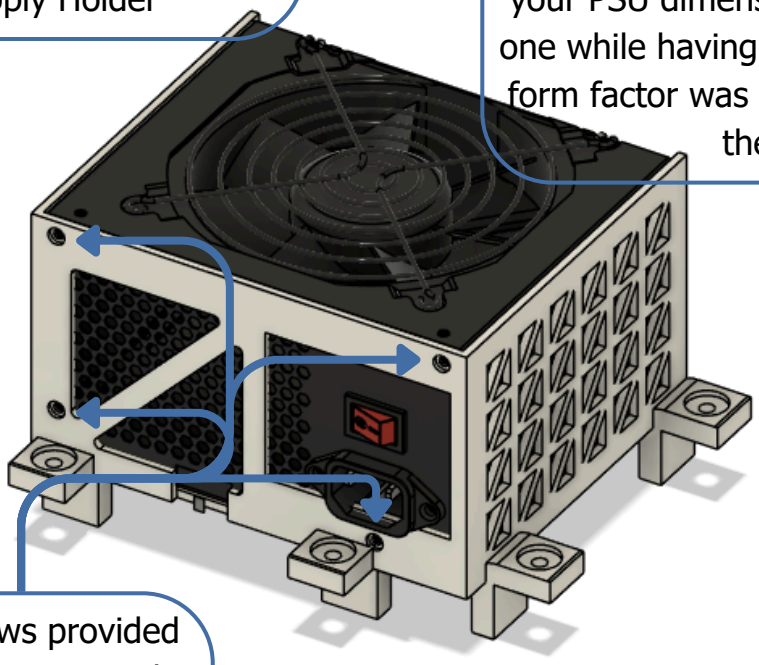


3D Print the part "Feet" and attach it with the aluminium extrusions.



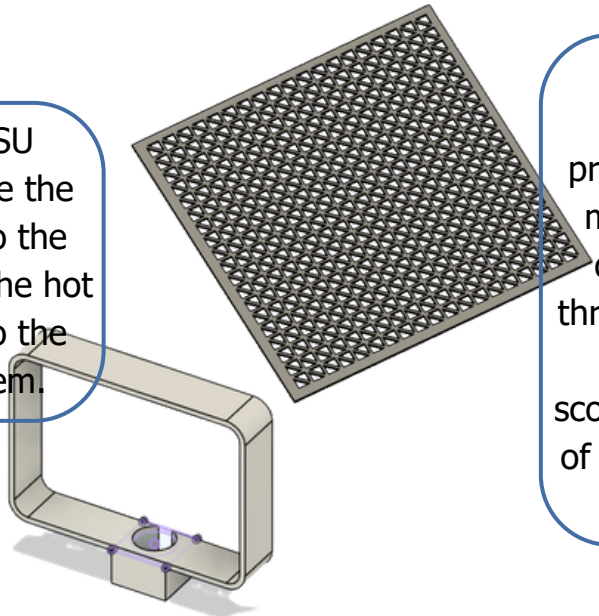
First we will setup the PSU - Power Supply Unit, for this we have to 3D Print the "Power Supply Holder"

Remmeber this psu holder was designed for the Corsair CX650 PSU (125mm in length, 150mm in width, and 86mm in height), so if you are using this holder for your build make sure that your PSU dimension is smaller or equal to this one while having the ATX form factor. The ATX form factor was choosen to reduce the cost of the overall setup.



Use Screws provided with PSU to securely connect the "Power Supply Holder"

Next 3D Print the "PSU Exhust" which will guide the exhust from the PSU to the outside and make sure the hot air doesn't flow back to the other parts of the system.



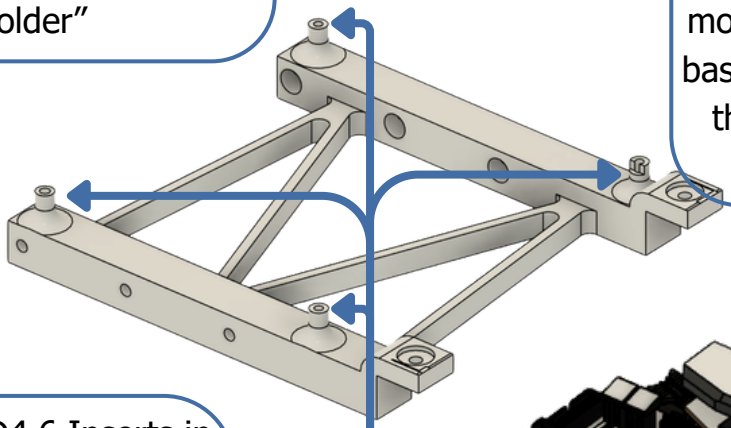
Also 3D printe the "PSU Shield" which acts like a protective mesh. This part is made so that large screws don't can't enter the PSU through the large openings of the PSU inlet. Just use a scotch tape or a small amount of glue to fit the mesh on top of the PSU

Next, Attach 10mm M5 Socket Cap Screws on top and M5 Drop In Nut on the other side of tthe screw and do the same for all of foure holes in PSU holder. Place the "Power Supply Holder" (which now have the PSU too) in one side of corner and tighten the screws and make sure a secure fit.

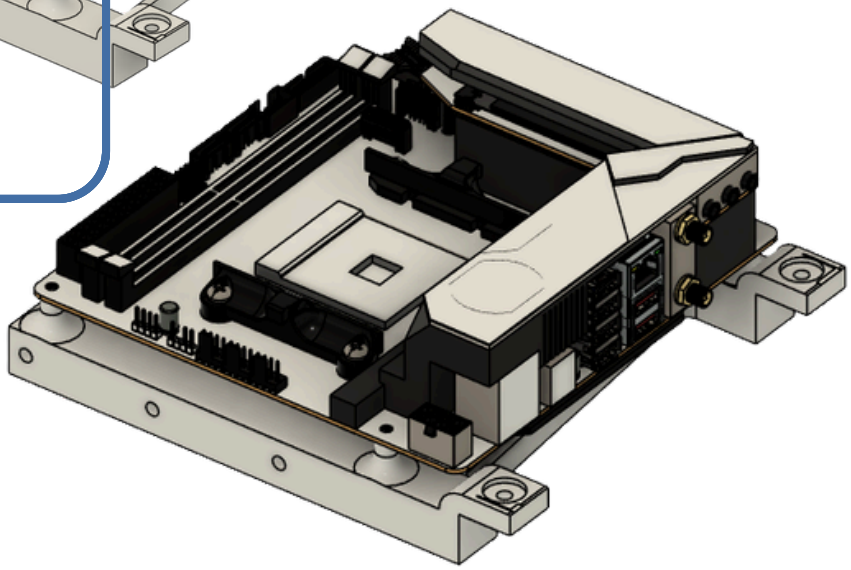
Any corner will be suitable as the base is symetric. This is what the final setup should look like:

Now the PSU is set, it is time for the motherboard. Here we are using a MAXSUN B850ITX ICE (Wi-Fi 6) any ITX motherboard will fit inside the base suing the same mount as the screws are of standard sizes.

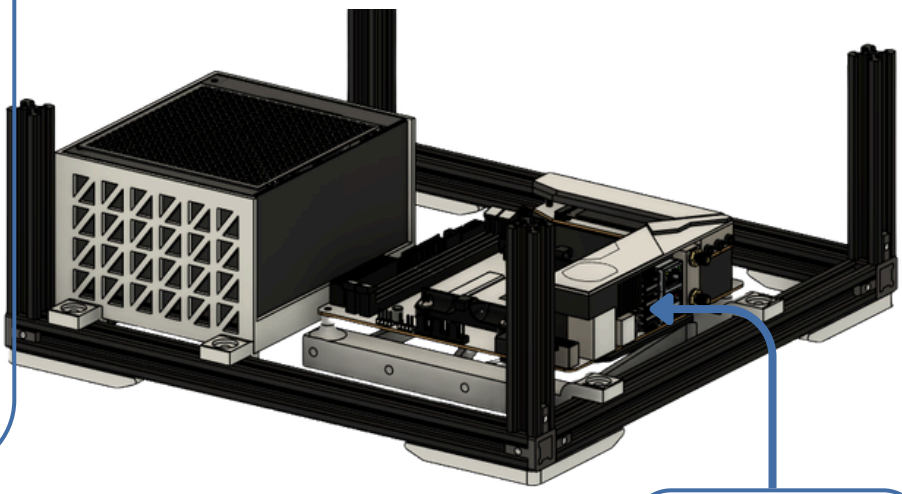
3D print the "Motherboard Holder"



Add M3*5.7-OD4.6 Inserts in the four holes and then fit the motherboard and screw using the screws provided with the motherboard or case.

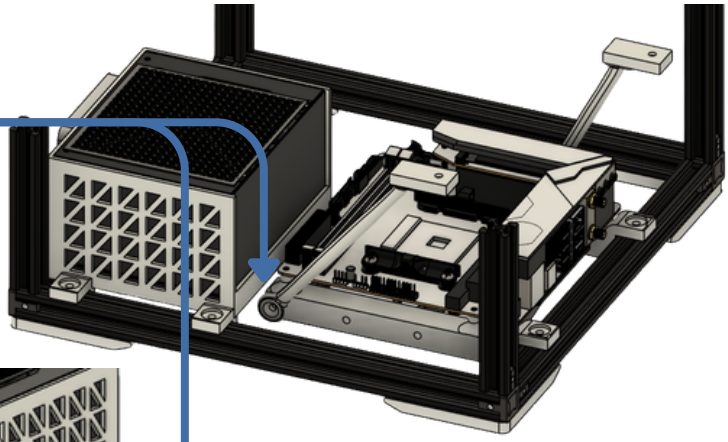


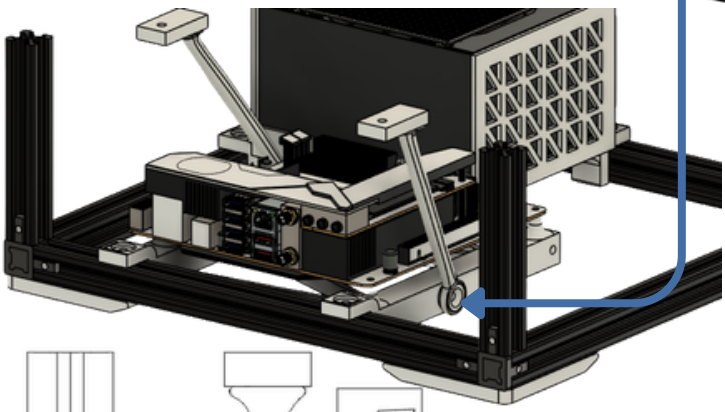
Now fit the mothboard along with the "Motherboard Holder" in the base. It will look like this:
(Make sure to follow the proper measurement while you screw dwon the motherboard as the IO output of the motherboard has to be aligned with the laser cut acrylic)

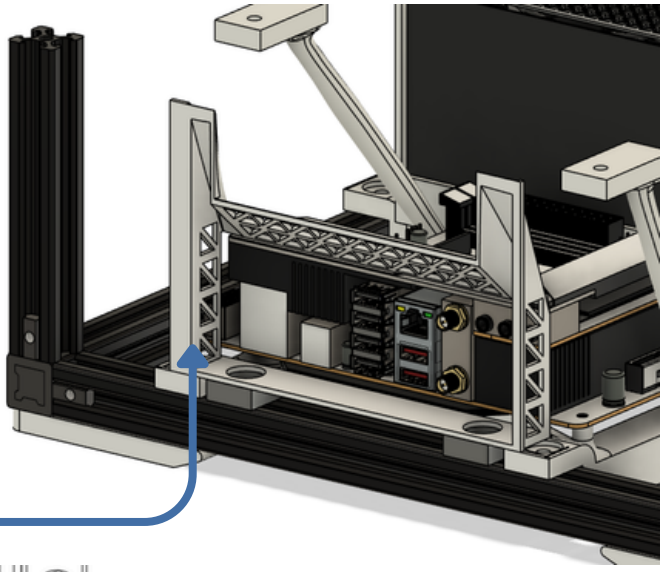


IO Outputs

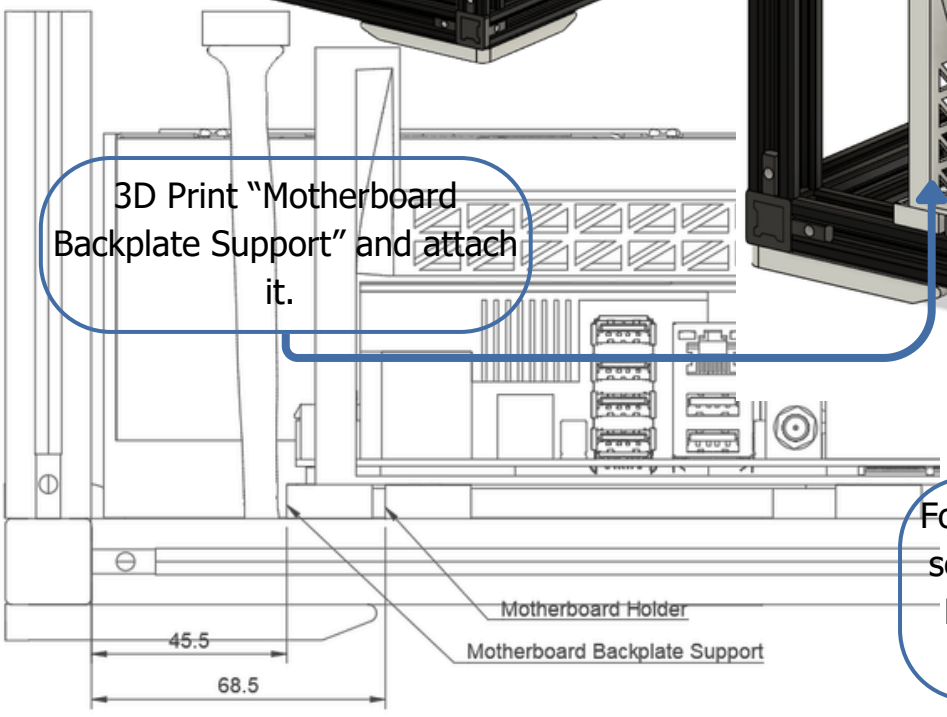
Print two mothboard supports "Motherboard Support 1" and "Motherboard Support 2" which will pull the end of motherboard from top. So it will prevent any kinda of sag in the motherboard holder and keep the motherboard safe.





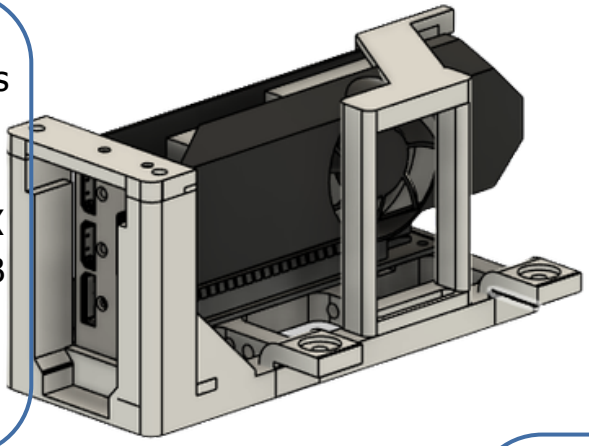


3D Print "Motherboard Backplate Support" and attach it.

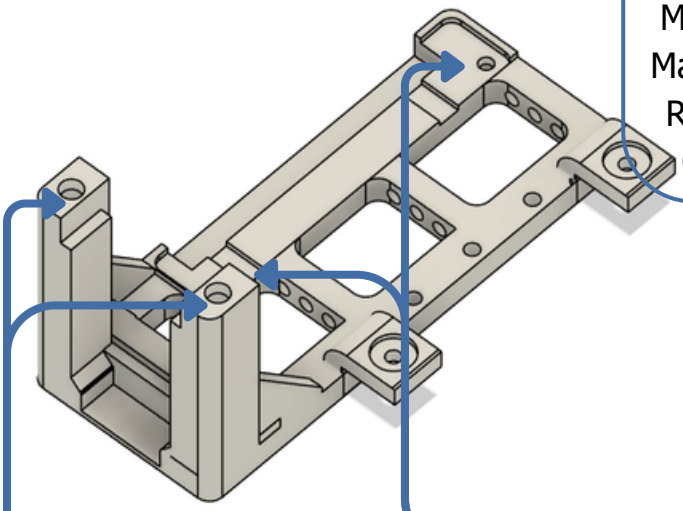


Follow these parameter while setting up the "Motherboard Holder" and "Motherboard Backplate Support"

Next we will print parts that holds the GPU. This design is can be used for any Low Profile GPU. Here, we are using Gigabyte GeForce RTX 3050 OC Low Profile 6G 6GB GDDR6 and Cooler Master PCIe 4.0 x16 300mm Riser Cable.

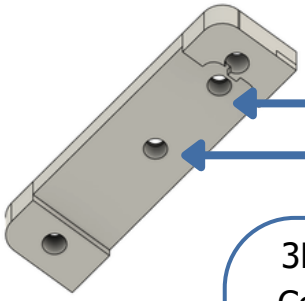


First 3D print the "GPU Holder Main Part", here the Cooler Master PCIe 4.0 x16 300mm Riser Cable will sit and the GPU will sit on top of it.



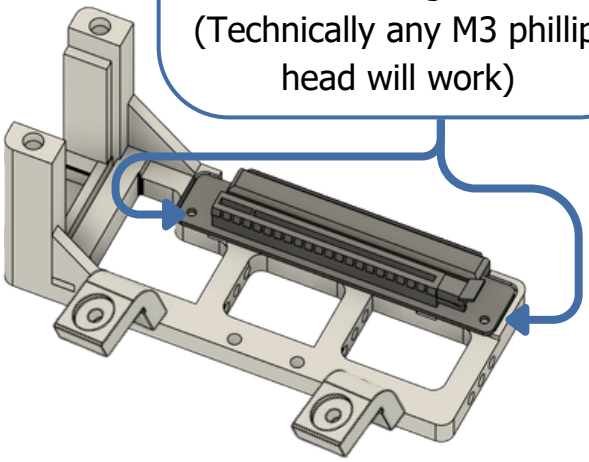
After it is printed then insert the M5 Inserts (Height: 8mm, Outer Diameter: 7mm) into these two holes.

Then insert the M3*5.7-OD4.6 Inserts here into the small holes that will hold the Riser Female

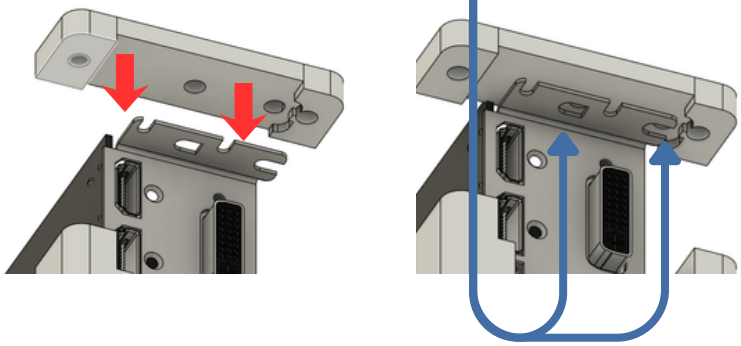


3D print the "GPU Shield Connector" and after ti is printed insert M3*5.7-OD4.6 Inserts in these holes.

Now attack the Riser Cable (PCIe Port) into the "GPU Holder Main Part" and secure them using proper screws from the Casing Screw Set (Technically any M3 phillips head will work)

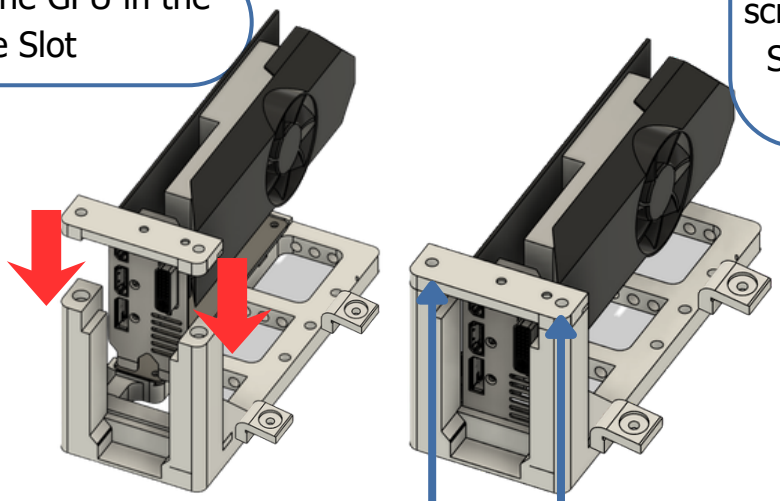


Connect GPU with the "GPU Shield Connector" using M3 Screws



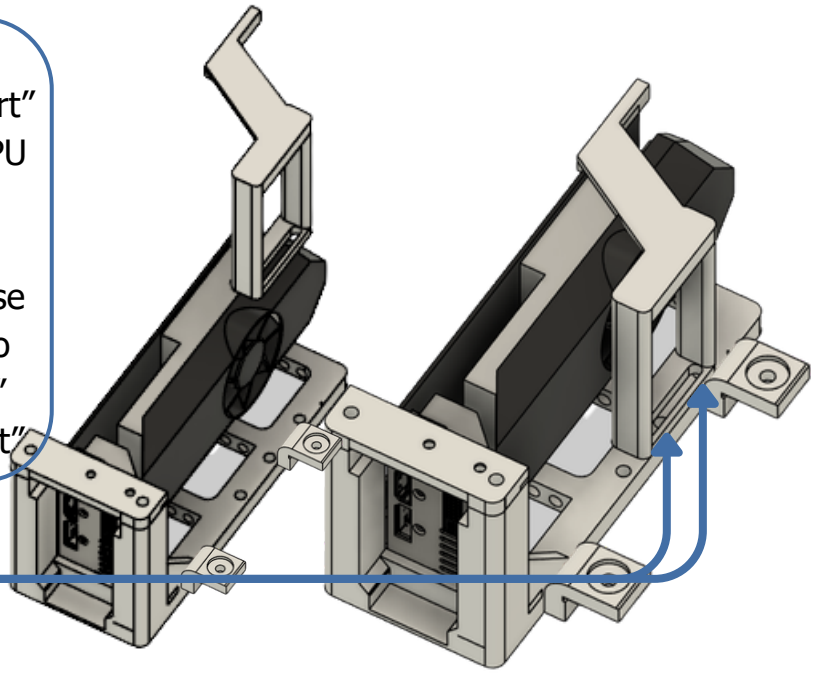
Now attach the GPU in the PCIe Slot

Secure the GPU with M5 screws (10mm) with the "GPU Shield Connecto" and "GPU Main Part"



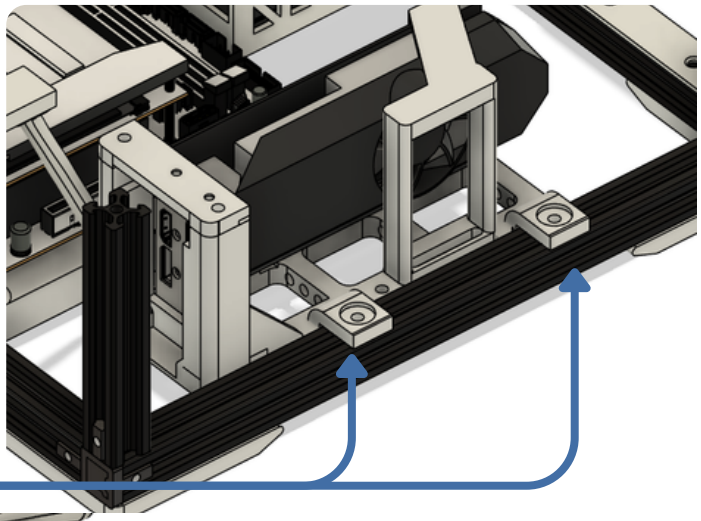
Connect the additional (optional) part "GPU Support" if you want to give your GPU more support.

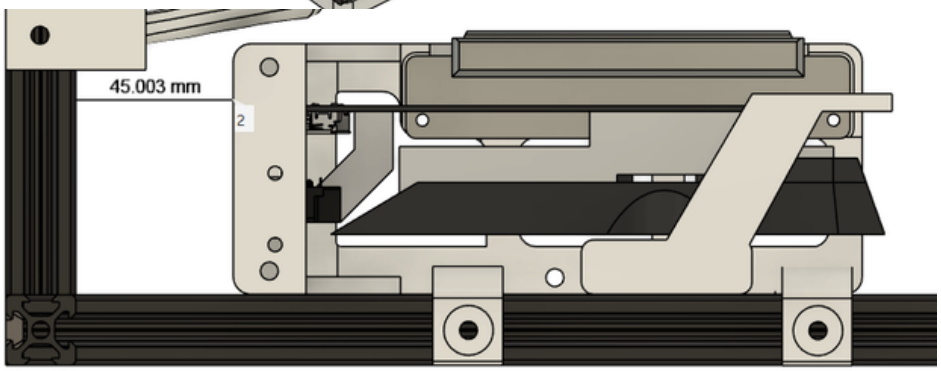
Align the holes and then use M5 Screw and Hex Nuts to Secure the "GPU Support" with "GPU Holder Main Part"



Finally connect the entire GPU Setup in teh base of the robot. It should look like something like this.

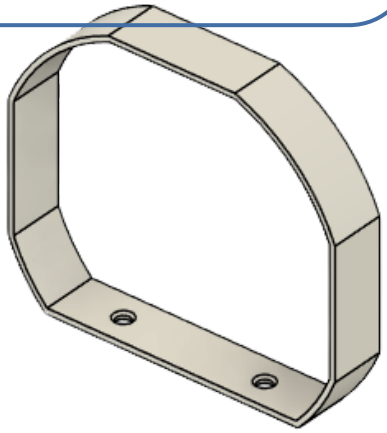
Give enough space for HDMI cable and tighten the screws with 2020 profiles.



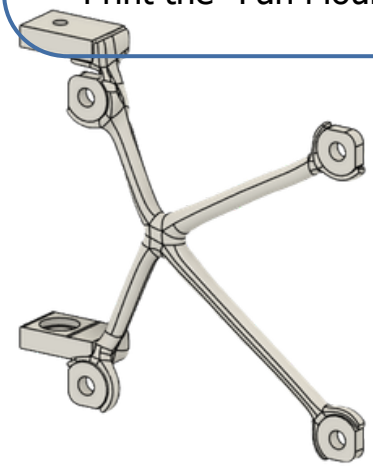


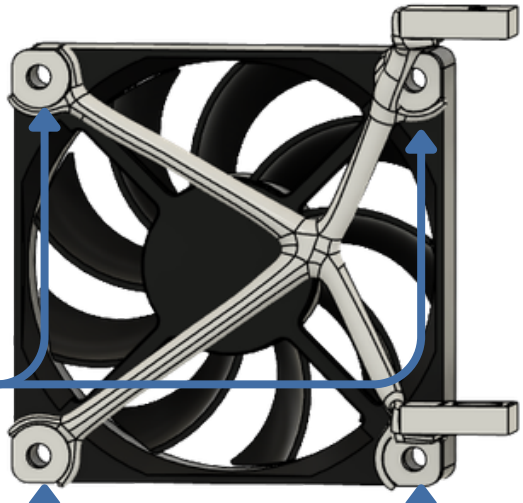
Now we will attach the 120mm FAN

Print the "Air Guide"



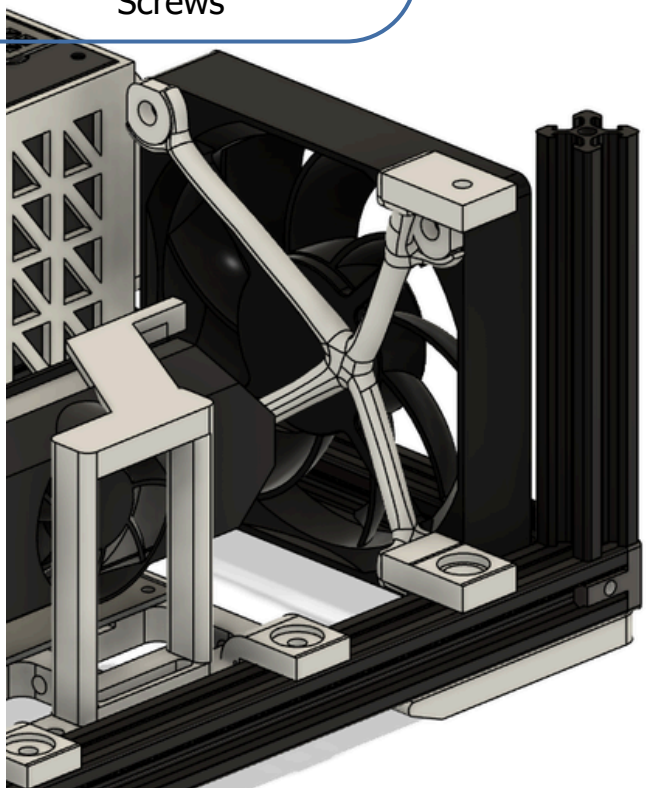
Print the "Fan Mount"



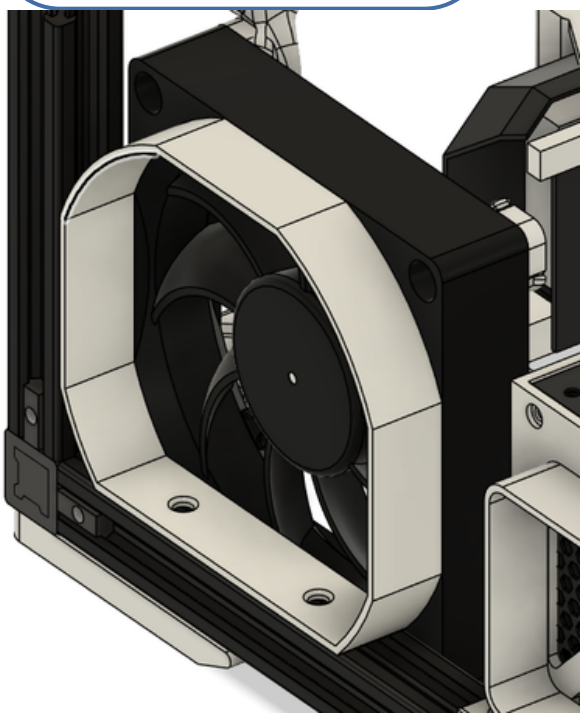


Attach "Fan Mount" with the 120mm Fan using the screws that comes with the Fan or Motherboard.

Attach "Fan Mount" and Fan in the base of the robt. Using M5 12mm Countersunk Screws

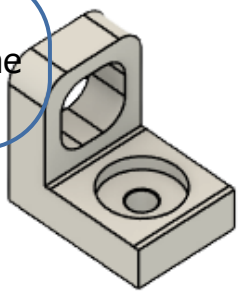


Attach the "Air Guide" Too



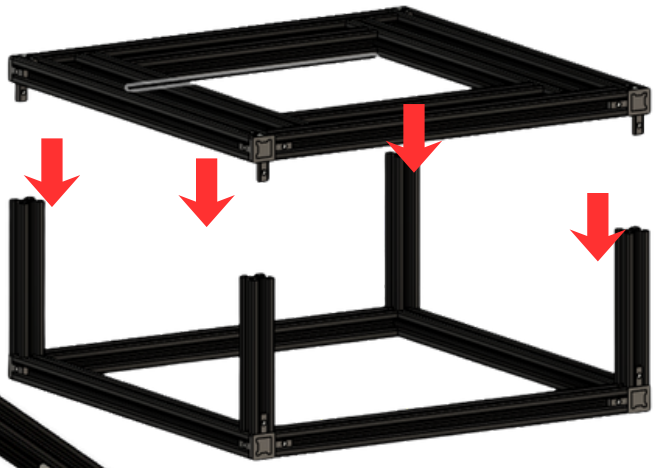
Now most of our components printing is complete.

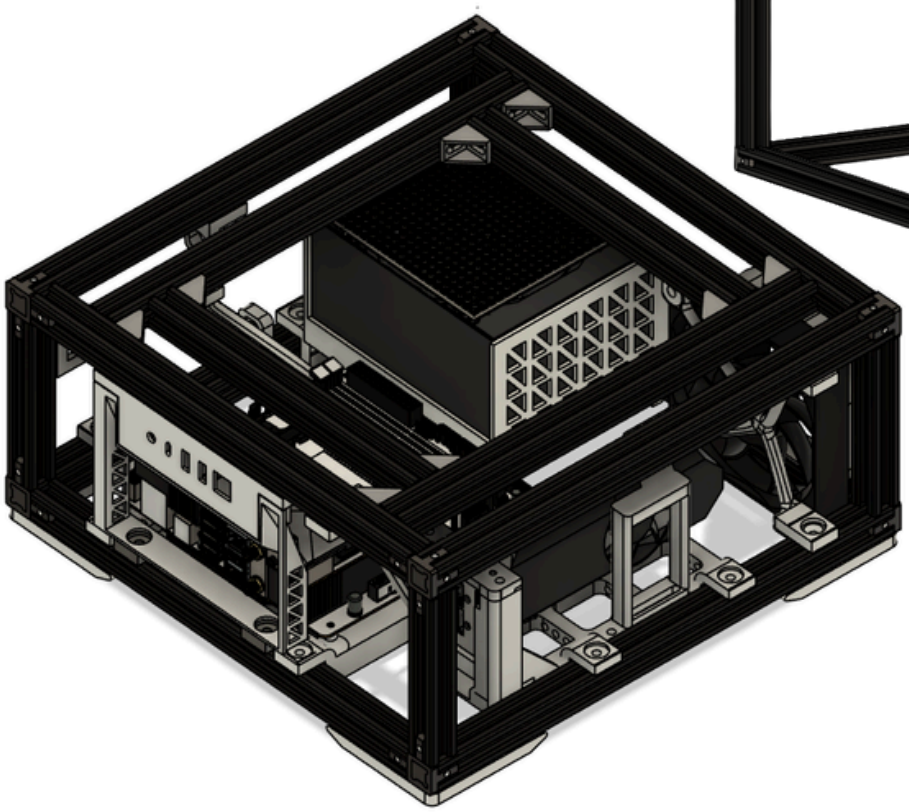
To make the cable tying process easy, 3D print some of these "Wire Guide"



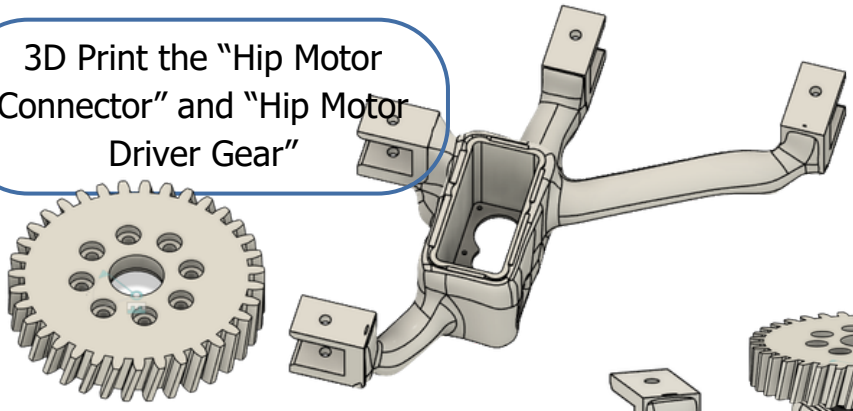
Now route the cables and test if everything powers on correctly or not. I suggest having an exper infront of you when you start routing your cable is very important. This is a lot harder task.

Attach the top alumunium profile assembly again, and screw everything.

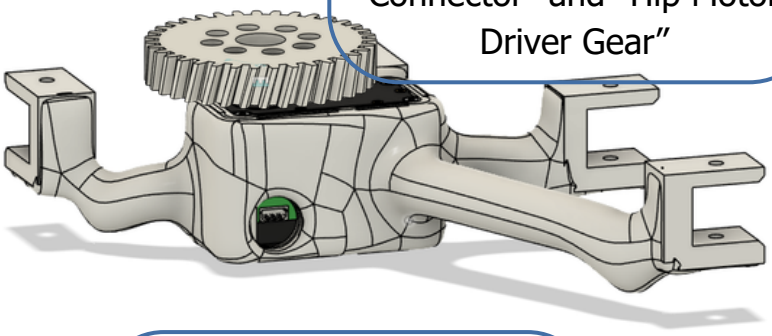




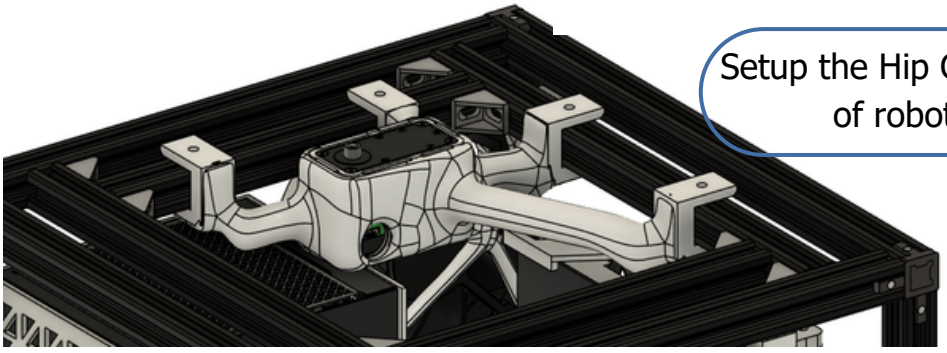
3D Print the "Hip Motor Connector" and "Hip Motor Driver Gear"



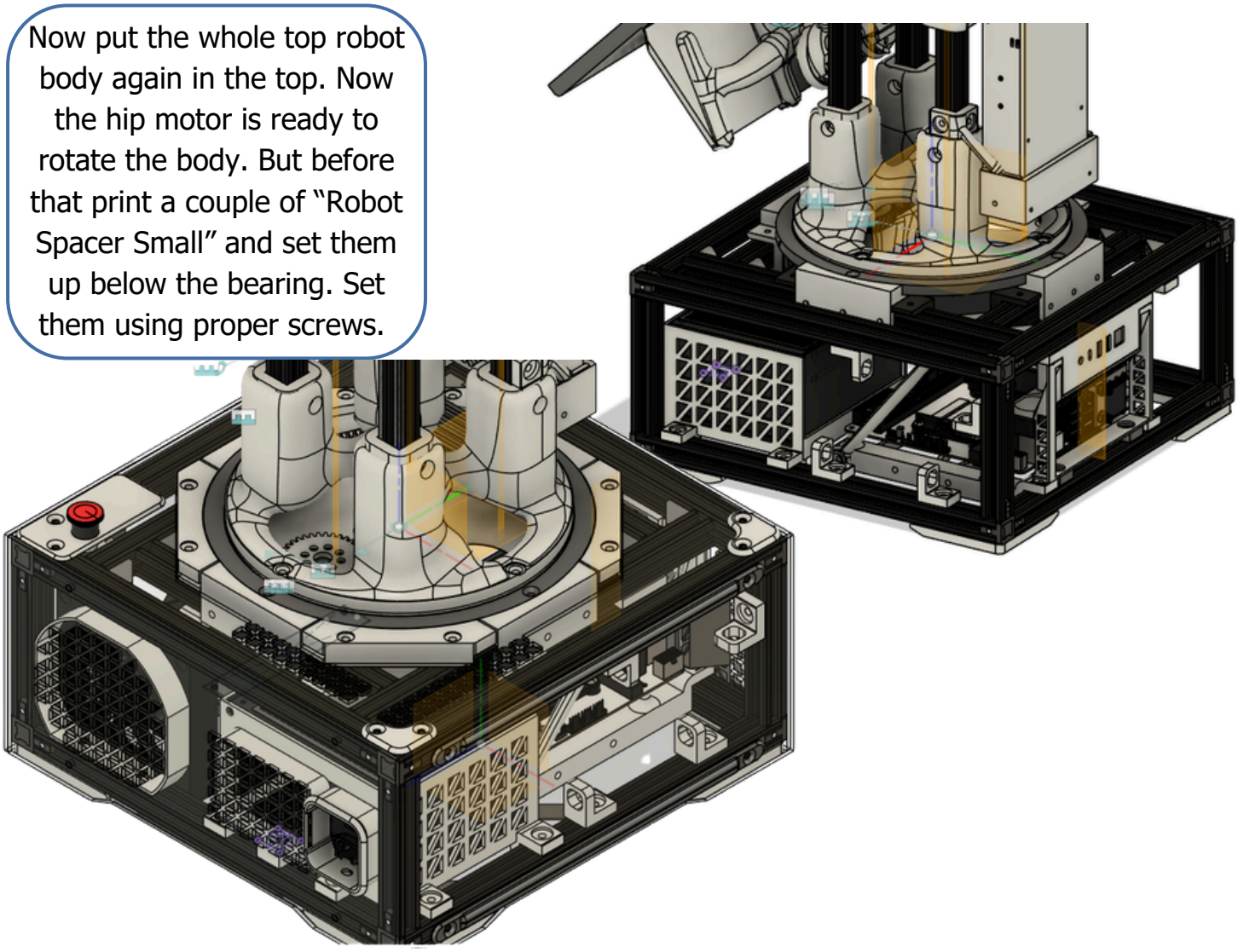
Connect the "Hip Motor Connector" and "Hip Motor Driver Gear"



Setup the Hip Gear in the top of robot base.



Now put the whole top robot body again in the top. Now the hip motor is ready to rotate the body. But before that print a couple of "Robot Spacer Small" and set them up below the bearing. Set them using proper screws.

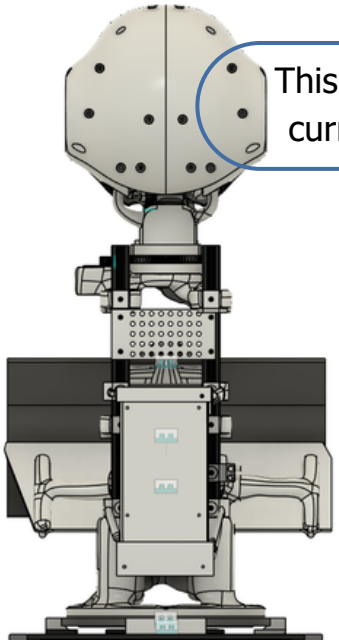


Once body is again in position, connect all the wires

Test and see everything works like Servo, Eye Animation, Camera (Currently in the chest)

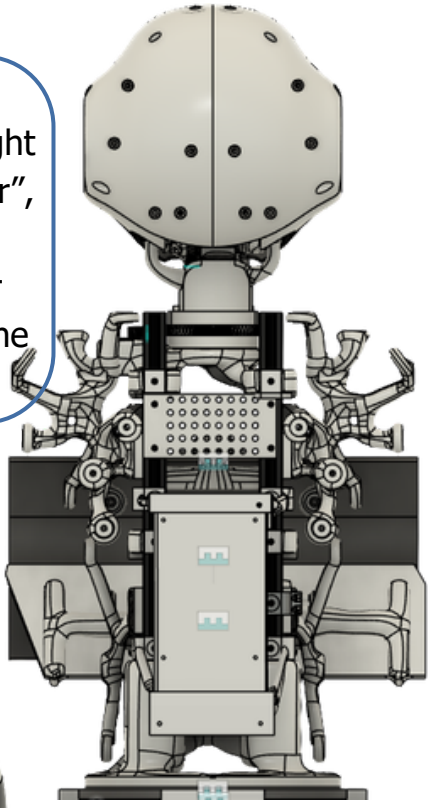
If everything works properly, we can go on to our next step, which is completing the full body.

So far we have a complete robot but without any appearance. Let's change that.

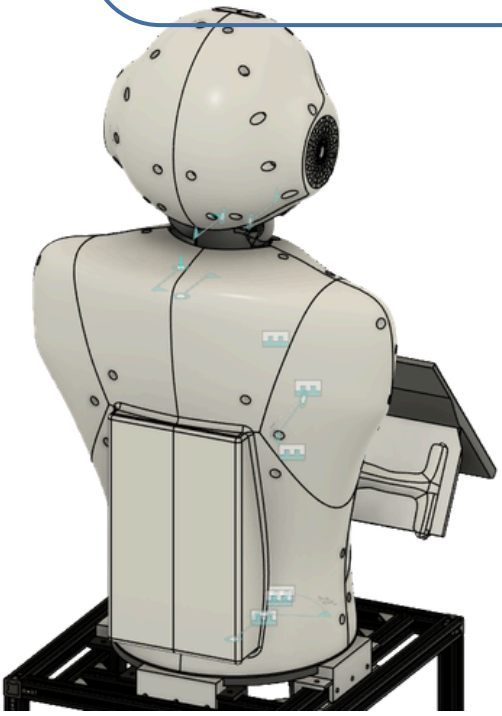


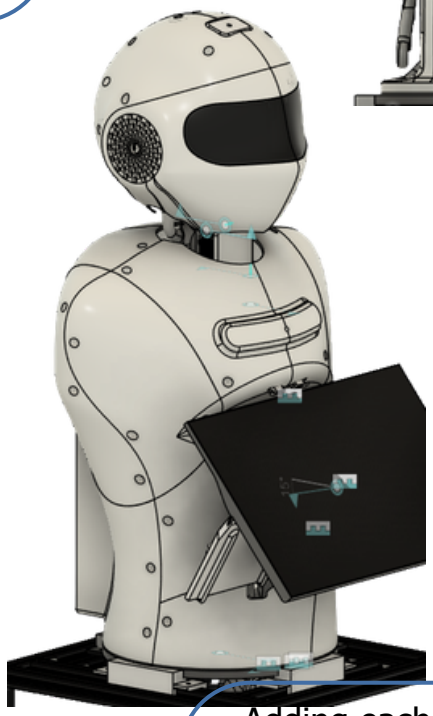
This is what the robot currently looks like.

Adding the "Left Upper Generated Connector", "Right Upper Generated Connector", "Left Lower Generated Connector", "Right Lower Generated Connector" to the core.



Adding each of the nine pannels one by one using M5 Countersunk 16mm screws

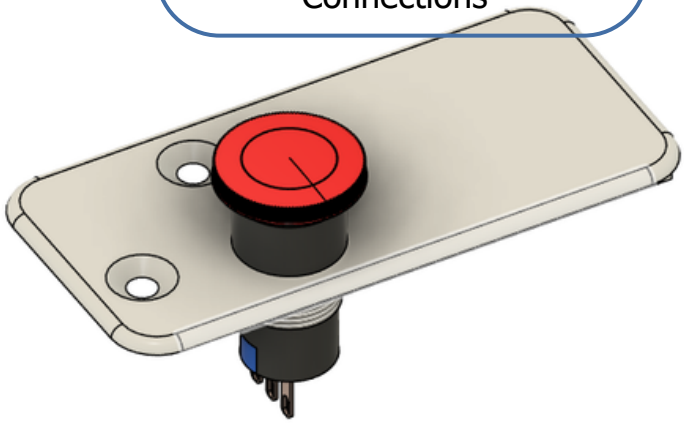




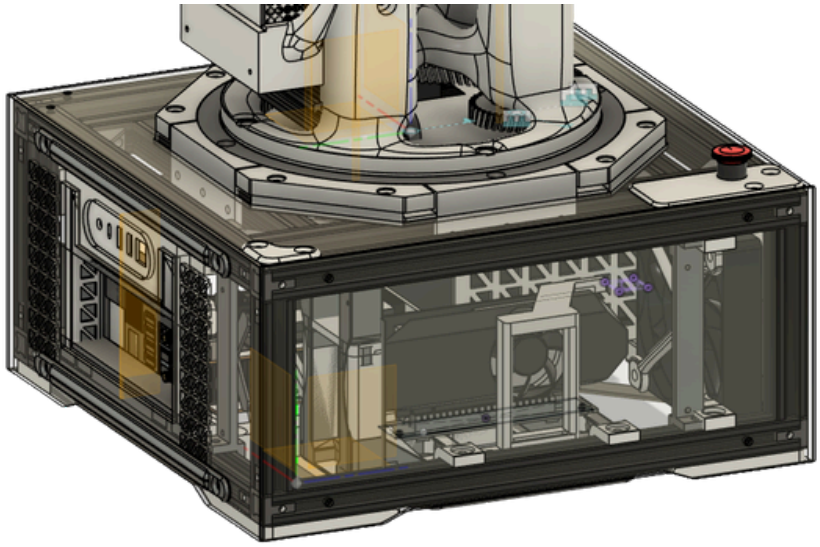
Adding each of the nine pannels one by one. And now CogniCare is ready. But the outside board aren't ready yeat.

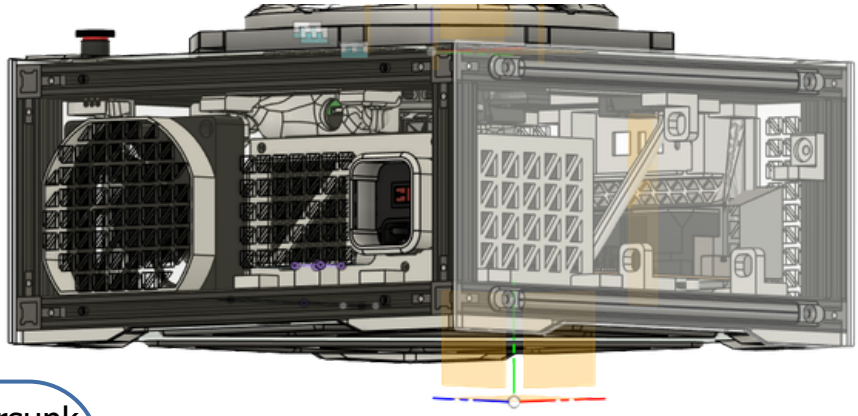
The outside board needs to be laser cut before can be attached. With this guide the

3D print the "Switch Connections"



3D print the "Top Support" about 4 times, "Side Panel Support 1" about 4 times, "Side Pannel Support 2" about 4 times





Then use 12mm Countersunk and drop in nuts to connect them all.

Our Task of Building CogniCare is Complete